

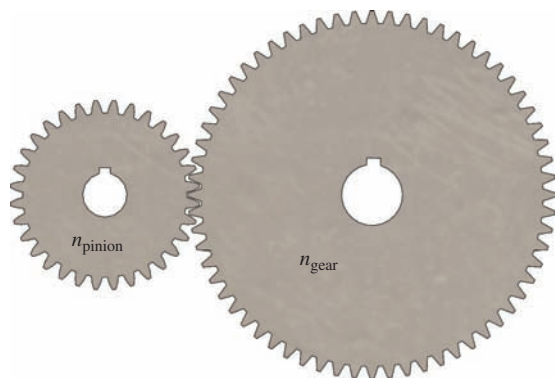


FIGURE 8-39 Two spur gears in mesh with the pinion driving the gear

of rotation and rotating with an angular velocity, ω , is found from

⇒ Pitch Line Speed of a Gear

$$v_t = R \omega \quad (8-32)$$

Using the subscript P for the pinion and G for the gear for two gears in mesh, we have

$$v_t = R_P \omega_P \quad \text{and} \quad v_t = R_G \omega_G$$

This set of equations says that the pitch line speeds of the pinion and the gear are the same. Equating these two and solving for ω_P/ω_G gives our definition for the velocity ratio, VR :

$$VR = \omega_P/\omega_G = R_G/R_P$$

In general, it is convenient to express the velocity ratio in terms of the pitch diameters, the rotational speeds, or

the numbers of teeth of the two gears in mesh. Remember that

$$R_G = D_G/2$$

$$R_P = D_P/2$$

$$D_G = N_G/P_d$$

$$D_P = N_P/P_d$$

$$n_P = \text{rotational speed of the pinion (in rpm)}$$

$$n_G = \text{rotational speed of the gear (in rpm)}$$

The velocity ratio can then be defined in any of the following ways:

⇒ Velocity Ratio for Gear Pair

$$VR = \frac{\omega_P}{\omega_G} = \frac{n_P}{n_G} = \frac{R_G}{R_P} = \frac{D_G}{D_P} = \frac{N_G}{N_P} = \frac{\text{speed}_P}{\text{speed}_G} = \frac{\text{size}_G}{\text{size}_P} \quad (8-33)$$

Another useful form for velocity ratio is shown next and is used in following example problems.

$$VR = \frac{\omega_{\text{input}}}{\omega_{\text{output}}} = \frac{N_{\text{output}}}{N_{\text{input}}}$$

Most gear drives are *speed reducers*; that is, their output speed is lower than their input speed. This results in a velocity ratio greater than 1. If a *speed increaser* is desired, then VR is less than 1. Note that not all books and articles use the same definition for velocity ratio. Some define it as the ratio of the output speed to the input speed, the inverse of our definition. It is thought that the use of VR greater than 1 for the reducer—that is, the majority of the time—is more convenient.

Let's look at examples of both a speed decreaser and a speed increaser.

Example Problem 8-6

- a. Figure 8-40 shows a gear pair in which a 20-tooth pinion drives a 40-tooth gear. The pinion has an angular velocity of 1000 rpm. Determine the velocity ratio and the angular velocity of the gear.

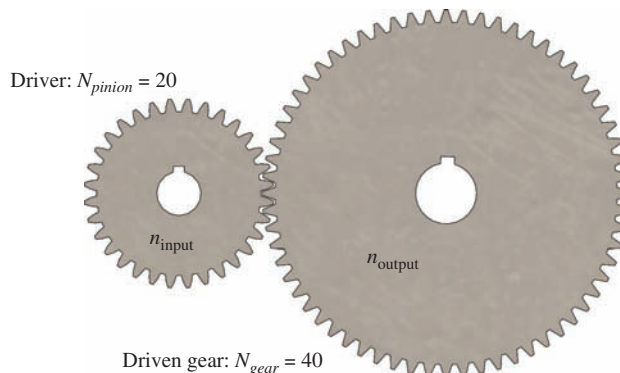


FIGURE 8-40 Speed decreaser gear drive

The pinion is the input that drives the gear. The gear is the output that is driven by the pinion. The velocity ratio is:

$$VR = \frac{\omega_{\text{input}}}{\omega_{\text{output}}} = \frac{N_{\text{output}}}{N_{\text{input}}} = \frac{40}{20} = \frac{2}{1}$$

This gear drive system is a speed reducer where the input angular velocity is twice the output angular velocity. The output angular velocity can be determined by:

$$VR = \frac{\omega_{\text{input}}}{\omega_{\text{output}}}$$

$$\omega_{\text{output}} = \frac{\omega_{\text{input}}}{VR} = \frac{1000 \text{ rpm}}{2} = 500 \text{ rpm}$$

The input angular velocity is 1000 rpm and the velocity ratio is 2 which gives us an output angular velocity of 500 rpm.

- b. Let's take the gear and place it on the input drive shaft and place the pinion on the output drive shaft as shown in Figure 8–41. The 40-tooth gear is now the input and drives the 20-tooth pinion which is now the output. Determine the velocity ratio and the output angular velocity if the input shaft rotates at 1000 rpm.

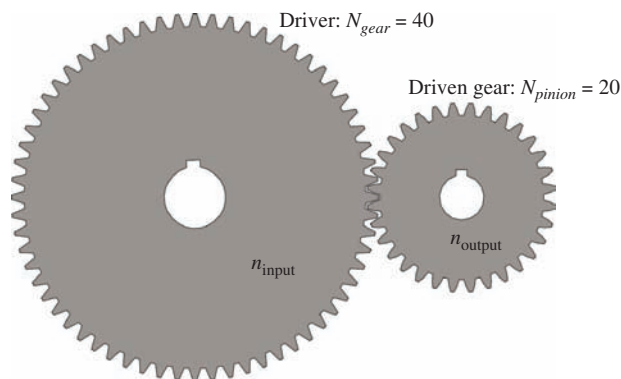


FIGURE 8–41 Speed increaser gear drive

In this example, the gear is the input and drives the pinion. The pinion is the output and is driven by the gear. The velocity ratio is:

$$VR = \frac{\omega_{\text{input}}}{\omega_{\text{output}}} = \frac{N_{\text{output}}}{N_{\text{input}}} = \frac{20}{40} = \frac{1}{2}$$

This gear drive system is a speed increaser. The output angular velocity can be determined:

$$VR = \frac{\omega_{\text{input}}}{\omega_{\text{output}}}$$

$$\omega_{\text{output}} = \frac{\omega_{\text{input}}}{VR} = \frac{1000 \text{ rpm}}{0.5} = 2000 \text{ rpm}$$

Keeping the input angular velocity at 1000 rpm, the velocity ratio is 1/2 which gives us an output angular velocity of 2000 rpm. While most gear drives are speed reducers, some gear drives are speed increasers.

Train Value

When more than two gears are in mesh, the term train value (TV) refers to the ratio of the input speed (for the first gear in the train) to the output speed (for the last gear in the train). By definition the train value is the product of the values of VR for

each gear pair in the train. In this definition, a gear pair is any set of two gears with a driver and a follower (driven) gear.

Again, TV will be greater than 1 for a reducer and less than 1 for an increaser. For example, consider the gear train shown in Figure 8–42. The input is through the

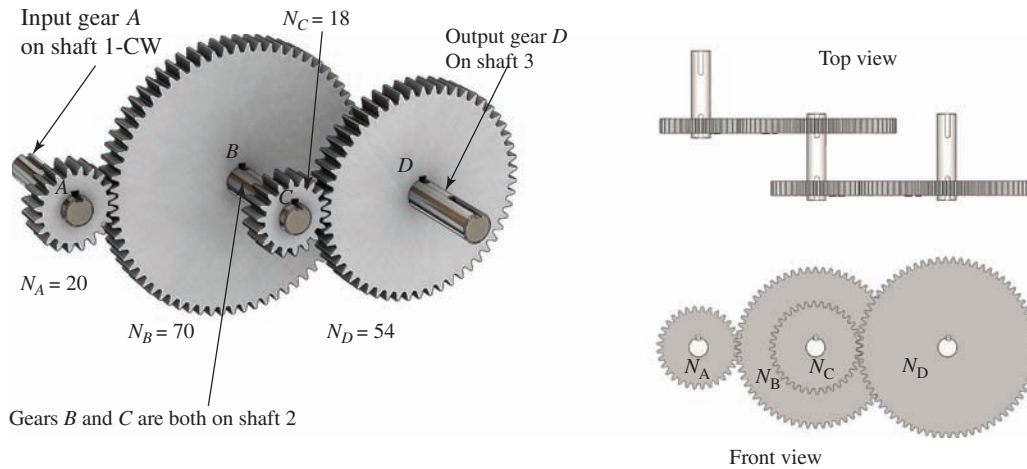


FIGURE 8-42 Double-reduction gear train

shaft carrying gear A. Gear A drives gear B. Gear C is on the same shaft with gear B and rotates at the same speed. Gear C drives gear D, which is connected to the output shaft. Then gears A and B constitute the first gear pair, and gears C and D constitute the second pair. The velocity ratios are

$$VR_1 = n_A/n_B \quad VR_2 = n_C/n_D$$

The train value is

$$TV = (VR_1)(VR_2) = \frac{n_A n_C}{n_B n_D}$$

But, because they are on the same shaft, $n_B = n_C$, and the preceding equation reduces to

$$TV = n_A/n_D$$

This is the input speed divided by the output speed, the basic definition of the train value. This process can be expanded to any number of stages of reduction in a gear train.

Remember that any of the forms for velocity ratio shown in Equation (8-33) can be used for computing the train value. In design, it is often most convenient to express the velocity ratio in terms of the numbers of teeth in each gear because they must be integers. Then, once the diametral pitch or module is defined, the values of the diameters or radii can be determined.

The train value of the double-reduction gear train in Figure 8-42 can be expressed in terms of the numbers of teeth in the four gears as follows:

$$VR_1 = N_B/N_A$$

Note that this is the number of teeth in the *driven* gear B divided by the number of teeth in the *driving* gear A. This is the typical format for velocity ratio. Then VR_2 can be found similarly:

$$VR_2 = N_D/N_C$$

Thus, the train value is

$$TV = (VR_1)(VR_2) = (N_B/N_A)(N_D/N_C)$$

This is usually shown in the form

Train Value

$$TV = \frac{N_B N_D}{N_A N_C}$$

$$TV = \frac{\text{product of number of teeth in the driven gears}}{\text{product of number of teeth in the driving gears}} \quad (8-34)$$

This is the form for train value that we will use most often.

The direction of rotation can be determined by observation, noting that there is a direction reversal for each pair of external gears.

We will use the term positive train value to refer to one in which the input and output gears rotate in the same direction. Conversely, if they rotate in the opposite direction, the train value will be negative.

Example Problem 8-7

For the gear train shown in Figure 8-42, if the input shaft rotates at 1750 rpm clockwise, compute the speed of the output shaft and its direction of rotation.

Solution

We can find the output speed if we can determine the train value:

$$TV = n_A/n_D = \text{input speed/output speed}$$

Then

$$n_D = n_A/TV$$

But

$$TV = (VR_1)(VR_2) = \frac{N_B}{N_A} \frac{N_D}{N_C} = \frac{70}{20} \frac{54}{18} = \frac{3.5}{1} \frac{3.0}{1} = \frac{10.5}{1} = 10.5$$

Now

$$n_D = n_A/TV = (1750 \text{ rpm})/10.5 = 166.7 \text{ rpm}$$

Gear *A* rotates clockwise; gear *B* rotates counterclockwise.

Gear *C* rotates counterclockwise; gear *D* rotates clockwise.

Thus, the train in Figure 8–42 is a positive train.

Idler Gear

The gear train shown in Figure 8–42 is sometimes referred to as a compound gear train. A compound gear has two gears mounted on one shaft. Gears *B* and *C* are mounted on the same shaft and would be considered compound gears. Since gears *B* and *C* are mounted on the same shaft, they also have the same angular velocity. Figure 8–43 shows a gear train that has only one gear mounted on each shaft and is referred to as a simple gear train.

The train value for the simple gear train can be expressed as the ratio of the number of teeth of the output gear to the number of teeth of the input gear. Looking at the first two gears in the drive train, gear *A* is the driving gear and gear *B* is the driven gear. The velocity ratio for these two gears in mesh is:

$$VR_1 = \frac{N_B}{N_A}$$

The next set of gears in mesh are gear *B* which is the driving gear and gear *C* which is the driven gear. The velocity ratio for this gear set is:

$$VR_2 = \frac{N_C}{N_B}$$

Similarly, the velocity ratios can be determined for each of the remaining gear sets in mesh.

$$VR_3 = \frac{N_D}{N_C}$$

$$VR_4 = \frac{N_E}{N_D}$$

Thus the train value for the simple gear train is:

$$TV = \frac{\omega_A}{\omega_E} = (VR_1)(VR_2)(VR_3)(VR_4)$$

Substituting the number of gear teeth into the equation, the train value is:

$$TV = \frac{\omega_A}{\omega_E} = \frac{N_B}{N_A} \cdot \frac{N_C}{N_B} \cdot \frac{N_D}{N_C} \cdot \frac{N_E}{N_D}$$

This equation can be reduced by cancelling the number of teeth of gears *B*, *C*, and *D*. This leaves the following train value:

$$TV = \frac{\omega_A}{\omega_E} = \frac{N_E}{N_A}$$

The train value or velocity ratio of the entire gear drive system is reduced to the ratio of the number of teeth on the output gear to the number of teeth on the input gear. Gears *B*, *C*, and *D*, do not have any effect on the ratio. These gears are classified as *idler gears*.

The example above introduces the concept of the idler gear, defined as follows:

Any gear in a gear train that performs as both a driving gear and a driven gear is called an idler gear or simply an idler.

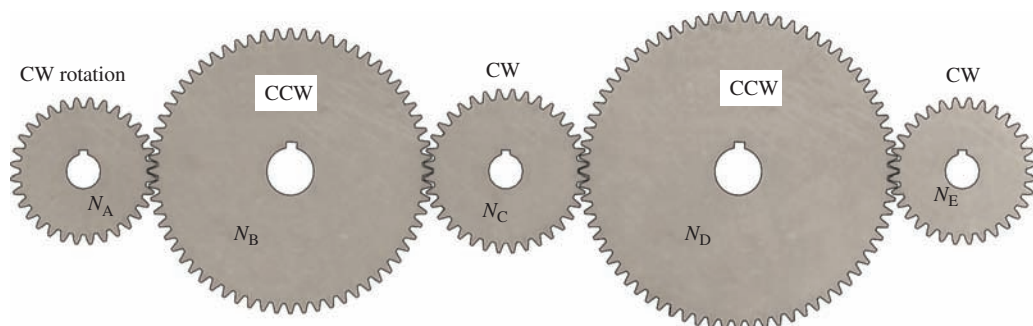


FIGURE 8–43 Simple gear train

The main features of an idler are as follows:

1. An idler does not affect the train value of a gear train because, since it is both a driver and driven gear, its number of teeth appears in both the numerator and denominator of the train value equation, Equation (8-34)
2. Placing an idler in a gear train causes a direction reversal of the output gear. In Figure 8-43, input gear A rotates clockwise and output gear E also rotates clockwise. If any one of the idler gears B, C, or D is removed, the direction of rotation of the output gear E will be counter-clockwise.
3. An idler gear may be used to fill a space between two gears in a gear train when the desired distance between their centers is greater than the center distance for the two gears alone.

Example Problem 8-8

Determine the train value for the train shown in Figure 8-44. If the shaft carrying gear A rotates at 1750 rpm clockwise, compute the speed and the direction of the shaft carrying gear E.

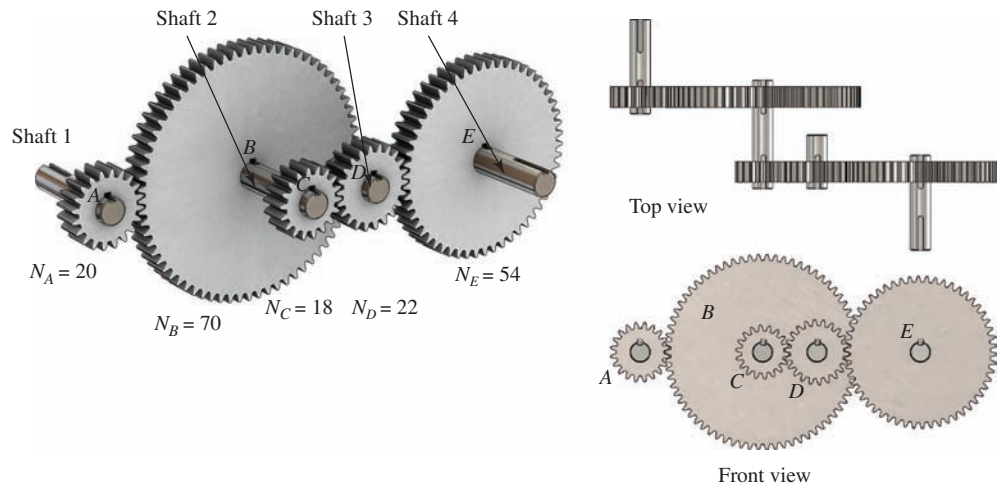


FIGURE 8-44 Double-reduction gear train with an idler. Gear D is an idler

Solution Look first at the direction of rotation. Remember that a gear pair is defined as any two gears in mesh (a driver and a follower). There are actually three gear pairs:

Gear A drives gear B: A rotates clockwise; B, counterclockwise.

Gear C drives gear D: C rotates counterclockwise; D, clockwise.

Gear D drives gear E: D rotates clockwise; E, counterclockwise.

Because gears A and E rotate in opposite directions, the train value is negative. Now

$$TV = -(VR_1)(VR_2)(VR_3)$$

In terms of the number of teeth,

$$TV = -\frac{N_B}{N_A} \frac{N_D}{N_C} \frac{N_E}{N_D}$$

Note that the number of teeth in gear D appears in both the numerator and the denominator and thus can be canceled. The train value then becomes

$$TV = -\frac{N_B}{N_A} \cdot \frac{N_E}{N_C} = -\frac{70}{20} \cdot \frac{54}{18} = -\frac{3.5}{1} \frac{3.0}{1} = -10.5$$

Gear D is called an *idler*. As demonstrated here, it has no effect on the magnitude of the train value, but it does cause a direction reversal. The output speed is then found from

$$TV = n_A/n_E$$

$$n_E = n_A/TV = (1750 \text{ rpm})/(-10.5) = -166.7 \text{ rpm (counterclockwise)}$$

Velocity of a Rack

Figure 8–45 shows the basic configuration of a *rack-and-pinion* drive. The function of such a drive is to produce a linear motion of the rack from the rotational motion of the driving pinion. The opposite is also true: If the rack is the driver having a linear motion, it produces a rotational motion of the pinion.

The linear velocity of the rack, v_R , must be the same as the pitch line velocity of the pinion, v_t , as defined by Equation (8–32), repeated here. Recall that ω_P is the angular velocity of the pinion:

$$v_R = v_t = R_P \omega_P = (D_P/2) \omega_P$$

Note on Units for Pitch Line Speed. In general, any velocity units may be used for pitch line speed at the discretion of the designer as long as careful manipulation of units is exercised. However, there are certain units that are preferred for later calculations in this book and for gear design in particular. The preferred units for pitch line speed are as follows:

U.S. Standard Diametral Pitch System: ft/min or fpm

Metric Module System: m/s

Certain data, design guidelines, and subsequent calculations are keyed to pitch line speed in these units. Because such calculations appear often in the following chapters, we develop here some unit-specific relationships to facilitate calculations in these units. Other assumptions for typical units for related quantities are included in these developments.

U.S. Standard Diametral Pitch System: Assumptions: Dimensions are in inches; Rotational speed is in rpm

$$v_t = (D/2) \omega = \frac{D(\text{in})}{2} \frac{n(\text{rev})}{(\text{min})} \frac{2\pi(\text{rad})}{(\text{rev})} \frac{1.0(\text{ft})}{12(\text{in})}$$

$$v_t = \frac{\pi D n}{12} \text{ft/min} \quad (8-35)$$

Metric Module System: Assumptions: Dimensions are in mm; Rotational speed is in rpm

$$v_t = (D/2) \omega$$

$$v_t = \frac{D(\text{mm})}{2} \frac{n(\text{rev})}{(\text{min})} \frac{2\pi(\text{rad})}{(\text{rev})} \frac{1.0(\text{min})}{60(\text{s})} \frac{1.0\text{ m}}{1000\text{ mm}}$$

$$v_t = \frac{\pi D n}{60\,000} \text{m/s} \quad (8-36)$$

Provided that data are supplied in the given units, the final form of Equations (8–35) or (8–36) can be used for problems and designs for this book. We refer to this type of equation as *unit specific*, meaning that it is only valid for input data in the proper units.

The linear displacement or position of the rack can be related to the angular displacement of the pinion, θ , by the following equation:

$$s_{\text{RACK}} = \frac{D_P}{2} \cdot \theta_P \quad (8-37)$$

The concept of center distance does not apply directly for a rack-and-pinion set because the center of the rack is at infinity. But it is critical that the pitch circle of the pinion be tangent to the pitch line of the rack as

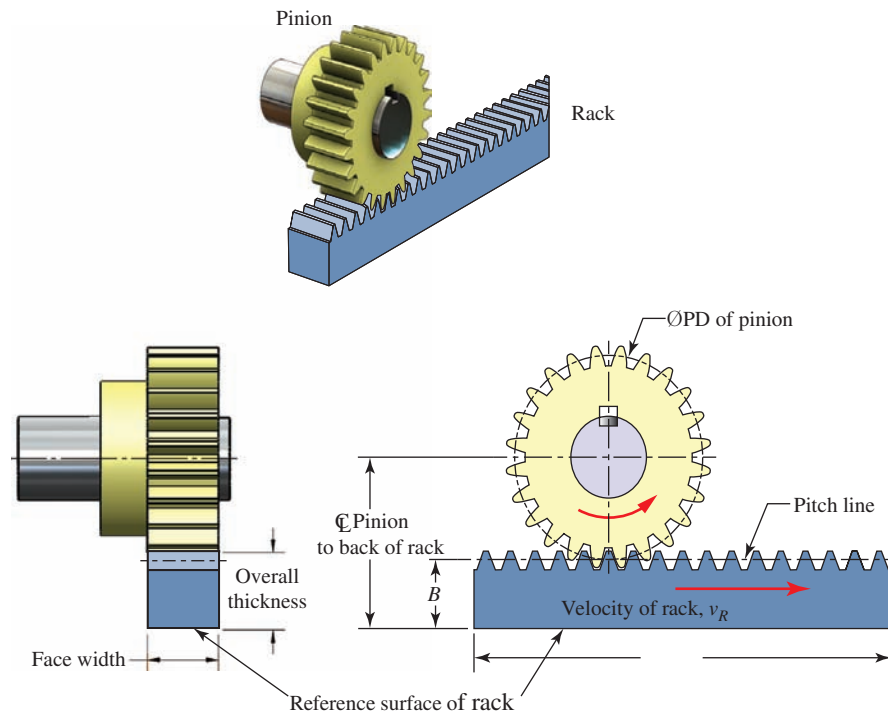


FIGURE 8–45 Rack driven by a pinion

TABLE 8-10 Example rack specifications

Diametral pitch	Pitch line to back (B)	Overall thickness	Face width	Nominal length [ft]
64	0.109	0.125	0.125	2
48	0.104	0.125	0.125	2
32	0.156	0.187	0.187	4
24	0.208	0.250	0.25	4
20	0.450	0.500	0.5	6
16	0.688	0.750	0.75	6
12	0.917	1.000	1	6
10	1.150	1.250	1.25	6
8	1.375	1.500	1.5	6
6	1.333	1.500	2	6
5	1.300	1.500	2.5	6
4	1.750	2.000	3.5	6

shown in Figure 8-28. The rack will be machined so that there is a specified dimension between the pitch line and a reference surface, typically the back of the rack. This is dimension B in Figure 8-28. Then the location of the center of the pinion can be computed using the relationships shown in the figure.

Table 8-10 gives examples of the basic rack dimensional information for different diametral pitches that can be used for problems in this book. This includes the pitch line to back of rack dimension, B , which is used to locate the rack from the centerline of the pinion. Data from each specific manufacturer should be used.

Example Problem 8-9

A rack is driven by pinion shown in Figure 8-46. The pinion rotates at 125 rpm, has 24 teeth, and a diametral pitch of 6.

- Determine the pitch diameter of the pinion.
- Determine the dimension from the pitch line to the back of the rack, B .
- Calculate the distance of the pinion center line to the back of the rack.
- Determine the linear velocity of the rack.
- How long would it take to move a rack that has a length of 20 ft?
- How many revolutions would the pinion turn in moving the rack 20 ft?

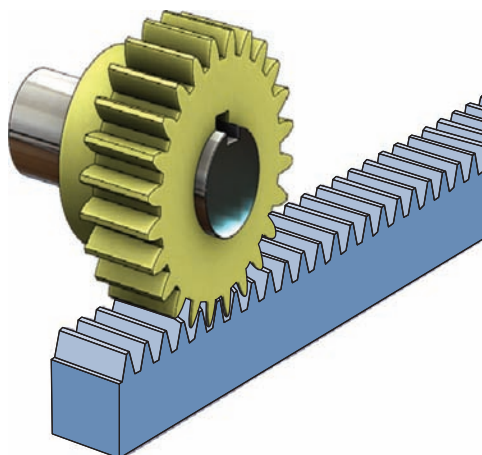


FIGURE 8-46 Pinion and rack for Example Problem 8-9

Solution Given: $n_{\text{pinion}} = 125$ rpm; Pinion $P_d = 6$; $N_{\text{pinion}} = 24$; Length of rack = $L = 20$ ft

- $D_p = N_p/P_d = 24/6 = 4.000$ in
- Distance from the pitch line to the back of the rack: $B = 1.333$ in [From Table 8-10]

- c. Distance from back of the rack to the pinion centerline: [Call this value $B-C$]

$$B-C = B + \text{Pinion radius} = B + D_P/2 = 1.333 \text{ in} + (4.000 \text{ in})/2 = 3.333 \text{ in}$$

- d. Linear velocity of the rack = $V_{\text{rack}} = r\omega = (D_P/2)(n_P)$

$$V_{\text{rack}} = (2.000 \text{ in})(125 \text{ rev/min})(2\pi \text{ rad/rev})(1.0 \text{ ft}/12 \text{ in}) = 130.9 \text{ ft/min} = 130.9 \text{ fpm}$$

- e. Time to move rack 20 ft: Using $V = s/t$, then $t = s/V$

$$t = \frac{s}{V} = \frac{20 \text{ ft}}{130.9 \text{ ft/min}} \cdot \frac{60 \text{ sec}}{\text{min}} = 9.167 \text{ sec}$$

- f. Number of pinion revolutions, θ_P , to move rack 20 ft:

From Equation 8-37:

$$s_{\text{RACK}} = \frac{D_P}{2} \cdot \theta_P$$

Then

$$\theta_P = \frac{s_{\text{RACK}}}{D_P/2} = \frac{20 \text{ ft}}{2.0 \text{ in}} \cdot \frac{12 \text{ in}}{\text{ft}} = 120 \text{ rad} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} = 19.09 \text{ rev}$$

8-14 DEVISING GEAR TRAINS

Now we will show several methods for devising gear trains to produce a desired train value. The result will typically be the specification of the number of teeth in each gear and the general arrangement of the gears relative to each other. The determination of the types of gears will generally not be considered except for how they may affect the direction of rotation or the general alignment of the shafts. Additional details can be specified after completion of the study of the design procedures in later chapters.

Hunting Tooth

Some designers recommend that integer velocity ratios be avoided, if possible, because the same two teeth would come into contact frequently and produce uneven wear patterns on the teeth. For example when using a velocity ratio of exactly 2.0, a given tooth on the pinion would contact the same two teeth on the gear with every two revolutions. In Chapter 9 you will learn that the pinion teeth are often made harder than the gear because the pinion experiences higher stresses. As the gears rotate, the pinion teeth tend to smooth any inherent roughness of the gear teeth, a process sometimes called *wearing in*. Each tooth on the pinion then has a slightly different geometry causing unique wear patterns on the few teeth with which it mates.

A more uniform wear pattern will result if the velocity ratio is not an integer. Adding or subtracting one tooth from the number of teeth in the gear produces the result that each pinion tooth would contact a different gear tooth with each revolution and the wear pattern would be more uniform. The added or subtracted tooth is called the *hunting tooth*. Obviously the velocity ratio

for the gear pair will be slightly different, but that is often not a concern unless precise timing between the driver and driven gears is required. Consider the following example.

An initial design for a gear pair calls for the pinion to be mounted to the shaft of an electric motor having a nominal speed of 1750 rpm. The pinion has 18 teeth and the gear has 36 teeth, resulting in a velocity ratio of 36/18 or 2.000. The output speed would then be:

$$\begin{aligned} \text{Initial design: } n_2 &= n_1(N_P/N_G) \\ &= 1750 \text{ rpm} (18/36) = 875 \text{ rpm} \end{aligned}$$

Now consider adding or subtracting one tooth from the gear. The output speeds would be,

$$\begin{aligned} \text{Modified design: } n_2 &= n_1(N_P/N_G) \\ &= 1750 \text{ rpm} (18/35) = 900 \text{ rpm} \end{aligned}$$

$$\begin{aligned} \text{Modified design: } n_2 &= n_1(N_P/N_G) \\ &= 1750 \text{ rpm} (18/37) = 851 \text{ rpm} \end{aligned}$$

The output speeds for the modified designs are less than 3.0 percent different from the original design. You would have to decide if that is acceptable in a given design project. However, be aware that the motor speed is typically not exactly 1750 rpm. As discussed in Chapter 21, 1750 rpm is a typical *full load speed* of a four-pole alternating current electric motor. When operating at a torque less than the full load torque the speed would be greater than 1750 rpm. Conversely, a greater torque would result in a slower speed. When precise speeds are required, a variable speed drive that can be adjusted according to actual loads is recommended.

A few general principles that were discussed earlier in this chapter are reviewed next.

GENERAL PRINCIPLES FOR DEVISING GEAR TRAINS ▼

1. The velocity ratio for any pair of gears can be computed in a variety of ways as indicated in Equation (8–33).
2. The number of teeth in any gear must be an integer.
3. Mating gears must have the same tooth form, pressure angle, and pitch.
4. When external gears mesh, there is a direction reversal of their shafts.
5. When an external pinion meshes with an internal gear, their shafts rotate in the same direction.
6. An idler is a gear that performs as both a driver and a driven gear in the same train. Its size and number of teeth have no effect on the magnitude of the train value, but the direction of rotation is changed.
7. Spur and typical helical gears operate on parallel shafts.
8. Bevel gears, crossed helical gears, and a worm/wormgear set operate on shafts perpendicular to each other.
9. The number of teeth in the pinion of a gear pair should not be such that it causes interference with the teeth of its mating gear. Refer to Table 8–7.
10. In general, the number of teeth in the gear should not be larger than about 150. This is somewhat arbitrary, but it is typically more desirable to use a double-reduction gear train rather than a very large, single-reduction gear pair.
11. Any given gear train design problem may require one or more pairs of gears depending on the overall train value required, space available, and practical gear size.
12. It is generally desirable to design the gear train with the fewest practical number of gears. Each added gear requires its own shaft that must be supported by bearings.
13. In general, it is expected that the overall gear train be small in size, compact, and arranged in a manner that facilitates assembly.
14. You should determine the required directions of the driver for the gear train and for the output shaft; at least whether the input and output should rotate in the same direction (a positive train) or the opposite direction (a negative train).
15. The use of the “hunting tooth” concept described above for any gear pair is desirable. However, in this book we will consider any practical ratio for a given gear pair to be acceptable.
16. It will be shown in Chapters 9 and 10 that gear pairs operating at slower speeds will be subjected to higher torque and tooth loads as compared with those operating at higher speed. This results in the slower gear teeth having a larger tooth size and circular pitch for a given velocity ratio. For this reason, it is desirable to allocate a higher percentage of the total train value to the higher speed pairs of gears to obtain a more optimal overall design for the train.

Three different design procedures are outlined next and demonstrated in example problems. It is recommended that all three be studied and understood so you can select the method most suitable to any given design situation. Introductory comments are given here on the

types of design situations where each procedure should be applied.

1. **Single pair of gears to produce a desired velocity ratio:** This is the fundamental process required to define the number of teeth in the pinion and the gear to produce a desired ratio.
2. **Residual ratio:** This process is used when two or more pairs of gears in a train are required. It calls for specifying all but one of the required ratios to produce an overall train value. Then you will be able to compute the required value of the final ratio. *This is the most general approach and the most frequently used approach in this book.*
3. **Factoring approach:** When two or more pairs of gears in a train are required and an exact ratio for the overall train value is also required, then the velocity ratio of each gear pair must be a factor of the overall train value. Determining the factors of the desired ratio is a necessary skill to apply this method.

Getting started: For any gear train design problem, first determine the minimum number of pairs of gears necessary to produce the overall train value. Here is an outline of the recommended approach.

1. Determine the overall train value, TV , required from the application data.
2. Determine the maximum velocity ratio, $VR_{\max}$, that can be achieved with one pair of gears, considering a reasonable maximum number of teeth for the gear and a number of teeth for the pinion that will not result in interference according to Table 8–7.
3. If the train value is greater than the ratio found from Step 2, divide $TV/VR_{\max}$. This will aid in determining how many pairs of gears are required.
4. Note that the overall train value is the product of the velocity ratio for each pair of gears. That is,

$$TV = VR_1 \times VR_2 \times VR_3 \dots$$

5. Determine if an idler gear is necessary to achieve the required output shaft direction.
6. Specify each individual velocity ratio using the guidelines listed above.
7. Specify the numbers of teeth in each gear of each pair of gears.
8. Sketch the arrangement of the gears to show how they are placed on shafts in proper relation to each other. At this stage, a schematic diagram is adequate.

Three methods are given in Example Problems 8–10, 8–11, and 8–12 that give tools that can be used to devise gear trains for a variety of applications.

- Example Problem 8–10 shows a method that is fundamental to the process of deciding on the number of teeth in each of two mating gears that will produce a desired velocity ratio.

- Example Problem 8–11 describes the *Residual Ratio Method*, used when two or more stages of speed reduction are employed. All but one of the component ratios are decided first and then the remaining part, called the residual ratio, is computed.
- The *Factoring Approach* is used when an exact train value is desired that can be achieved by using individual stages of reduction that are factors of the overall train value.

Design of a Single Pair of Gears to Produce a Desired Velocity Ratio

Example Problem 8–10

Devise a gear train to reduce the speed of rotation of a drive from an electric motor shaft operating at 3450 rpm to approximately 650 rpm. Use $N_{\max} = 150$ teeth.

Solution First we will compute the nominal train value:

$$TV = (\text{input speed})/(\text{output speed}) = 3450/650 = 5.308$$

If a single pair of gears is used then the train value is equal to the velocity ratio for that pair. That is, $TV = VR = N_G/N_P$.

Let's decide that spur gears having 20° , full-depth, involute teeth are to be used. Then we can refer to Table 8–7 and determine that no fewer than 16 teeth should be used for the pinion in order to avoid interference. We can specify the number of teeth in the pinion and use the velocity ratio to compute the number of teeth in the gear:

$$N_G = (VR)(N_P) = (5.308)(N_P)$$

All possible examples are given in Table 8–11.

Conclusion and Comments

The combination of $N_P = 26$ and $N_G = 138$ gives the most ideal result for the output speed. But all of the trial values give output speeds reasonably close to the desired value. Only two are more than 2.0 rpm off the desired value. It remains a design decision as to how close the output speed must be to the stated value of 650 rpm. Note that the input speed is given as 3450 rpm, the full load speed of an electric motor. But how accurate is that? The actual speed of the input will vary depending on the load on the motor. Therefore, it is not likely that the ratio must be precise.

TABLE 8–11 All Possible Values for N_P and N_G to Produce the Desired Velocity Ratio

N_P	Computed $N_G = (5.308)(N_P)$	Nearest integer N_G	Actual VR: $VR = N_G/N_P$	Actual output speed (rpm): $n_G = n_P/VR = n_P(N_P/N_G)$
16	84.92	85	$85/16 = 5.31$	649.4
17	90.23	90	$90/17 = 5.29$	651.7
18	95.54	96	$96/18 = 5.33$	646.9
19	100.85	101	$101/19 = 5.32$	649.0
20	106.15	106	$106/20 = 5.30$	650.9
21	111.46	111	$111/21 = 5.29$	652.7
22	116.77	117	$117/22 = 5.32$	648.7
23	122.08	122	$122/23 = 5.30$	650.4
24	127.38	127	$127/24 = 5.29$	652.0
25	132.69	133	$133/25 = 5.32$	648.5
26	138.00	138	$138/26 = 5.308$	650.0 Exact
27	143.31	143	$143/27 = 5.30$	651.4
28	148.61	149	$149/28 = 5.32$	648.3
29	153.92	154	Too large	

Residual Ratio Method

Example Problem 8–11

Devise a gear train for a conveyor drive. The drive motor rotates at 1150 rpm, and it is desired that the output speed for the shaft that drives the conveyor be in the range of 24 to 28 rpm. Use a double-reduction gear train. Power transmission analysis indicates that it would be desirable for the reduction ratio for the first pair of gears to be somewhat greater than that for the second pair.

Solution We use the *getting started* recommendations given earlier to start the solution.

Permissible Train Values

First let's compute the nominal train value that will produce an output speed of 26.0 rpm at the middle of the allowable range:

$$TV_{\text{nom}} = (\text{input speed})/(\text{nominal output speed}) = 1150/26 = 44.23$$

Now we can compute the minimum and maximum allowable speed ratio:

$$TV_{\text{min}} = (\text{input speed})/(\text{maximum output speed}) = 1150/28 = 41.07$$

$$TV_{\text{max}} = (\text{input speed})/(\text{minimum output speed}) = 1150/24 = 47.92$$

Possible Ratio for Single Pair

The maximum ratio that any one pair of gears can produce occurs when the gear has 150 teeth and the pinion has 17 teeth (see Table 8–7). Then

$$VR_{\text{max}} = N_G/N_P = 150/17 = 8.82 \text{ (too low)}$$

Possible Train Value for Double-Reduction Train

$$TV = (VR_1)(VR_2)$$

But the maximum value for either VR is 8.82. Then the maximum train value is

$$TV_{\text{max}} = (8.82)(8.82) = (8.82)^2 = 77.9$$

A double-reduction train is practical.

Optional Designs

The general layout of the proposed train is shown in Figure 8–47. Its train value is

$$TV = (VR_1)(VR_2) = (N_B/N_A)(N_D/N_C)$$

We need to specify the number of teeth in each of the four gears to achieve a train value within the range just computed. Our approach is to specify two ratios, VR_1 and VR_2 , such that their product is within the desired range. If the two ratios were equal, each would be the square root of the target ratio, 44.23. That is,

$$VR_1 = VR_2 = \sqrt{44.23} = 6.65$$

But, as described in Item 16 of the *General Principles for Devising Gear Trains*, we want the first ratio to be somewhat larger than the second. Let's specify

$$VR_1 = 8.0 = (N_B/N_A)$$

If we let pinion A have 17 teeth, the number of teeth in gear B must be

$$N_B = (N_A)(8) = (17)(8) = 136$$

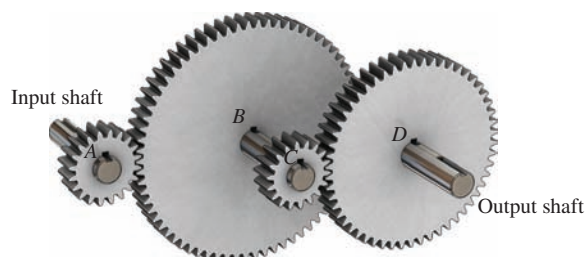


FIGURE 8–47 General layout of the proposed gear train for Example Problem 8–11

Then the second ratio should be approximately

$$VR_2 = TV/(VR_1) = 44.23/8.0 = 5.53$$

This is the *residual ratio* left after the first ratio has been specified. Now if we specify 17 teeth for pinion *C*, gear *D* must be

$$VR_2 = 5.53 = N_D/N_C = N_D/17$$

$$N_D = (5.53)(17) = 94.01$$

Rounding this off to 94 is likely to produce an acceptable result. Finally,

$$N_A = 17 \quad N_B = 136 \quad N_C = 17 \quad N_D = 94$$

We should check the final design:

$$TV = (136/17)(94/17) = 44.235 = n_A/n_D$$

The actual output speed is

$$n_D = n_A/TV = (1150 \text{ rpm})/44.235 = 26.0 \text{ rpm}$$

This is right in the middle of the desired range.

Factoring Approach for Compound Gear Trains

Example Problem 8–12

Devise a gear train for a recorder for a precision measuring instrument. The input is a shaft that rotates at exactly 3600 rpm. The output speed must be exactly 11.25 rpm. Use 20°, full-depth, involute teeth; no fewer than 17 teeth; and no more than 150 teeth in any gear.

Solution

Target *TV*

$$TV_{\text{nom}} = 3600/11.25 = 320$$

Maximum Single *VR*

$$VR_{\text{max}} = 150/17 = 8.824$$

Maximum *TV* for Double Reduction

$$TV_{\text{max}} = (8.824)^2 = 77.8 \text{ (too low)}$$

Maximum *TV* for Triple Reduction

$$TV_{\text{max}} = (8.824)^3 = 687 \text{ (okay)}$$

Design a triple-reduction gear train such as that shown in Figure 8–48. The train value is the product of the three individual velocity ratios:

$$TV = (VR_1)(VR_2)(VR_3)$$

If we can find three factors of 320 that are within the range of the possible ratio for a single pair of gears, they can be specified for each velocity ratio.

Factors of 320

One method is to divide by the smallest prime numbers that will divide evenly into the given number, typically 2, 3, 5, or 7. For example,

$$320/2 = 160$$

$$160/2 = 80$$

$$80/2 = 40$$

$$40/2 = 20$$

$$20/2 = 10$$

$$10/2 = 5$$

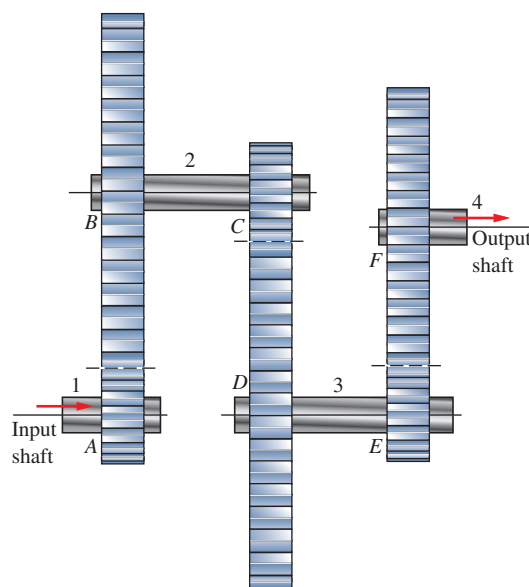


FIGURE 8-48 Triple-reduction gear train

Then the prime factors of 320 are 2, 2, 2, 2, 2, and 5. We desire a set of three factors, which we can find by combining each set of three “2” factors into their product. That is,

$$(2)(2)(2) = 8$$

Then the three factors of 320 are

$$(8)(8)(5) = 320$$

Now let the number of teeth in the pinion of each pair be 17. The number of teeth in the gears will then be $(8)(17) = 136$ or $(5)(17) = 85$. Finally, we can specify

$$\begin{array}{lll} N_A = 17 & N_C = 17 & N_E = 17 \\ N_B = 136 & N_D = 136 & N_F = 85 \end{array}$$

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INTERNET SITES RELATED TO KINEMATICS OF GEARS

1. **American Gear Manufacturers Association (AGMA)**. Develops and publishes voluntary, consensus standards for gears and gear drives. Some standards are jointly published with the American National Standards Institute (ANSI).
2. **Boston Gear Company**. A manufacturer of gears and complete gear drives. Part of Altra Industrial Motion, Inc. Data provided for spur, helical, bevel, and worm gearing.
3. **Regal-Beloit Corporation**. The Browning and Morse Divisions produce spur, helical, bevel, and worm gearing and complete gear drives.
4. **Gear Industry Home Page**. Information source for many companies that manufacture or use gears or gearing systems. Includes gear machinery, gear cutting tools, gear materials, gear drives, open gearing, tooling & supplies, software, training and education. Publishes *Gear Technology Magazine*, *The Journal of Gear Manufacturing*.
5. **Power Transmission Home Page**. Clearinghouse on the Internet for buyers, users, and sellers of power transmission-related products and services. Included are gears, gear drives, and gearmotors.
6. **Baldor/Dodge**. Manufacturer of many power transmission components, including complete gear-type speed reducers, bearings, and components such as belt drives, chain drives, clutches, brakes, and couplings.
7. **Gear—Wikipedia site** General discussion of the kinematics and gears, including an animated drawing of meshing gears.
8. **Gear tooth engagement animation** Search the Internet on "gears meshing animation." Several sites appear showing involute gear teeth engaging.
9. **Maryland Metrics Co.** Distributor of a wide variety of metric hardware and power transmission products, including many types of gears. On the home page, search on "Mechanical Power Transmission Products."
10. **Stock Drive Products/Sterling Instruments Co.** Distributor of a wide variety of U.S. styles and metric hardware and mechanical power transmission products, including many types of gears.
11. **W. M. Berg Co.** Distributor of a wide variety of U.S. styles of hardware and power transmission products, including many types of gears.
12. **Profilator**. Germany-based manufacturer of a variety of machine tools for the gear-cutting industry, using several processes, including scudding. U.S. based representation is by the German Machine Tools of America (GMTA).

PROBLEMS

Gear Geometry

1. A gear has 44 teeth of the 20° , full-depth, involute form and a diametral pitch of 12. Compute the following:
 - (a) Pitch diameter
 - (b) Circular pitch
 - (c) Equivalent module
 - (d) Nearest standard module
 - (e) Addendum
 - (f) Dedendum
 - (g) Clearance
 - (h) Whole depth
 - (i) Working depth
 - (j) Tooth thickness
 - (k) Outside diameter
 Repeat Problem 1 for the following gears:
 2. $N = 34; P_d = 24$
 3. $N = 45; P_d = 2$
 4. $N = 18; P_d = 8$
 5. $N = 22; P_d = 1.75$
 6. $N = 20; P_d = 64$
 7. $N = 180; P_d = 80$
 8. $N = 28; P_d = 18$
 9. $N = 28; P_d = 20$
 For Problems 10–17, repeat Problem 1 for the following gears in the metric module system. Replace Part (c) with equivalent P_d and Part (d) with nearest standard P_d .
 10. $N = 34; m = 3$
 11. $N = 45; m = 1.25$
 12. $N = 18; m = 12$
 13. $N = 22; m = 20$
 14. $N = 20; m = 1$
 15. $N = 180; m = 0.4$
 16. $N = 28; m = 1.5$

17. $N = 28$; $m = 0.8$
18. Define *backlash*, and discuss the methods used to produce it.
19. For the gears of Problems 1 and 12, recommend the amount of backlash.

Velocity Ratio

20. An 8-pitch pinion with 18 teeth mates with a gear having 64 teeth as shown in Figure P8-20. The pinion rotates at 2450 rpm. Compute the following:
 - (a) Center distance
 - (b) Velocity ratio
 - (c) Speed of gear
 - (d) Pitch line speed

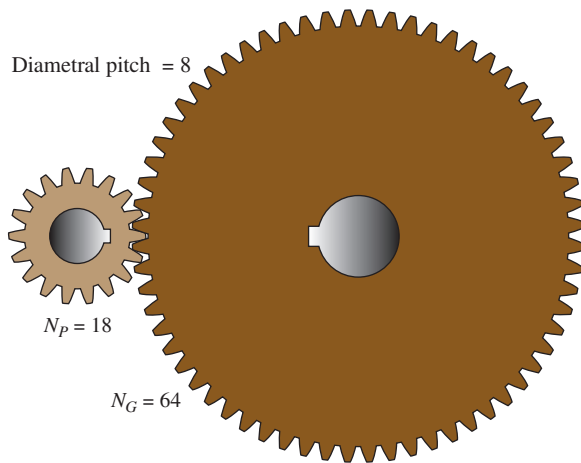


FIGURE P8-20 Gear pair for Problem 8-20

Repeat Problem 20 for the following data:

21. $P_d = 4$; $N_P = 20$; $N_G = 92$; $n_P = 225$ rpm
22. $P_d = 20$; $N_P = 30$; $N_G = 68$; $n_P = 850$ rpm
23. $P_d = 64$; $N_P = 40$; $N_G = 250$; $n_P = 3450$ rpm
24. $P_d = 12$; $N_P = 24$; $N_G = 88$; $n_P = 1750$ rpm
25. $m = 2$; $N_P = 22$; $N_G = 68$; $n_P = 1750$ rpm
26. $m = 0.8$; $N_P = 18$; $N_G = 48$; $n_P = 1150$ rpm
27. $m = 4$; $N_P = 36$; $N_G = 45$; $n_P = 15$ rpm
28. $m = 12$; $N_P = 15$; $N_G = 36$; $n_P = 480$ rpm

For Problems 29–32, all gears are made in standard 20° , full-depth, involute form. Tell what is wrong with the following statements:

29. An 8-pitch pinion having 24 teeth mates with a 10-pitch gear having 88 teeth. The pinion rotates at 1750 rpm, and the gear at approximately 477 rpm. The center distance is 5.900 in.
30. A 6-pitch pinion having 18 teeth mates with a 6-pitch gear having 82 teeth. The pinion rotates at 1750 rpm, and the gear at approximately 384 rpm. The center distance is 8.3 in.
31. A 20-pitch pinion having 12 teeth mates with a 20-pitch gear having 62 teeth. The pinion rotates at 825 rpm, and the gear at approximately 160 rpm. The center distance is 1.850 in.
32. A 16-pitch pinion having 24 teeth mates with a 16-pitch gear having 45 teeth. The outside diameter of the pinion is 1.625 in. The outside diameter of the gear is 2.938 in. The center distance is 2.281 in.

Housing Dimensions

33. The gear pair described in Problem 20 is to be installed in a rectangular housing. Specify the dimensions X and Y as sketched in Figure P8-33 that would provide a minimum clearance of 0.10 in.

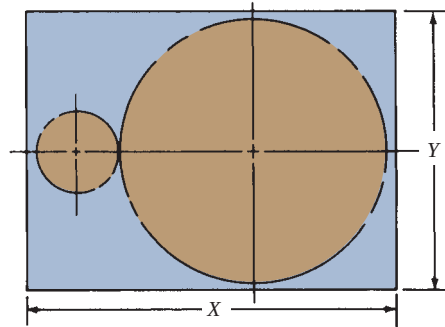


FIGURE P8-33 (Problems 33, 34, 35, and 36)

34. Repeat Problem 33 for the data of Problem 23.
35. Repeat Problem 33 for the data of Problem 26, but make the clearance 2.0 mm.
36. Repeat Problem 33 for the data of Problem 27, but make the clearance 2.0 mm.

Analysis of Simple Gear Trains: Problems 37–40

For the gear trains sketched in the given figures, compute the output speed and the direction of rotation of the output shaft if the input shaft rotates at 1750 rpm clockwise.

37. Use Figure P8-37.
38. Use Figure P8-38.
39. Use Figure P8-39.
40. Use Figure P8-40.

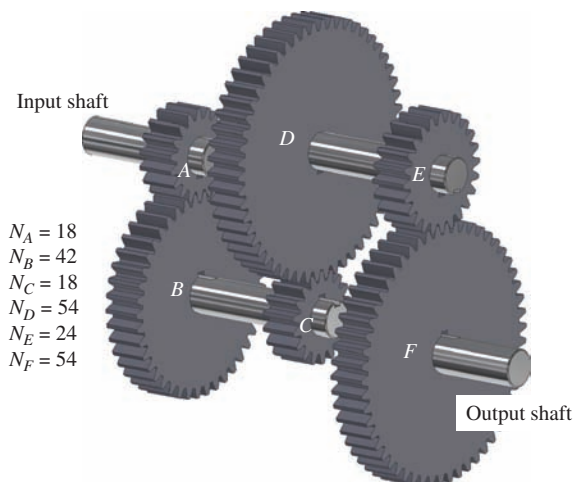


FIGURE P8-37 Gear train layout for Problem 8-37

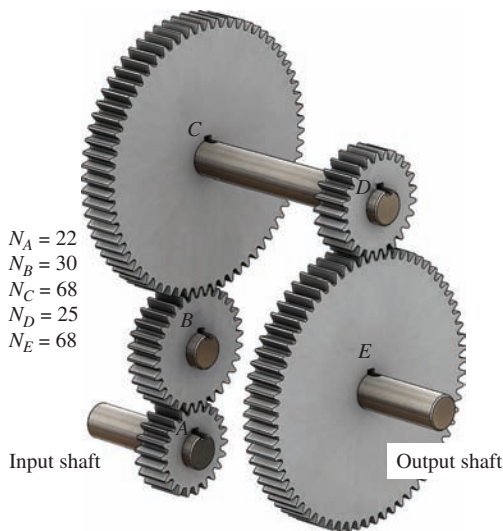


FIGURE P8-38 Gear train layout for Problem 8-38

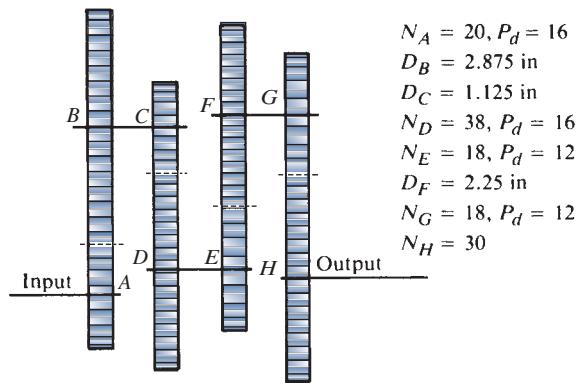


FIGURE P8-39 Gear train layout for Problem 8-39

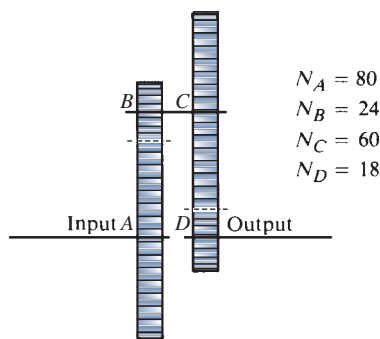


FIGURE P8-40 Gear train layout for Problem 8-40

Helical Gearing

41. A helical gear has a transverse diametral pitch of 8, a transverse pressure angle of $14\frac{1}{2}^\circ$, 45 teeth, a face width of 2.00 in, and a helix angle of 30° . Compute the circular pitch, normal circular pitch, normal diametral pitch, axial pitch, pitch diameter, and normal pressure angle. Then compute the number of axial pitches in the face width.

42. See Figure P8-42. A helical gear has a normal diametral pitch of 12, a normal pressure angle of 20° , 48 teeth, a face width of 1.50 in, and a helix angle of 45° . Compute the circular pitch, normal circular pitch, transverse diametral pitch, axial pitch, pitch diameter, and transverse pressure angle. Then compute the number of axial pitches in the face width.

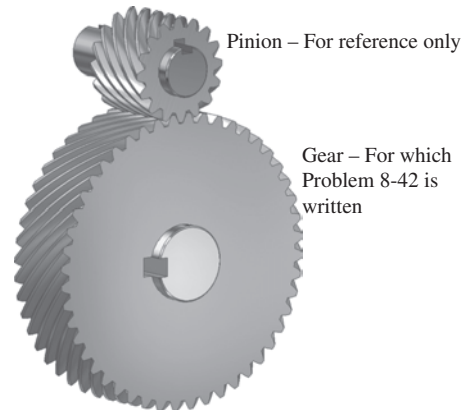


FIGURE P8-42 Helical gear for Problem 8-42

43. A helical gear has a transverse diametral pitch of 6, a transverse pressure angle of $14\frac{1}{2}^\circ$, 36 teeth, a face width of 1.00 in, and a helix angle of 45° . Compute the circular pitch, normal circular pitch, normal diametral pitch, axial pitch, pitch diameter, and normal pressure angle. Then compute the number of axial pitches in the face width.
44. A helical gear has a normal diametral pitch of 24, a normal pressure angle of $14\frac{1}{2}^\circ$, 72 teeth, a face width of 0.25 in, and a helix angle of 45° . Compute the circular pitch, normal circular pitch, transverse diametral pitch, axial pitch, pitch diameter, and transverse pressure angle. Then compute the number of axial pitches in the face width.

Bevel Gears

45. A straight bevel gear pair has the following data: $N_p = 15$; $N_G = 45$; $P_d = 6$; 20° pressure angle. Compute all of the geometric features from Table 8-8.
46. Draw the gear pair of Problem 45 to scale. The following additional dimensions are given (refer to Figure 8-17). Mounting distance (M_{dp}) for the pinion = 5.250 in; M_{dG} for the gear = 3.000 in; face width = 1.250 in. Supply any other needed dimensions.
47. A straight bevel gear pair has the following data: $N_p = 25$; $N_G = 50$; $P_d = 10$; 20° pressure angle. Compute all of the geometric features from Table 8-8.
48. Draw the gear pair of Problem 47 to scale. The following additional dimensions are given (refer to Figure 8-17). Mounting distance (M_{dp}) for the pinion = 3.375 in; M_{dG} for the gear = 2.625 in; face width = 0.700 in. Supply any other needed dimensions.
49. A straight bevel gear pair has the following data: $N_p = 18$; $N_G = 72$; $P_d = 12$; 20° pressure angle. Compute all of the geometric features from Table 8-8.

50. A straight bevel gear pair has the following data:
 $N_P = 16$; $N_G = 64$; $P_d = 32$; 20° pressure angle.
 Compute all of the geometric features from Table 8–8.
51. A straight bevel gear pair has the following data:
 $N_P = 12$; $N_G = 36$; $P_d = 48$; 20° pressure angle.
 Compute all of the geometric features from Table 8–8.

Wormgearing

52. A wormgear set has a single-thread worm with a pitch diameter of 1.250 in, a diametral pitch of 10, and a normal pressure angle of 14.5° . If the worm meshes with a wormgear having 40 teeth and a face width of 0.625 in, compute the lead, axial pitch, circular pitch, lead angle, addendum, dedendum, worm outside diameter, worm root diameter, gear pitch diameter, center distance, and velocity ratio.
53. Three designs are being considered for a wormgear set to produce a velocity ratio of 20 when the wormgear rotates at 90 rpm. All three have a diametral pitch of 12, a worm pitch diameter of 1.000 in, a gear face width of 0.500 in, and a normal pressure angle of 14.5° . One has a single-thread worm and 20 wormgear teeth; the second has a double-thread worm and 40 wormgear teeth; the third has a four-thread worm and 80 wormgear teeth. For each design, compute the lead, axial pitch, circular pitch, lead angle, gear pitch diameter, and center distance.
54. A wormgear set has a double-threaded worm with a normal pressure angle of 20° , a pitch diameter of 0.625 in, and a diametral pitch of 16. Its mating wormgear has 100 teeth and a face width of 0.3125 in. Compute the lead, axial pitch, circular pitch, lead angle, addendum, dedendum, worm outside diameter, center distance, and velocity ratio.
55. A wormgear set has a four-threaded worm with a normal pressure angle of $14\frac{1}{2}^\circ$, a pitch diameter of 2.000 in, and a diametral pitch of 6. Its mating wormgear has 72 teeth and a face width of 1.000 in. Compute the lead, axial pitch, circular pitch, lead angle, addendum, dedendum, worm outside diameter, center distance, and velocity ratio.
56. A wormgear set has a single-threaded worm with a normal pressure angle of $14\frac{1}{2}^\circ$, a pitch diameter of 4.000 in, and a diametral pitch of 3. Its mating wormgear has 54 teeth and a face width of 2.000 in. Compute the lead, axial pitch, circular pitch, lead angle, addendum, dedendum, worm outside diameter, center distance, and velocity ratio.
57. A wormgear set has a four-threaded worm with a normal pressure angle of 25° , a pitch diameter of 0.333 in, and a diametral pitch of 48. Its mating wormgear has 80 teeth and a face width of 0.156 in. Compute the lead, axial pitch, circular pitch, lead angle, addendum, dedendum, worm outside diameter, center distance, and velocity ratio.

Analysis of Complex Gear Trains

58. The input shaft for the gear train shown in Figure P8–58 rotates at 3450 rpm cw. Compute the rotational speed and direction of the output shaft.
59. The input shaft for the gear train shown in Figure P8–59 rotates at 12 200 rpm. Compute the rotational speed of the output shaft.
60. The input shaft for the gear train shown in Figure P8–60 rotates at 6840 rpm. Compute the rotational speed of the output shaft.

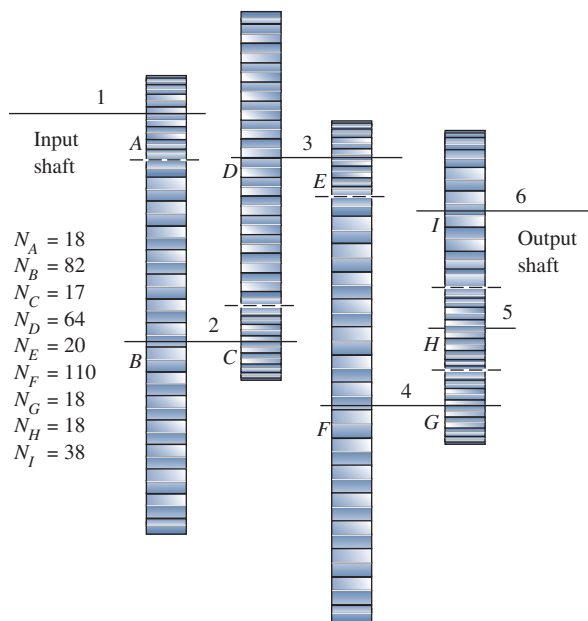


FIGURE P8–58 Gear train for Problem 8–58

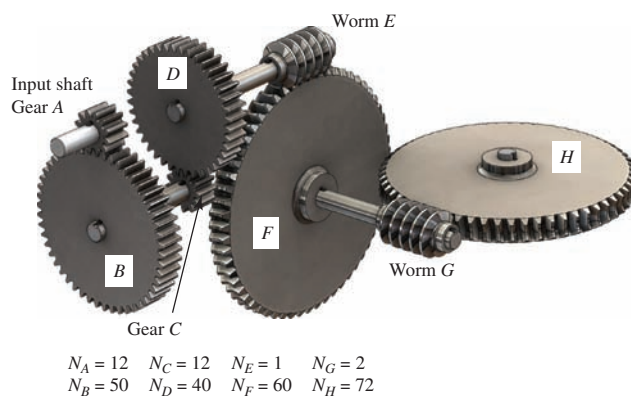


FIGURE P8–59 Gear train for Problem 8–59

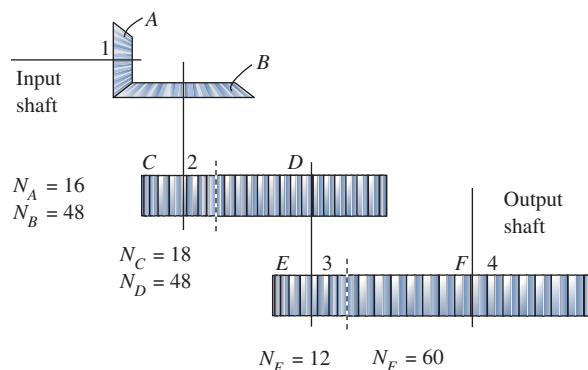


FIGURE P8–60 Gear train for Problem 8–60

61. The input shaft for the gear train shown in Figure P8–61 rotates at 2875 rpm. Compute the rotational speed of the output shaft.

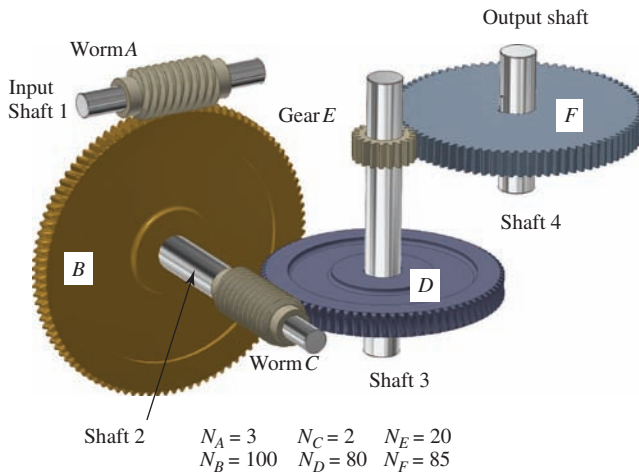


FIGURE P8–61 Gear train for Problem 8–61

Kinematic Design of a Single Gear Pair

62. Specify the numbers of teeth for the pinion and gear of a single gear pair to produce a velocity ratio of π as closely as possible. Use no fewer than 16 teeth nor more than 24 teeth in the pinion.
63. Specify the numbers of teeth for the pinion and gear of a single gear pair to produce a velocity ratio of $\sqrt{3}$ as closely as possible. Use no fewer than 16 teeth nor more than 24 teeth in the pinion.
64. Specify the numbers of teeth for the pinion and gear of a single gear pair to produce a velocity ratio of $\sqrt{38}$ as closely as possible. Use no fewer than 18 teeth nor more than 24 teeth in the pinion.
65. Specify the numbers of teeth for the pinion and gear of a single gear pair to produce a velocity ratio of 7.42 as closely as possible. Use no fewer than 18 teeth nor more than 24 teeth in the pinion.

Kinematic Design of Gear Trains

For Problems 66–75, devise a gear train using all external gears on parallel shafts. Use 20° full-depth involute teeth and no more than 150 teeth in any gear. Ensure that there is no interference. Sketch the layout for your design.

Problem no.	Input speed (rpm)	Output speed range (rpm)
66.	1800	2.0 Exactly
67.	1800	21.0 to 22.0
68.	3360	12.0 Exactly
69.	4200	13.0 to 13.5
70.	5500	221 to 225
71.	5500	13.0 to 14.0
72.	1750	146 to 150
73.	850	40.0 to 44.0
74.	3000	548 to 552 Use two pairs
75.	3600	3.0 to 5.0

For Problems 76–80, devise a gear train using any type of gears. Try for the minimum number of gears while avoiding interference and having no more than 150 teeth in any gear. Sketch your design.

Problem no.	Input speed (rpm)	Output speed (rpm)
76.	3600	3.0 to 5.0
77.	1800	8.0 Exactly
78.	3360	12.0 Exactly
79.	4200	13.0 to 13.5
80.	5500	13.0 to 14.0

Rack and Pinion Analysis

81. In Figure P8–81, the rack is driven by the pinion that has a rotational speed of 50 rpm. The diametral pitch is 12, $N_p = 30$, and the length of the rack is 7.00 ft. Find the following:
- The pitch diameter of the pinion
 - The distance from the pitch line to the back of the rack
 - The center distance
 - The linear velocity of the rack
 - The time required to move the rack 7.00 ft.
 - The number of revolutions of the pinion as the rack moves 7.00 ft.

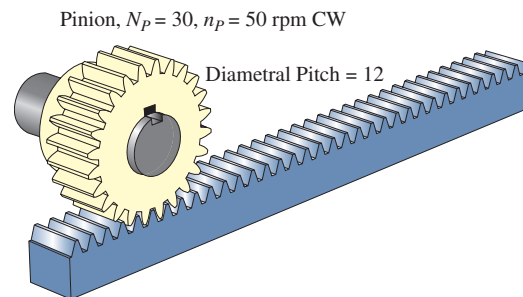


FIGURE P8–81 Rack and pinion for Problem 8–81

SPUR GEAR DESIGN

The Big Picture

You Are the Designer

- 9–1 Objectives of This Chapter
- 9–2 Concepts from Previous Chapters
- 9–3 Forces, Torque, and Power in Gearing
- 9–4 Introduction to Stress Analysis for Gears
- 9–5 Bending Stress in Gear Teeth
- 9–6 Contact Stress in Gear Teeth
- 9–7 Metallic Gear Materials
- 9–8 Selection of Gear Materials
- 9–9 Design of Spur Gears to Specify Suitable Materials
- 9–10 Gear Design for the Metric Module System
- 9–11 Computer-Aided Spur Gear Design and Analysis
- 9–12 Use of the Spur Gear Design Spreadsheet
- 9–13 Power-Transmitting Capacity
- 9–14 Plastics Gearing
- 9–15 Practical Considerations for Gears and Interfaces with Other Elements

THE BIG PICTURE

Spur Gear Design

Discussion Map

- A spur gear has involute teeth that are straight and parallel to the axis of the shaft that carries the gear.
- You need to understand how to design spur gears, specifying the form and size of the teeth, the face width, the material and its heat treatment.

Discover

Describe the action of the teeth of the driving gear on those of the driven gear. What kinds of stresses are produced?

How do the geometry of the gear teeth, the materials from which they are made, and the operating conditions affect the stresses and the life of the drive system?

This chapter will help you acquire the skills to perform the necessary analyses and to design safe spur gear drive systems that demonstrate long life.

The goal of this chapter is to help you gain the knowledge and skills necessary to design spur gears to transmit power from a source such as an electric motor, gasoline engine, fluid power motor, turbine, or other prime mover to a driven machine while changing the speed of the input shaft to some other speed at the output shaft. Most gear-type power transmissions are *speed reducers* that deliver the power to the driven machine at a lower speed and a higher torque. Examples of such transmissions are:

1. Electric motor drives through the transmission to a conveyor in a factory.

2. Gasoline engine for a vehicle drives through the transmission to the drive wheels.
3. Fluid power motor drives through the transmission to a winch on a tractor.
4. Water turbine drives through the transmission to an electric generator.
5. Gas turbine (jet) engine drives through the transmission to the rotor of a helicopter.

Although more rare than speed reducers, transmissions may also be used as *speed increasers*. An important example is a wind turbine that rotates relatively

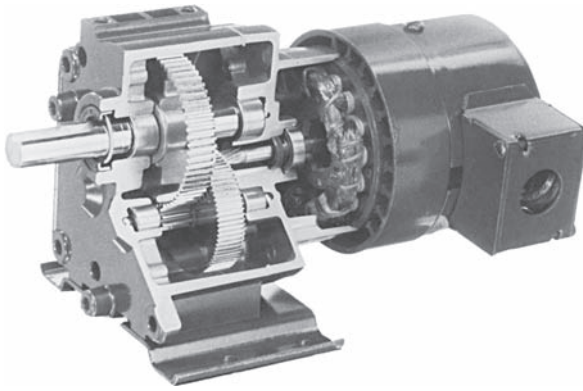


FIGURE 9-1 Double-reduction spur gear reducer (Bison Gear & Engineering Corporation, St. Charles, IL)

slowly, say 20.0 rpm, and the speed must be increased to 1800 rpm to drive an electric generator.

Figure 9-1 shows a commercially available double-reduction, spur gear type speed reducer driven by an electric motor that can be applied to many kinds of applications. This is an example of the type of reducer addressed in this chapter and about which much of design process discussed in Part II of this book (Chapters 7–15) is directed. Make note of how the gears are positioned and how they are mounted on shafts that are then supported by bearings mounted in a rigid housing.

A *spur gear* is one of the most fundamental types of gears. Its teeth are straight and parallel to the axis of the shaft that carries the gear. The teeth have the involute form described in Chapter 8. So, in general, the action of one tooth on a mating tooth is like that of two convex, curved members in contact: As the driving gear rotates, its teeth exert a force on the mating gear that is tangential to the pitch circles of the two gears. Because this force acts at a distance equal to the pitch radius of the gear, a torque is developed in the shaft that carries the gear. When the two gears rotate, they transmit power that is proportional to the torque. Indeed, that is the primary purpose of the spur gear drive system.

Consider the action described in the preceding paragraph:

- How does that action relate to the design of the gear teeth? Look back at Figure 8-1 as you consider this question and those that follow.

- As the force is exerted by the driving tooth on the driven tooth, what kinds of stresses are produced in the teeth? Consider both the point of contact of one tooth on the other and the whole tooth. Where are stresses a maximum?
- How could the teeth fail under the influence of these stresses?
- What material properties are critical to allow the gears to carry such loads safely and with a reasonable life span?
- What important geometric features affect the level of stress produced in the teeth?
- How does the precision of the tooth geometry affect its operation?
- How does the nature of the application affect the gears? What if the machine that the gears drive is a rock crusher that takes large boulders and reduces them to gravel made up of small stones? How would that loading compare with that of a gear system that drives a fan providing ventilation air to a building?
- What is the influence of the driving machine? Would the design be different if an electric motor were the driver or if a gasoline engine were used?
- The gears are typically mounted on shafts that deliver power from the driver to the input gear of a gear train and that take power from the output gear and transmit it to the driven machine. Describe various ways that the gears can be attached to the shafts and located with respect to each other. How can the shafts be supported?

This chapter contains the kinds of information that you can use to answer such questions and to complete the analysis and design of spur gear power transmission systems.

Chapters 10–15 cover similar topics for helical gears, bevel gears, and worm gearing, along with the design and specification of keys, couplings, seals, shafts, and bearings—all of which are needed to design a complete mechanical drive.

YOU ARE THE DESIGNER

You have already made the design decision that a spur gear type of speed reducer is to be used for a particular application. How do you complete the design of the gears themselves?

This is a continuation of a design scenario that was started in Chapter 1 of this book when the original goals were stated and when an overview of the entire book was given. The introduction to Part II continued this theme by indicating that the arrangement of the

chapters is aligned with the design process that you could use to complete the design of the speed reducer.

Then in Chapter 8, you, as the designer, dealt with the kinematics of a gear reducer that would take power from the shaft of an electric motor rotating at 1750 rpm and deliver it to a machine that was to operate at approximately 292 rpm. There you limited your interest to the decisions that affected motion and the basic

geometry of the gears. It was decided that you would use a double-reduction gear train to reduce the speed of rotation of the drive system in two stages using two pairs of gears in series. You also learned how to specify the layout of the gear train, along with key design decisions such as the numbers of teeth in all of the gears and the relationships among the diametral pitch, the number of teeth in the gears, the pitch diameters, and the distance between the centers of the shafts that carry those gears. For a chosen diametral pitch, you learned how to compute the dimensions of key features of the gear teeth such as the addendum, dedendum, and tooth width.

But the design is not complete until you have specified the material from which the gears are to be made and until you have verified that the gears will withstand the forces exerted on the gears

as they transmit power and the corresponding torque. The teeth must not break, and they must have a sufficiently long life to meet the needs of the customer who uses the reducer.

To complete the design, you need more data: How much power is to be transmitted? To what kind of machine is the power from the output of the reducer being delivered? How does that affect the design of the gears? What is the anticipated duty cycle for the reducer in terms of the number of hours per day, days per week, and years of life expected? What options do you have for materials that are suitable for gears? Which material will you specify, and what will be its heat treatment?

You are the designer. The information in this chapter will help you complete the design. ■

9-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to demonstrate the competencies listed below. They are presented in the order that they are covered in this chapter. The primary objectives are numbers 6, 7, and 8, which involve (a) the calculation of the bending strength and the ability of the gear teeth to resist pitting and (b) the design of gears to be safe with regard to both strength and pitting resistance. The competencies are as follows:

1. Compute the forces exerted on gear teeth as they rotate and transmit power.
2. Describe various methods for manufacturing gears and the levels of precision and quality to which they can be produced.
3. Specify a suitable level of quality for gears according to the use to which they are to be put.
4. Describe suitable metallic materials from which to make the gears, in order to provide adequate performance for both strength and pitting resistance.
5. Use the standards of the American Gear Manufacturers Association (AGMA) as the basis for completing the design of the gears.
6. Use appropriate stress analyses to determine the relationships among the applied forces, the geometry of the gear teeth, the precision of the gear teeth, and other factors specific to a given application, in order to make final decisions about those variables.
7. Perform the analysis of the tendency for the contact stresses exerted on the surfaces of the teeth to cause pitting of the teeth, in order to determine an adequate hardness of the gear material that will provide an acceptable level of pitting resistance for the reducer.
8. Complete the design of the gears, taking into consideration both the stress analysis and the analysis of pitting resistance. The result will be a complete specification of the gear geometry, the material for the gear, and the heat treatment of the material.

Outline of the Chapter: The process of designing spur gear drives contains numerous steps and requires

the determination of several quantities that affect the performance of the drives. This chapter builds that process within the following sections, all of which contribute to the final design process described in Section 9-9. The list below gives the primary steps in the process for the design of a drive consisting of two steel gears.

1. Define the goals of the gear drive design: power to be transmitted, speed of the input gear, desired speed of the output gear, the kind of driver that provides the power, the kind of driven machine, and any special design features.
2. Determine the forces applied to the gear (Section 9-3).
3. Define the precision required for the gears by specifying the quality number (Section 9-5).
4. Understand the types of steel alloys typically used for gears and the kinds of heat treatment available (Section 9-7).
5. Propose the geometry of the gears including the pitch, number of teeth in each gear, pitch diameter, form of the teeth, and face width (Section 9-9).
6. Determine the expected stress due to bending in the gear teeth. This step requires specification of several factors that are functions of the manner of use and the manufacturing processes that will be used to produce the gears (Section 9-5).
7. Determine the expected actual contact stress experienced by the face of the teeth (Section 9-6).
8. Using the bending stress from Step 6 and the contact stress from Step 7, determine the strength and hardness required for the materials from which the gears are to be made to ensure adequate safety and life and specify the steel alloy and heat treatment that will meet these requirements (Section 9-8).
9. Summarize the design details.

9-2 CONCEPTS FROM PREVIOUS CHAPTERS

As you study this chapter, it is assumed that you are familiar with the geometry of gear features and the kinematics of one gear driving another as presented in Chapter 8.

(See also References 4 and 25.) Key relationships that you should be able to use include the following:

$$\text{Pitch line speed} = v_t = R\omega = (D/2)\omega$$

where R = radius of the pitch circle

D = pitch diameter

ω = angular velocity of the gear

Because the pitch line speed is the same for both the pinion and the gear, values for R , D , and ω can be for either. In the computation of stresses in gear teeth, in the U.S. system, it is usual to express the pitch line speed in the units of ft/min, while the size of the gear is given as its pitch diameter expressed in inches. Speed of rotation is typically given as n rpm—that is, n rev/min. Let's compute the unit-specific equation that gives pitch line speed in ft/min:

⇨ Pitch Line Speed U.S. Units

$$\begin{aligned} v_t &= (D/2)\omega = \frac{D \text{ in}}{2} \cdot \frac{n \text{ rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \\ &= (\pi D n / 12) \text{ ft/min} \end{aligned} \quad (9-1)$$

In the SI metric system, diameter is typically expressed in mm and speed in rpm while the pitch line speed is in m/s. Using a process similar to that used above for Equation (9-1), the following unit-specific expression was developed in Chapter 8.

⇨ Pitch Line Speed SI Units

$$v_t = (\pi D n / 60\,000) \text{ m/s} \quad (9-2)$$

The velocity ratio can be expressed in many ways. For the particular case of a pinion driving a larger gear,

⇨ Velocity Ratio

The general definition of velocity ratio, VR , for gear drives is:

$$VR = (\omega_{\text{input}})/(\omega_{\text{output}}) = N_{\text{output}}/N_{\text{input}}$$

When the pinion is the driver and the gear is driven, the velocity ratio can be written as:

$$\text{Velocity ratio} = VR = \frac{\omega_P}{\omega_G} = \frac{n_P}{n_G} = \frac{R_G}{R_P} = \frac{D_G}{D_P} = \frac{N_G}{N_P} \quad (9-3)$$

A related ratio, m_G , called the *gear ratio*, is often used in analysis of the performance of gears. It is always defined as the ratio of the number of teeth in the larger gear to the number of teeth in the pinion, regardless of which is the driver. Thus, m_G is always greater than or equal to 1.0. When the pinion is the driver, as it is for a speed reducer, m_G is equal to VR . That is,

⇨ Gear Ratio

$$\text{Gear ratio} = m_G = N_G/N_P \geq 1.0 \quad (9-4)$$

In the U.S. system the diametral pitch, P_d , characterizes the physical size of the teeth of a gear. It is related to the pitch diameter and the number of teeth as follows:

⇨ Diametral Pitch

$$P_d = N_G/D_G = N_P/D_P \quad (9-5)$$

In the SI metric system, the metric module, m , characterizes the size of the teeth. It is defined as

⇨ Metric Module

$$m = D_G/N_G = D_P/N_P \quad (9-6)$$

The pressure angle, ϕ , is an important feature that characterizes the form of the involute curve that makes up the active face of the teeth of standard gears. See Figure 8-13. Also notice in Figure 8-12 that the angle between a normal to the involute curve and the tangent to the pitch circle of a gear is equal to the pressure angle.

9-3 FORCES, TORQUE, AND POWER IN GEARING

To understand the method of computing stresses in gear teeth, consider the way power is transmitted by the gear system shown in Figure 9-2. Part (a) is a perspective view and part (b) is the side view. The electric motor delivers power at the speed of its shaft through a flexible coupling to the input gear. The input gear drives the larger output gear to achieve a speed reduction. The shaft carrying the output gear transmits the power through a second flexible coupling to drive a roller in a printing press. It is important to determine the torque in both the input and output shafts using the following equation:

⇨ Torque

$$\text{Torque} = \text{power/rotational speed} = P/n \quad (9-7)$$

Because the speed of the pinion is higher than the speed of the gear, the output torque is higher than the input torque. The power is transmitted from the input shaft to the pinion through a key. The teeth of the pinion drive the teeth of the gear and thus transmit the power to the gear. But again, power transmission actually involves the application of a torque during rotation at a given speed. The torque is the product of the force acting tangent to the pitch circle of the pinion times the pitch radius of the pinion. We will use the symbol W_t to indicate the *tangential force*. As described, W_t is the force exerted *by the pinion teeth on the gear teeth*. This force is used to transmit power from one gear to the other and is referred to as the *driving force*. But if the gears are rotating at a constant speed and are transmitting a uniform level of power, the system is in equilibrium. Therefore, there must be an equal and opposite tangential force exerted by the gear teeth back on the pinion teeth as shown in Figure 9-2(c). This is an application of the principle of action and reaction.

To complete the description of the power flow, the tangential force on the gear teeth produces a torque on the gear equal to the product of W_t times the pitch radius of the gear. Because W_t is the same on the pinion and the gear, but the pitch radius of the gear is larger than that of the pinion, the torque on the gear (the output torque)

is greater than the input torque. However, note that the power transmitted to the gear is only slightly less because of the high mechanical inefficiency of the gearing. The power then flows from the gear through the key to the output shaft, the second flexible coupling, and finally to the driven machine.

From this description of power flow, we can see that gears transmit power by exerting a force by the driving teeth on the driven teeth while the reaction force acts back on the teeth of the driving gear. Figure 9–3 shows a single gear tooth with the tangential force W_t acting on it. But this is not the total force on the tooth as shown in Figure 9–2(d). Because of the involute form of the

tooth, the total force transferred from one tooth to the mating tooth acts normal to the involute profile. This action is shown as W_n . The tangential force W_t is actually the horizontal component of the total force. To complete the picture, note that there is a vertical component of the total force acting radially on the gear tooth, indicated by W_r . The radial force tends to separate the gear set and is called the *separating force*.

The normal and radial forces on the gears must be resisted by the bearings that support the shafts as shown in Figure 9–2(e). This observation is important for the design of the shafts carrying the gears as will be demonstrated in Chapter 12.

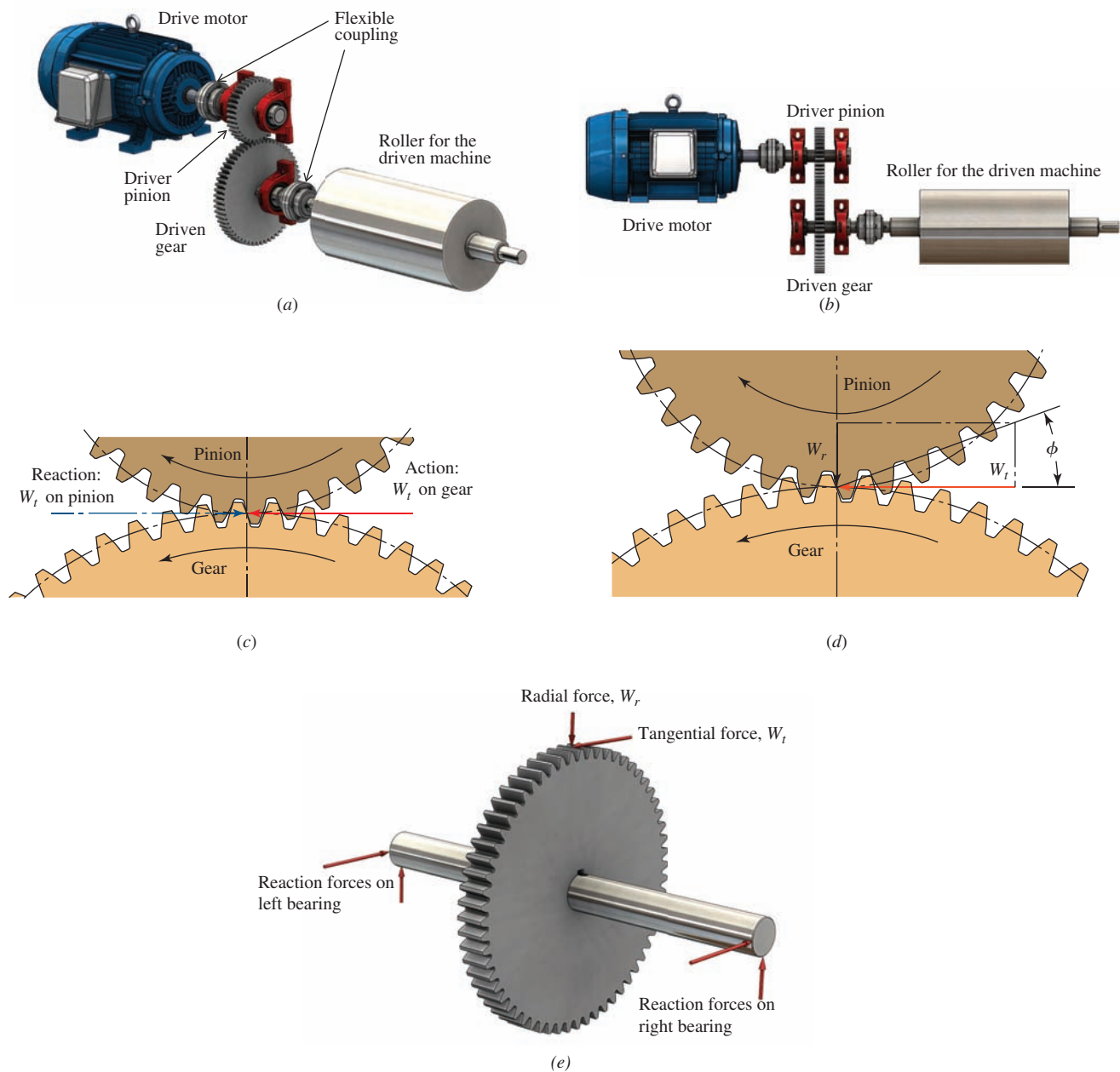


FIGURE 9–2 (a) Power flow through a gear pair (b) Side view of drive train (c) View of gear teeth in mesh showing tangential forces on both gears (d) Tangential and radial forces exerted by the pinion tooth on the gear tooth (e) Forces on the shaft carrying the driven gear

The discussion about power flow and forces existing in gears thus far is generic and independent of units. Following discussion is developed primarily for the U.S. unit system based on the diametral pitch, P_d . Later, we adapt the results to the SI metric unit system and module, m .

We will start the computation of forces with the transmitted force, W_t , because its value is based on the given data for power and speed. It is convenient to develop unit-specific equations for W_t because standard practice typically calls for the following units for key quantities pertinent to the analysis of gear sets:

Forces in pounds (lb)

Power in horsepower (hp) (Note that
1.0 hp = 550 lb · ft/s.)

Rotational speed in rpm, that is, rev/min

Pitch line speed in ft/min

Torque in lb · in

The torque exerted on a gear is the product of the transmitted load, W_t , and the pitch radius of the gear. The torque is also equal to the power transmitted divided by the rotational speed. Then

$$T = W_t(R) = W_t(D/2) = P/n$$

Then we can solve for the force, and the units can be adjusted as follows:

⇨ Tangential Force

$$W_t = \frac{2P}{Dn} = \frac{2P(\text{hp})}{D(\text{in}) \cdot n(\text{rev/min})} \cdot \frac{550 \text{ lb} \cdot \text{ft/s}}{(\text{hp})} \cdot \frac{1.0 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{60 \text{ s/min}}{12 \text{ in}} \cdot \frac{12 \text{ in}}{\text{ft}}$$

$$W_t = (126\,000)(P)/(nD) \text{ lb} \quad (9-8)$$

Data for either the pinion or the gear can be used in this equation. Other relationships are now developed because they are needed in other parts of the process of analyzing the gears or the shafts that carry them.

Power is also the product of transmitted force, W_t , and the pitch line velocity:

$$P = W_t \cdot v_t$$

Then, solving for the force and adjusting units, we have

⇨ Tangential Force

$$W_t = \frac{P}{v_t} = \frac{P(\text{hp})}{v_t(\text{ft/min})} \cdot \frac{550 \text{ lb} \cdot \text{ft/s}}{1.0 \text{ hp}} \cdot \frac{60 \text{ s/min}}{12 \text{ in}} \cdot \frac{12 \text{ in}}{\text{ft}}$$

$$= 33\,000 (P)/(v_t) \text{ lb} \quad (9-9)$$

We may also need to compute torque in lb · in:

⇨ Torque

$$T = \frac{P}{\omega} = \frac{P(\text{hp})}{n(\text{rev/min})} \cdot \frac{550 \text{ lb} \cdot \text{ft/s}}{1.0 \text{ hp}} \cdot \frac{1.0 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{60 \text{ s/min}}{12 \text{ in}} \cdot \frac{12 \text{ in}}{\text{ft}}$$

$$T = 63\,000(P)/n \text{ lb} \cdot \text{in} \quad (9-10)$$

These values can be computed for either the pinion or the gear by appropriate substitutions. Remember that the pitch line speed is the same for the pinion and the gear and that the transmitted loads on the pinion and the gear are the same, except that they act in opposite directions.

The normal force, W_n , and the radial force, W_r , can be computed from the known W_t by using the right triangle relations evident in Figure 9-3:

⇨ Radial Force

$$W_r = W_t \tan \phi \quad (9-11)$$

⇨ Normal Force

$$W_n = W_t / \cos \phi \quad (9-12)$$

where ϕ = pressure angle of the tooth form

In addition to causing the stresses in the gear teeth, these forces act on the shaft. In order to maintain equilibrium, the bearings that support the shaft must provide the reactions. Figure 9-2(e) shows the free-body diagram of the output shaft of the reducer.

Let's look at an example of a pinion driving a gear to see how the rotational speed, torque, and forces are determined.

Power Flow and Efficiency

The discussion thus far has focused on power, torque, and forces for a single pair of gears. For compound gearing having two or more pairs of gears, the flow of power and the overall efficiency become increasingly important.

Power losses in gear drives made from spur, helical, and bevel gears depend on the action of each tooth on its mating tooth, a combination of rolling and sliding. For accurate, well-lubricated gears, the power loss ranges from 0.5% to 2.0% and is typically taken to be approximately 1.0%. (See Reference 20.) *Because this is quite small, it is customary to neglect it in sizing individual gear pairs; we do that in this book.*

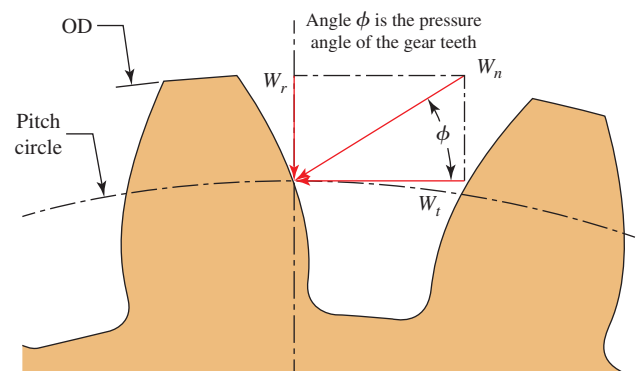


FIGURE 9-3 Forces on gear teeth: Tangential force, W_t ; Radial force, W_r ; Normal force, W_n

Example Problem 9-1

A pair of spur gears with a 20° pressure angle, full depth, involute teeth transmit 30 hp. The pinion is mounted on a jack shaft assembly that is directly coupled to the shaft of the electric motor operating at 600 rpm as shown in Figure 9-2(a) and (b). The pinion has 36 teeth and a diametral pitch of 5. The gear has 60 teeth and is mounted on a jack shaft assembly that is directly coupled to the output roller. The output roller is part of the driven machinery. Compute the following:

- The velocity ratio and the gear ratio for the gear pair.
- The rotational speed of the gear.
- The pitch diameter of the pinion and the gear.
- The center distance of the shafts carrying the pinion and the gear.
- The pitch line speed for both the pinion and the gear.
- The torque on the pinion shaft and gear shaft.
- The tangential force and on the teeth of each gear.
- The radial force acting on the teeth of each gear.
- The normal force acting on the teeth of each gear.

Solution

- From Equation (9-3), with the pinion as the driver and the gear being driven, the velocity ratio is:

$$\text{Velocity ratio} = VR = \frac{\omega_p}{\omega_g} = \frac{N_g}{N_p} = \frac{60}{36} = 1.667$$

- The velocity ratio can be rewritten to solve for the rotational speed of the gear:

$$\omega_g = \frac{\omega_p}{VR} = \frac{600 \text{ rpm}}{1.667} = 360 \text{ rpm}$$

- The pitch diameters of the pinion and gear are:

$$D_p = \frac{N_p}{P_d} = \frac{36}{5} = 7.200 \text{ in} \quad D_g = \frac{N_g}{P_d} = \frac{60}{5} = 12.000 \text{ in}$$

- The center distance for the pinion and gear:

$$CD = \frac{D_p + D_g}{2} = \frac{7.200 \text{ in} + 12.000 \text{ in}}{2} = 9.600 \text{ in}$$

- The pitch line speed can be solved using either the pinion or the gear:

$$v_t = \frac{D}{2} \cdot \omega$$

$$v_{tp} = \frac{D_p}{2} \cdot \omega_p = \frac{7.200 \text{ in}}{2} \cdot \frac{600 \text{ rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 1130 \text{ fpm}$$

$$v_{tg} = \frac{D_g}{2} \cdot \omega_g = \frac{12.000 \text{ in}}{2} \cdot \frac{360 \text{ rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 1130 \text{ fpm}$$

- The torque on the pinion shaft and gear shaft can be determined by solving the power equation in terms of torque

$$T = \frac{P}{\omega}$$

$$T_p = \frac{P}{\omega_p} = \frac{30 \text{ hp}}{600 \text{ rpm}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 3151.3 \text{ lb} \cdot \text{in}$$

$$T_g = \frac{P}{\omega_g} = \frac{30 \text{ hp}}{360 \text{ rpm}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 5252.1 \text{ lb} \cdot \text{in}$$

You will notice the ratio of the torque on the gear shaft to the torque on the pinion shaft is the same as the gear ratio: The power through each shaft is 30 hp. If the rotational speed of the shaft decreases, the torque will increase proportionally keeping the power constant.

$$m_g = \frac{T_g}{T_p} = \frac{5252.1 \text{ lb} \cdot \text{in}}{3151.3 \text{ lb} \cdot \text{in}} = 1.67$$

- g. The tangential force is solved using the pinion torque and the pitch circle radius or the gear torque and gear pitch circle radius. This is the drive force the pinion tooth applies to the gear tooth and equal and opposite reaction force the gear tooth applies to the pinion tooth.

$$W_t = \frac{T_p}{\left(\frac{D_p}{2}\right)} = \frac{3151.3 \text{ lb} \cdot \text{in}}{\left(\frac{7.200 \text{ in}}{2}\right)} = 875.4 \text{ lb}$$

$$W_t = \frac{T_g}{\left(\frac{D_g}{2}\right)} = \frac{5252.1 \text{ lb} \cdot \text{in}}{\left(\frac{12.000 \text{ in}}{2}\right)} = 875.4 \text{ lb}$$

- h. The radial force is calculated using the tangential force and the pressure angle. As you can see the direction of the force is toward the centerline of the shaft and tends to separate the gear set.

$$W_r = W_t \cdot \tan(\phi) = 875.4 \text{ lb} \cdot \tan(20^\circ) = 318.6 \text{ lb}$$

- i. The normal force to the tooth profile is solved using Pythagorean theorem. This force is along the line of action and is normal to the tooth profile as shown in Figure 9–4.

$$W_n = \sqrt{(W_t)^2 + (W_r)^2} = \sqrt{(875.4 \text{ lb})^2 + (318.6 \text{ lb})^2} = 931.5 \text{ lb}$$

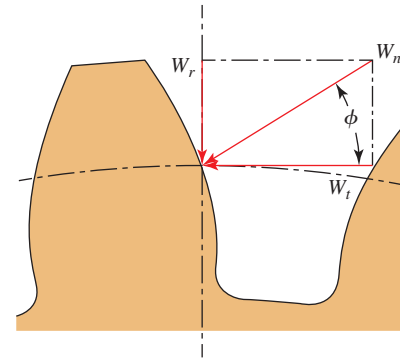


FIGURE 9–4 Forces on the gear tooth

Example Problem 9–2 follows, in which the power from a drive motor is delivered through a gear train to four separate output shafts, demonstrating the concept of

power flow through a gear train. Tracking the power, torque, and speed of each gear in the train is critical to proper analysis.

Example Problem 9–2

Figure 9–5 represents the power flow through a gear drive to multiple output shafts. Gear A is mounted on an electric motor that has a rotational speed of 1500 rpm (CW). Gear A drives a simple gear train consisting of gears B, C, D, and E. Gear A transfers the input power through each gear to the shaft on which it is mounted. The power levels delivered from shafts B, C, D, and E are 3 hp, 8 hp, 3 hp, and 3 hp, respectively. All gears have a diametral pitch of 10 and the following numbers of teeth:

$$N_A = 60 \quad N_B = 30 \quad N_C = 90 \quad N_D = 30 \quad N_E = 30 \quad P_d = 8$$

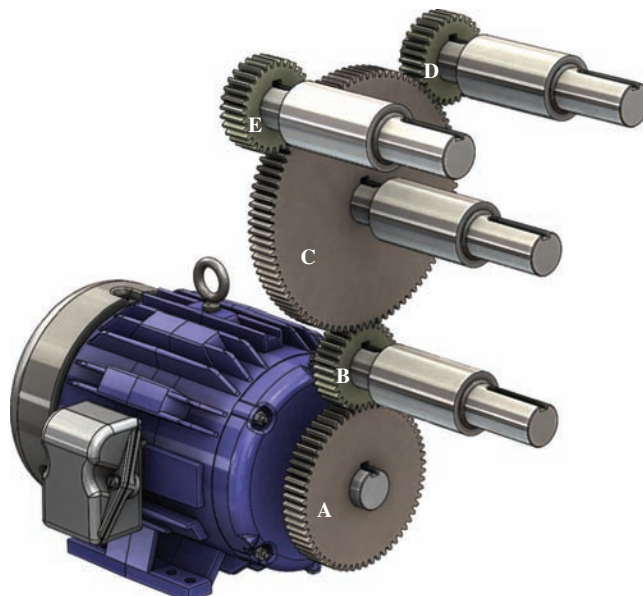


FIGURE 9–5 Power flow through gear drive system

Determine the following data for the drive system:

- The pitch diameter of each gear
- The center distance of each gear mesh
- The rotational speed of each output shaft
- Neglecting efficiency losses, find the power requirement of the electric motor
- The torque through each output shaft and the tangential force on the gear teeth

Solution a. Pitch Diameter for each gear:

$$D_A = \frac{N_A}{p_d} = \frac{60}{10} = 6.000 \text{ in} \quad D_B = \frac{N_B}{p_d} = \frac{30}{10} = 3.000 \text{ in} \quad D_C = \frac{N_C}{p_d} = \frac{90}{10} = 9.000 \text{ in}$$

$$D_D = \frac{N_D}{p_d} = \frac{30}{10} = 3.000 \text{ in} \quad D_E = \frac{N_E}{p_d} = \frac{30}{10} = 3.000 \text{ in}$$

- b. The center distance for each gear mesh:

$$CD_{A-B} = \frac{D_A + D_B}{2} = \frac{6.0 \text{ in} + 3.0 \text{ in}}{2} = 4.500 \text{ in} \quad CD_{B-C} = \frac{D_B + D_C}{2} = \frac{3.0 \text{ in} + 9.0 \text{ in}}{2} = 6.000 \text{ in}$$

$$CD_{C-D} = \frac{D_C + D_D}{2} = \frac{9.0 \text{ in} + 3.0 \text{ in}}{2} = 6.000 \text{ in} \quad CD_{C-E} = \frac{D_C + D_E}{2} = \frac{9.0 \text{ in} + 3.0 \text{ in}}{2} = 6.000 \text{ in}$$

- c. The next item we want to solve for is the rotational speed of each gear of the gear train. Let's step through one gear mesh at a time, starting with gear A that drives gear B:

$$VR_{A-B} = \frac{n_A}{n_B} = \frac{N_B}{N_A} = \frac{30}{60} = \frac{1}{2}$$

The equation $VR_{A-B} = \frac{n_A}{n_B}$ is used to solve for the rotational speed of gear B. The velocity ratio gives a speed increase from gear A to gear B:

$$n_B = \frac{n_A}{VR_{A-B}} = \frac{1500 \text{ rpm}}{0.5} = 3000 \text{ rpm}$$

Likewise, considering gear B driving gear C, the velocity ratio is:

$$VR_{B-C} = \frac{n_B}{n_C} = \frac{N_C}{N_B} = \frac{90}{30} = 3$$

$$\text{The rotation speed of gear C: } n_C = \frac{n_B}{VR_{B-C}} = \frac{3000 \text{ rpm}}{3} = 1000 \text{ rpm}$$

It is important to note, however, that using the velocity ratio equation from gear A to gear C would give us the same result:

$$VR_{A-C} = \frac{n_A}{n_C} = \frac{N_B}{N_A} \cdot \frac{N_C}{N_B} = \frac{30}{60} \cdot \frac{90}{30} = 1.5$$

$$\text{The rotational speed of gear C: } n_C = \frac{n_A}{VR_{A-C}} = \frac{1500 \text{ rpm}}{1.5} = 1000 \text{ rpm}$$

The rotational speed of gear D and gear E will be the same since both gears have the same number of teeth and are driven by gear C.

$$VR_{C-D} = \frac{n_C}{n_D} = \frac{N_D}{N_C} = \frac{30}{90} = \frac{1}{3}$$

$$n_E = n_D = \frac{n_C}{VR_{C-D}} = \frac{1000 \text{ rpm}}{0.333} = 3000 \text{ rpm}$$

Again, we could have used the velocity ratio equation from gear A to gear D or E. That would have given us the same results:

$$VR_{A-D} = \frac{n_A}{n_D} = \frac{N_B}{N_A} \cdot \frac{N_C}{N_B} \cdot \frac{N_D}{N_C} = \frac{30}{60} \cdot \frac{90}{30} \cdot \frac{30}{90} = \frac{1}{2}$$

$$\text{The rotational speed of gear D and gear E } n_D = n_E = \frac{n_A}{VR_{A-D}} = \frac{1500 \text{ rpm}}{0.5} = 1000 \text{ rpm}$$

- d. The power required by the motor, assuming no losses in the system, is equal to the sum of the power required by each drive shaft. This can be written as:

$$Power_{in} = Power_{out}$$

$$P_{in} = P_B + P_C + P_D + P_E$$

$$P_{in} = 3 \text{ hp} + 8 \text{ hp} + 3 \text{ hp} + 3 \text{ hp} = 17 \text{ hp}$$

- e. The next part of the problem is asking us to find the torque transmitted through each output shaft and the tangential force on the gear teeth. Start with gear E and work through to the drive motor gear A. The torque transmitted through shaft E can be found using the required output power and the rotational speed of shaft.

$$P_{out E} = T_E \cdot n_E$$

$$T_E = \frac{P_{out E}}{n_E} = \frac{3.0 \text{ hp}}{3000 \text{ rev/min}} \cdot \frac{33000 \cdot \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 63.0 \text{ lb} \cdot \text{in}$$

The tangential gear force, $W_{tC/E}$, is the force gear C applies to gear E to obtain the torque required to transmit the required power through shaft E. The tangential force, $W_{tE/C}$ is the equal to but opposite the reaction force gear E applies to gear C. The tangential gear force can be determined:

$$W_{tD/C} = W_{tE/C} = \frac{T_E}{\left(\frac{D_E}{2}\right)} = \frac{63.0 \text{ lb} \cdot \text{in}}{\left(\frac{3.0 \text{ in}}{2}\right)} = 42.0 \text{ lb}$$

Since gear E has the same rotational speed, power requirement, and pitch diameter as gear D, the torque through shaft E and the tangential gear tooth force is the same. This is illustrated in Figure 9–6.

$$T_D = T_E = 63.0 \text{ lb} \cdot \text{in}$$

$$W_{tC/D} = W_{tD/C} = 42.0 \text{ lb}$$

8 hp is transmitted through shaft C at a rotational speed of 1000 rpm. The torque through shaft C can then be determined by rearranging this equation:

$$P_{out C} = T_C \cdot n_C$$

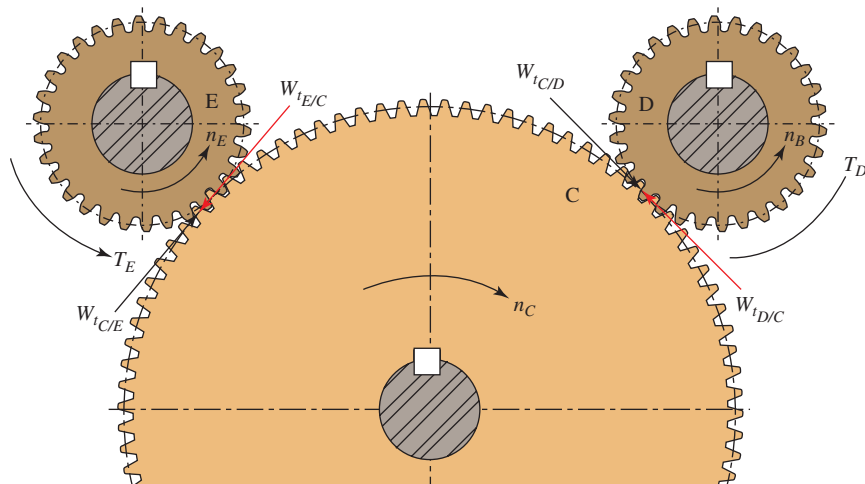


FIGURE 9–6 Torques and tangential gear force shown on gears C, D, and E

$$T_C = \frac{P_{out C}}{n_C} = \frac{8.0 \text{ hp}}{1000 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 504.2 \text{ lb} \cdot \text{in}$$

The total torque required to drive gears C, D, and E:

$$T_{C \text{ total}} = T_C + W_{tE/C} \cdot \left(\frac{D_C}{2}\right) + W_{tD/C} \cdot \left(\frac{D_C}{2}\right)$$

$$T_{C \text{ total}} = 504.2 \text{ lb} \cdot \text{in} + 42.0 \text{ lb} \cdot \left(\frac{9.0 \text{ in}}{2}\right) + 42.0 \text{ lb} \cdot \left(\frac{9.0 \text{ in}}{2}\right) = 882.3 \text{ lb} \cdot \text{in}$$

We can verify this torque, using the sum of the power required by shafts C, D, and E.

$$P_{C \text{ total}} = P_C + P_D + P_E = 8 \text{ hp} + 3 \text{ hp} + 3 \text{ hp} = 14 \text{ hp}$$

The total torque required to drive gears C, D, and E:

$$T_{C \text{ total}} = \frac{P_{C \text{ total}}}{n_C} = \frac{14 \text{ hp}}{1000 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 882.3 \text{ lb} \cdot \text{in}$$

We can see this gives us the same value as was shown above. This torque is used to find the tangential force gear C applies to gear B:

$$W_{tC/B} = \frac{T_{C \text{ total}}}{\left(\frac{D_C}{2}\right)} = \frac{882.3 \text{ lb} \cdot \text{in}}{\left(\frac{9.0 \text{ in}}{2}\right)} = 196.0 \text{ lb}$$

The tangential force gear C applies to gear B is equal to the reaction force gear B applies to gear C.

$$W_{tC/B} = W_{tB/C} = 196.0 \text{ lb}$$

This is shown in Figure 9–7.

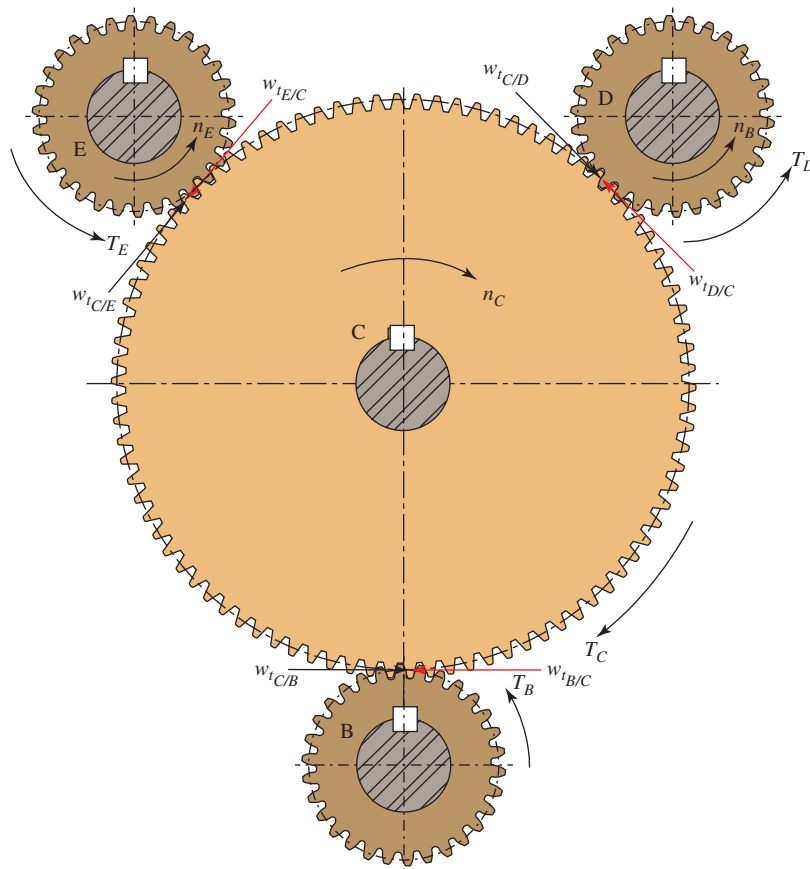


FIGURE 9–7 Tangential gear forces on gears B, C, D, and E

Shaft B transmits 3 hp at a rotational speed of 3000 rpm. The torque applied to shaft B to drive the output:

$$T_B = \frac{P_B}{n_B} = \frac{3.0 \text{ hp}}{3000 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 63.0 \text{ lb} \cdot \text{in}$$

The total torque required to drive gears B, C, D, and E

$$T_{B \text{ total}} = T_B + W_{t \text{ C/B}} \cdot \left(\frac{D_B}{2} \right) = 63.0 \text{ lb} \cdot \text{in} + 196.0 \text{ lb} \cdot \left(\frac{3.0 \text{ in}}{2} \right) = 357.1 \text{ lb} \cdot \text{in}$$

We can verify this torque using the sum of the power required by shafts B, C, D, and E.

$$P_{B \text{ total}} = P_B + P_C + P_D + P_E = 3 \text{ hp} + 8 \text{ hp} + 3 \text{ hp} + 3 \text{ hp} = 17 \text{ hp}$$

The total torque required to drive gears B, C, D, and E:

$$T_{B \text{ total}} = \frac{P_{B \text{ total}}}{n_C} = \frac{17 \text{ hp}}{1000 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 357.1 \text{ lb} \cdot \text{in}$$

This gives us the same result as was shown above. This torque is used to find the tangential force gear B applies to gear A:

$$W_{t \text{ B/A}} = \frac{T_{B \text{ total}}}{\left(\frac{D_B}{2} \right)} = \frac{357.1 \text{ lb} \cdot \text{in}}{\left(\frac{3.0 \text{ in}}{2} \right)} = 238.1 \text{ lb}$$

The tangential force gear B applies to gear A is equal to the reaction force gear A applies to gear B and is shown in Figure 9–8.

$$W_{t \text{ B/A}} = W_{t \text{ A/B}} = 238.1 \text{ lb}$$

The required torque of the motor is:

$$T_{\text{Motor}} = T_A = W_{t \text{ A/B}} \cdot \left(\frac{D_A}{2} \right) = 238.1 \text{ lb} \cdot \text{in} \cdot \left(\frac{6.0 \text{ in}}{2} \right) = 714.3 \text{ lb} \cdot \text{in}$$

The motor torque and rotational speed can be used to verify the input power requirement.

$$P_{\text{Motor}} = T_{\text{Motor}} \cdot n_{\text{Motor}} = 714.3 \text{ lb} \cdot \text{in} \cdot 1500 \text{ rpm} \cdot \frac{1 \text{ hp}}{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 17 \text{ hp}$$

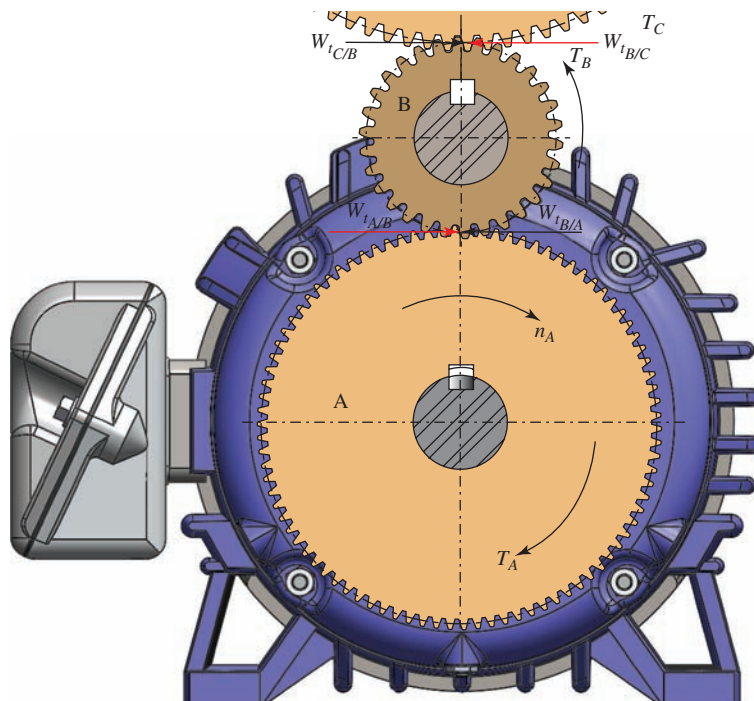


FIGURE 9–8 Tangential gear forces on gears A and B

Compound gear drives employ several pairs of gears in series to produce large speed reduction ratios. With 1.0% power loss in each pair, the accumulated power loss for the system can become significant, and it can affect the size of motor to drive the system or the ultimate power and torque available for use at the output. Furthermore, the power loss is transferred to the environment or into the gear lubricant and, for large power transmissions, the management of the heat generated is critical to the overall performance of the unit. The viscosity and load-carrying ability of lubricants is degraded with increasing temperature.

Tracking power flow in a simple or compound gear train is simple, the power is transferred from one gear pair to the next with only a small power loss at each mesh. More complex designs may split the power flow at some point to two or more paths. This is typical of planetary gear trains. In such cases, you should consider the basic relationship among power, torque, and rotational speed shown in Equation (9-7), $P = T \times n$. We can present this in another form. Let the rotational speed, n , that is typically taken to be in the units of rpm, be the more general term *angular velocity*, ω , in the units of rad/s. Now express the torque in terms of the transmitted forces, W_t , and the pitch radius of the gear, R . That is, $T = W_t R$. Equation (9-7) then becomes,

$$P = T \times n = W_t R \omega$$

But $R \omega$ is the pitch line velocity for the gears, v_t . Then,

$$P = W_t R \omega = W_t v_t$$

Knowing how the power splits enables the determination of the transmitted load at each mesh.

9-4 INTRODUCTION TO STRESS ANALYSIS FOR GEARS

Later in this chapter, design procedures are presented in which two forms of gear-tooth failure are considered.

A gear tooth acts like a cantilever beam in resisting the force exerted on it by the mating tooth. The point of highest tensile bending stress is at the root of the tooth where the involute curve blends with the fillet. The AGMA has developed a set of *allowable bending stress numbers*, called s_{at} , which are compared to computed bending stress levels in the tooth to rate the acceptability of a design.

A second, independent form of failure is the pitting of the surface of the teeth, usually near the pitch line, where high contact stresses occur. The transfer of force from the driving to the driven tooth theoretically occurs across a line contact because of the action of two convex curves on each other. Repeated application of these high contact stresses can cause a type of fatigue failure of the surface, resulting in local fractures and an actual loss of material. This is called *pitting*. The AGMA has developed a set of *allowable contact stress numbers*, called s_{ac} , which are compared to computed contact stress levels in the tooth to rate the acceptability of a design.

Both bending stress and contact stress must be at safe levels in any pair of gears. The processes involved in verifying those safe stress levels are developed and demonstrated in the following sections 9-5 through 9-12, as outlined here:

- Section 9-5 describes the method for calculating the bending stress in gear teeth and demonstrates that process in Example Problem 9-3.
- Section 9-6 describes the method for calculating the contact stress in gear teeth and demonstrates that process in Example Problem 9-4.
- Section 9-7 describes typical metallic materials that are used for gearing and provides sample data for allowable stresses that can be used for problem solving in this book. Additional data of this type can be found in References 6, 8, and 9.
- Section 9-8 describes the process for evaluating the safety of materials proposed for application to a particular pair of gears.
 - Example Problem 9-5 demonstrates the evaluation of three proposed materials for a given application with regard to bending stress.
 - Example Problem 9-6 then demonstrates a similar evaluation for contact stress.
 - The comparison of the results for Example Problems 9-3 to 9-6 is then presented and conclusions are drawn on which proposed material is most satisfactory.
- Section 9-9 then presents an alternate approach to the design of gear pairs in which the final step is the specification of the material that will satisfy both the bending stress and the contact stress. Example Problem 9-7 demonstrates that process.
- Section 9-10 modifies the process developed in Section 9-9 for design of gearing in the metric module system. Example Problem 9-8 demonstrates that process.
- Sections 9-11 and 9-12 show examples of the use of spreadsheets to perform the many calculations required for analysis and design of gear pairs, along with the use of the spreadsheets to work toward an optimum design.

9-5 BENDING STRESS IN GEAR TEETH

The stress analysis of gear teeth is facilitated by consideration of the orthogonal force components, W_t and W_r , as shown in Figure 9-3.

The tangential force, W_t , produces a bending moment on the gear tooth similar to that on a cantilever beam. The resulting bending stress is maximum at the base of the tooth in the fillet that joins the involute profile to the bottom of the tooth space. Taking the detailed geometry of the tooth into account, Wilfred Lewis developed the equation for the stress at the base of the involute profile, which is now called the *Lewis equation*:

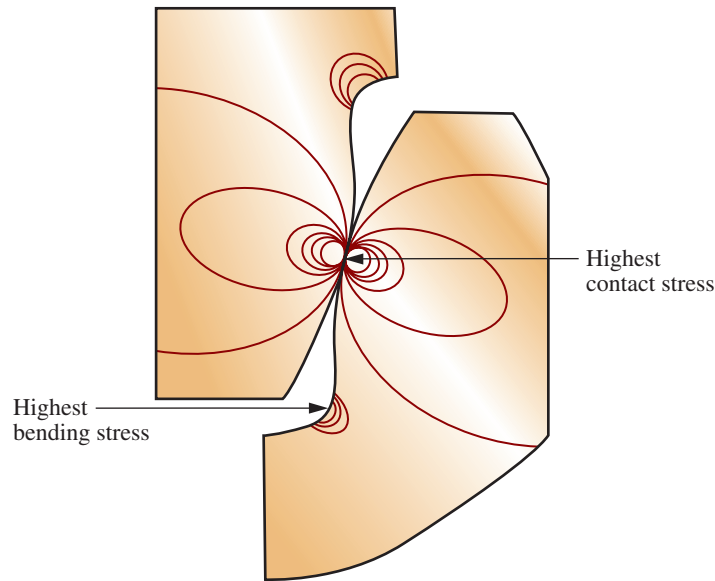


FIGURE 9-9 Photoelastic study of gear teeth under load

⇒ Lewis Equation for Bending Stress in Gear Teeth

$$\sigma_t = \frac{W_t P_d}{FY} \quad (9-13)$$

where

W_t = tangential force

P_d = diametral pitch of the tooth

F = face width of the tooth

Y = *Lewis form factor*, which depends on the tooth form, the pressure angle, the diametral pitch, the number of teeth in the gear, and the place where W_t acts

While the theoretical basis for the stress analysis of gear teeth is presented, the Lewis equation must be modified for practical design and analysis. One important limitation is that it does not take into account the stress concentration that exists in the fillet of the tooth. Figure 9-9 is a drawing made from an experimental stress analysis of the stress distribution in gear teeth. It indicates a stress concentration in the fillet at the root of the tooth as well as high contact stresses at the mating surface (contact stress is discussed in the following section). Comparing the actual stress at the root with that predicted by the Lewis equation enables us to determine the stress concentration factor, K_t , for the fillet area. Placing this into Equation (9-13) gives

$$\sigma_t = \frac{W_t P_d K_t}{FY} \quad (9-14)$$

The value of the stress concentration factor is dependent on the form of the tooth, the shape and size of the fillet at the root of the tooth, and the point of application of the force on the tooth. Note that the value of the Lewis form factor, Y , also depends on the tooth geometry. Therefore, the two factors are combined into one term, the *geometry factor*, J , where $J = Y/K_t$. The value of J also, of course, varies with the location of the point

of application of the force on the tooth because Y and K_t vary.

Figure 9-10 shows graphs giving the values for the geometry factor for 20° and 25°, full-depth, involute teeth. The safest value to use is the one for the load applied at the tip of the tooth. However, this value is overly conservative because there is some load sharing by another tooth at the time that the load is initially applied at the tip of a tooth. The critical load on a given tooth occurs when the load is at the highest point of single-tooth contact, when the tooth carries the entire load. The upper curves in Figure 9-10 give the values for J for this condition.

Using the geometry factor, J , in the stress equation gives

$$\sigma_t = \frac{W_t P_d}{FJ} \quad (9-15)$$

The graphs in Figure 9-10 are taken from the former AGMA Standard 218.01 which has been superseded by the two new standards: AGMA 2001, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, (Reference 6), and AGMA 908 *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical and Herringbone Gear Teeth*, (Reference 3). Standard 908 includes an analytical method for calculating the geometry factor, J . But the values for J are unchanged from those in the former standard. Rather than graphs, the new standard reports values for J for a variety of tooth forms in tables. The graphs from the former standard are shown in Figure 9-10 so that you can visualize the variation of J with the number of teeth in the pinion and the gear.

Note also that J factors for only two tooth forms are included in Figure 9-10 and that the values are valid only for those forms. Designers must ensure that J factors for the tooth form actually used, including the form of the fillet, are included in the stress analysis.

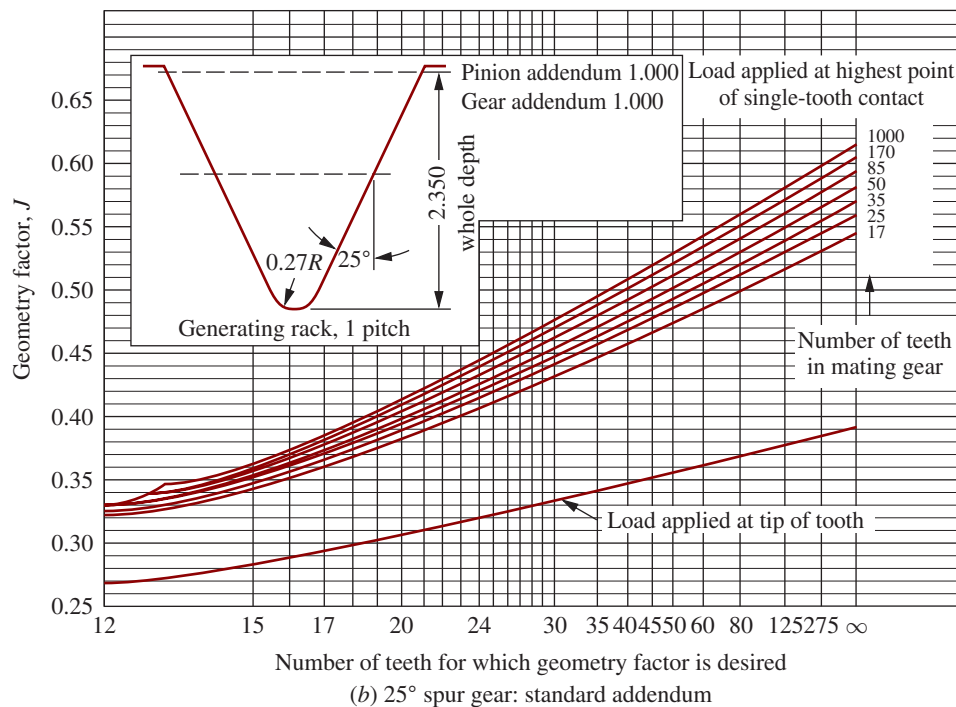
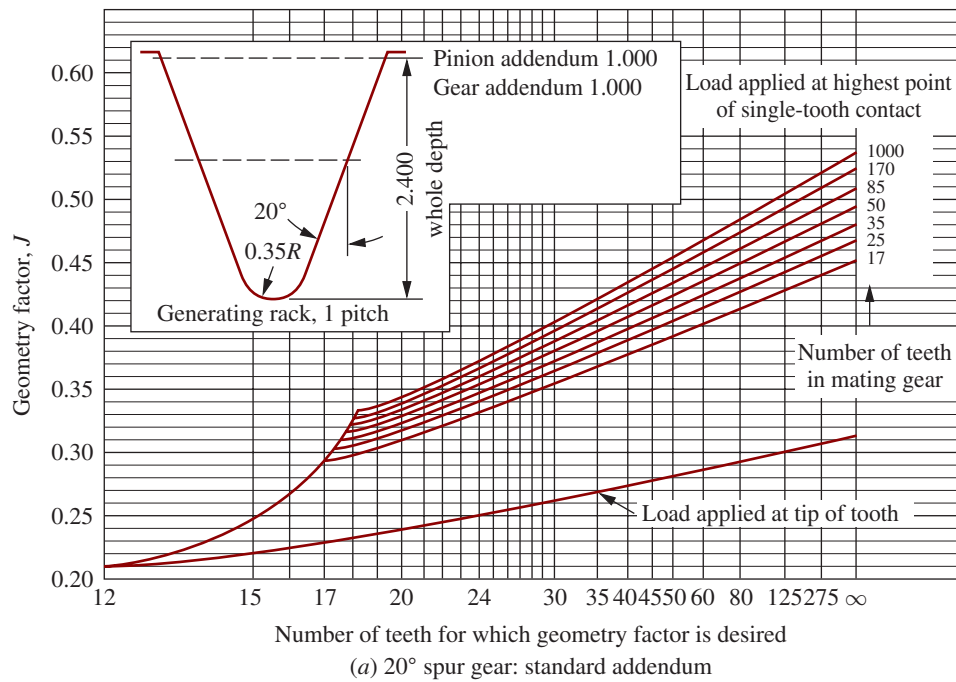


FIGURE 9-10 Geometry factor, J (Extracted from AGMA 218.01 Standard, *Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

Equation (9-15) can be called the *modified Lewis equation*. Other modifications to the equation are recommended by the AGMA in Standard 2001 for practical design to account for the variety of conditions that can be encountered in service.

The approach used by the AGMA is to apply a series of additional modifying factors to the bending stress from the modified Lewis equation to compute a value called the *bending stress number*, s_f . These factors

represent the degree to which the actual loading case differs from the theoretical basis of the Lewis equation. The result is a better estimate of the real level of bending stress that is produced in the teeth of the gear and the pinion.

Then, separately, the allowable bending stress number, s_{at} , is modified by a series of factors that affect that value when the environment is different from the nominal situation assumed when the values for s_{at} are set.

The result here is a better estimate of the real level of the bending strength of the material from which the gear or the pinion is made.

The design is completed in a manner that ensures that the bending stress number is less than the modified allowable bending stress number. This process should be completed for both the pinion and the gear of a given pair because materials may be different; the geometry factor, J , is different; and other operating conditions may be different. This is demonstrated in example problems later in this chapter.

Often the major decision to be made is the specification of suitable materials from which to make the pinion and the gear. In such cases, the required basic allowable bending stress number, s_{at} , will be computed. When steel is used, the required hardness of the material is found from the data described in Section 9–7. Finally, the material and its heat treatment are specified to ensure that it will have at least the required hardness.

We proceed now with the discussion of the bending stress number, s_t .

Bending Stress Number, s_t

The design analysis method used here is based primarily on AGMA Standard 2001. However, because values for some of the factors are not included in the standard, data from other sources are added. These data illustrate the kinds of conditions that affect the final design. The designer ultimately has the responsibility for making appropriate design decisions.

The following equation will be used in this book:

◇ Bending Stress Number, s_t

$$s_t = \frac{W_t P_d}{F J} K_o K_s K_m K_B K_v \quad (9-16)$$

where W_t = tangential force

P_d = diametral pitch

F = face width

J = geometry factor

K_o = overload factor for bending strength

K_s = size factor for bending strength

K_m = load-distribution factor for bending strength

K_B = rim thickness factor

K_v = dynamic factor for bending strength

Methods for specifying values for these factors are discussed below.

Tangential Force, W_t

The tangential force, W_t , is the driving force of the gear mesh. The pinion transmits a force tangent to the pitch circle on the gear. The method for calculating this force is described in detail in Section 9–3.

Diametral Pitch, P_d

The diametral pitch is the size of the gear tooth and is related to the number of teeth and the pitch diameter. If the diametral pitch is not given as a specific value in a problem, use Figure 9–11 to select a trial value. Figure 9–11 provides initial guidance when selecting a diametral pitch for a pair of steel gears. The graph of design power transmitted versus the pinion rotational speed was derived for selected pitches and pinion diameters. Steel that is through hardened to HB 300 is used to generate the graph. Because of the numerous variables involved, the value of P_d read from the figure is only an initial target value. Subsequent iterations may require considering a different value, either higher or lower.

The design power equation is:

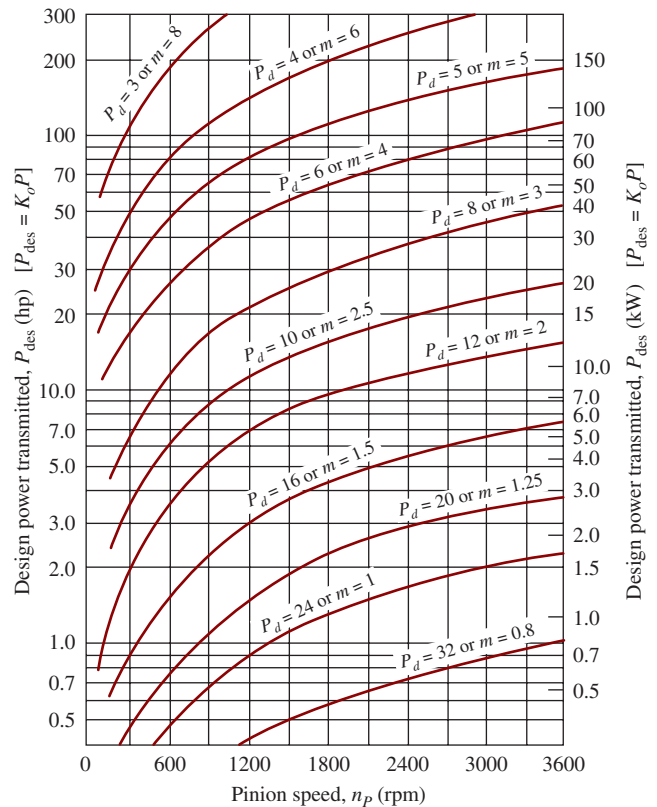
$$P_{\text{design}} = K_o \cdot P$$

where

K_o is the overload factor which will be explained in detail in this section

P is the power being transmitted by the gear pair

P_{design} is the power level used for the design of the gear drive system



For all curves: 20° full depth teeth;
 $N_p = 24$; $N_G = 96$; $m_G = 4.00$; $F = 12/P_d$; $A_v = 11$
 Steel gears, HB 300; $s_{at} = 36000$ psi (250 MPa); $s_{ac} = 126000$ psi (869 MPa)

FIGURE 9-11 Design power transmitted versus pinion speed for spur gears with different pitches and diameters

Face Width, F

The face width of the gear tooth should fall within this range

$$\frac{8}{P_d} < F < \frac{16}{P_d}$$

The upper limit given tends to minimize alignment problems and ensures reasonable loading across the face. When the face width is less than the lower limit, it is probable that a more compact design can be achieved with a smaller diametral pitch. Also, the face width normally is less than twice the pitch diameter of the pinion.

For problems solved in this book, either a face width will be given that falls within the range or the nominal face width value can be calculated using:

$$F = \frac{12}{P_d}$$

Geometry Factor, J

The geometry factor, as discussed in the previous section, is found for both the pinion and gear. Figure 9–10(a) and (b) are used to select the geometry factor, based on the pressure angle and the number of teeth of the meshing gear set. For a 30 tooth pinion and a 50 tooth gear with a pressure angle of 20°, the geometry factor from Figure 9–10(a) would be 0.38 for the pinion and 0.41 for the gear.

Overload Factor, K_o

Overload factors consider the probability that load variations, vibrations, shock, speed changes, and other application-specific conditions may result in peak loads greater than W_t being applied to the gear teeth during operation. A careful analysis of actual conditions should be made, and the AGMA Standard 2001-D04 gives no specific values for K_o . Reference 20 gives some recommended values, and many industries have established suitable values based on experience.

For problem solutions in this book, we will use the values shown in Table 9–1. The primary considerations are the nature of *both* the driving power source and the driven machine. An overload factor of 1.00 would be applied for a perfectly smooth electric motor driving a perfectly smooth generator through a gear-type speed

reducer. Any rougher conditions call for a value of K_o greater than 1.00. For power sources we will use the following:

Uniform: Electric motor or constant-speed gas turbine

Light shock: Water turbine, variable-speed drive

Moderate shock: Multicylinder engine

Examples of the roughness of driven machines include the following:

Uniform: Continuous-duty generator, paper, and film winders.

Light shock: Fans and low-speed centrifugal pumps, liquid agitators, variable-duty generators, uniformly loaded conveyors, rotary positive displacement pumps, and metal strip processing.

Moderate shock: High-speed centrifugal pumps, reciprocating pumps and compressors, heavy-duty conveyors, machine tool drives, concrete mixers, textile machinery, meat grinders, saws, bucket elevators, freight elevators, escalators, concrete mixers, plastics molding and processing, sewage disposal equipment, winches, and cable reels.

Heavy shock: Rock crushers, punch press drives, pulverizers, processing mills, tumbling barrels, wood chippers, vibrating screens, railroad car dumpers, log conveyors, lumber handling equipment, metal shears, hammer mills, commercial washers, heavy-duty hoists and cranes, reciprocating feeders, dredges, rubber processing, compactors, and plastics extruders.

Size Factor, K_s

The AGMA indicates that the size factor can be taken to be 1.00 for most gears. But for gears with large-size teeth or large face widths, a value greater than 1.00 is recommended. Reference 20 recommends a value of 1.00 for diametral pitches of 5 or greater or for a metric module of 5 or smaller. For larger teeth, the values shown in Table 9–2 can be used.

Load-Distribution Factor, K_m

The determination of the load-distribution factor is based on many variables in the design of the gears themselves

TABLE 9–1 Suggested Overload Factors, K_o

Driven Machine				
Power source	Uniform	Light shock	Moderate shock	Heavy shock
Uniform	1.00	1.25	1.50	1.75
Light shock	1.20	1.40	1.75	2.25
Moderate shock	1.30	1.70	2.00	2.75

TABLE 9–2 Suggested Size Factors, K_s

Diametral pitch, P_d	Metric module, m	Size factor, K_s
≥ 5	≤ 5	1.00
4	6	1.05
3	8	1.15
2	12	1.25
1.25	20	1.40

as well as in the shafts, bearings, housings, and the structure in which the gear drive is installed. Therefore, it is one of the most difficult factors to specify. Much analytical and experimental work is continuing on the determination of values for K_m .

If the intensity of loading on all parts of all teeth in contact at any given time were uniform, the value of K_m would be 1.00. However, this is seldom the case. Any of the following factors can cause misalignment of the teeth on the pinion relative to those on the gear:

1. Inaccurate gear teeth.
2. Misalignment of the axes of shafts carrying gears.
3. Elastic deformations of the gears, shafts, bearings, housings, and support structures.
4. Clearances between the shafts and the gears, the shafts and the bearings, or the bearings and the housing.
5. Thermal distortions during operation.
6. Crowning or end relief of gear teeth.

AGMA Standard 2001 presents extensive discussions of methods of determining values for K_m . One is empirical and considers gears up to 40 in (1000 mm) wide. The other method is analytical and considers the stiffness and mass of individual gears and gear teeth and the total mismatch between mating teeth. We will not provide so much detail. However, rough guidelines are given below.

The designer can minimize the load-distribution factor by specifying the following:

1. Accurate teeth (a low quality number from AGMA 2015).
2. Narrow face widths.
3. Gears centered between bearings (straddle mounting).
4. Short shaft spans between bearings.
5. Large shaft diameters (high stiffness).
6. Rigid, stiff housings.
7. High precision and small clearances on all drive components.

You are advised to study the details of AGMA Standard 2001 which covers a wide range of physical sizes for gear systems. But the gear designs discussed in this book are of moderate size, typical of power transmissions in light industrial and vehicular applications. A more limited set of data are reported here to illustrate the concepts that must be considered in gear design.

We will use the following equation for computing the value of the load-distribution factor:

$$K_m = 1.0 + C_{pf} + C_{ma} \quad (9-17)$$

where C_{pf} = pinion proportion factor (see Figure 9-12)

C_{ma} = mesh alignment factor (see Figure 9-13)

In this book, we are limiting designs to those with face widths of 15 in or less. Wider face widths call for additional factors. Also, some commercially successful designs employ modifications to the basic tooth form to achieve a more uniform meshing of the teeth. Such methods are not discussed in this book.

Figure 9-12 shows that the pinion proportion factor is dependent on the actual face width of the pinion and on the ratio of the face width to the pinion pitch diameter. Figure 9-13 relates the mesh alignment factor to expected accuracy of different methods of applying gears.

- *Open gearing* refers to drive systems in which the shafts are supported in bearings that are mounted on structural elements of the machine with the expectation that relatively large misalignments will result.
- In *commercial-quality enclosed gear units*, the bearings are mounted in a specially designed housing that provides more rigidity than for open gearing, but for which the tolerances on individual dimensions are fairly loose.
- *Precision enclosed gear units* are made to tighter tolerances.
- *Extra-precision enclosed gear units* are made to exacting precision and are often adjusted at assembly to achieve excellent alignment of the gear teeth.

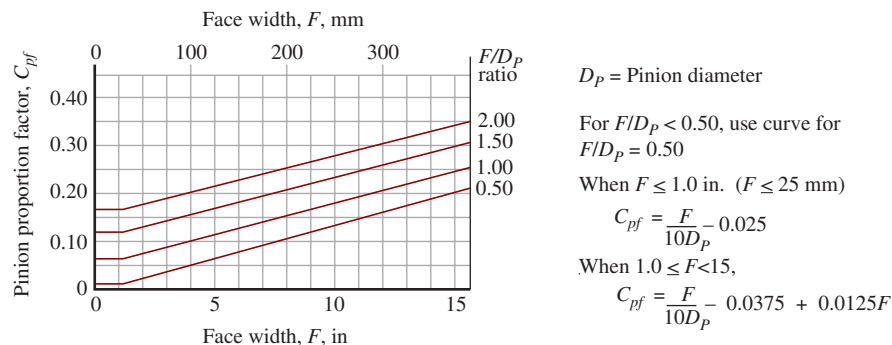


FIGURE 9-12 Pinion proportion factor, C_{pf} (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

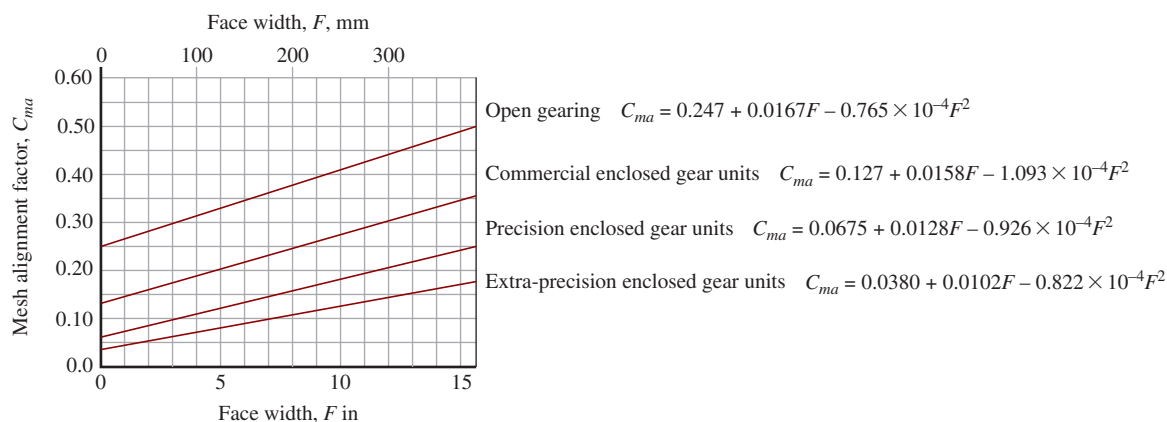


FIGURE 9-13 Mesh alignment factor, C_{ma} (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

Experience with similar units in the field will help you gain better understanding among the different types of designs. *Commercial* or *precision* types are recommended for this book.

Rim Thickness Factor, K_B

The basic analysis used to develop the Lewis equation assumes that the gear tooth behaves as a cantilever attached to a perfectly rigid support structure at its base. This is true if the gear is made from a solid blank as shown in Figure 8-4(b) or with a blank that has a thinned web as shown in Part (d) of that figure. These are typical of small to medium-sized gears. Larger gears are often made with the spoked design shown in Figure 8-4(a) in order to save material and produce a lighter gear. Commercially made spoked gears can be expected to have a well-supported rim. For these kinds of gears, $K_B = 1.0$ can be used.

However, when designing a spoked gear for a special application, care must be exercised that the rim is sufficiently stiff to support the gear teeth without dangerous

stresses created in the rim. Figure 9-14 should be used to estimate the influence of rim thickness. The key geometry parameter is called the *backup ratio*, m_B , where

$$m_B = t_R/h_t$$

t_R = rim thickness

h_t = whole depth of the gear tooth

For $m_B > 1.2$, the rim is sufficiently strong and stiff to support the tooth, and $K_B = 1.0$.

Special Case for Small Pinions Mounted to a Shaft with a Keyseat.

A frequent design option is to mount a small pinion onto a shaft as shown in Figure 9-15, with a key used to transmit the torque from the pinion to the shaft, and requiring a keyway to be machined into the bore of the pinion and into the shaft. Care must be taken to ensure that there is sufficient material above the keyway. It is recommended that the condition of $m_B > 1.2$ be applied in such cases where the t_R , dimension is measured above the top of the keyway. Then the factor $K_B = 1.0$ can be used.

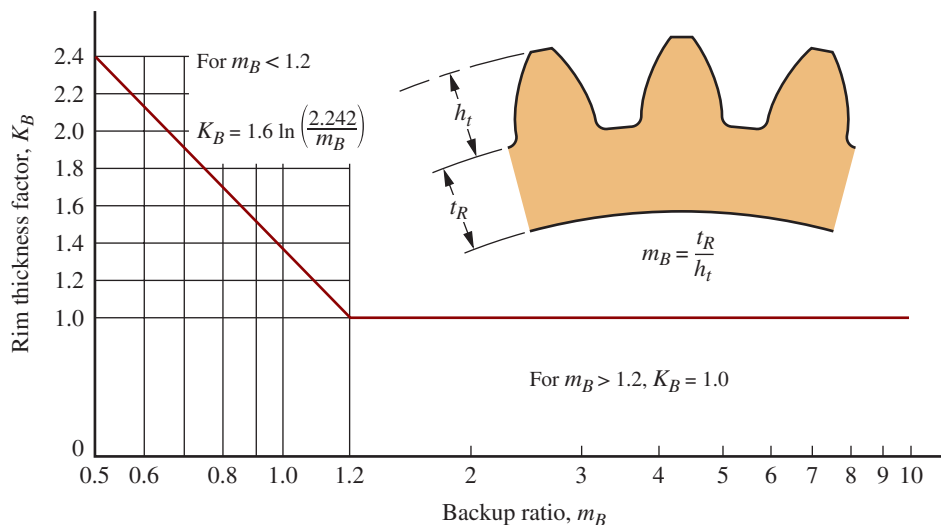


FIGURE 9-14 Rim thickness factor, K_B (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

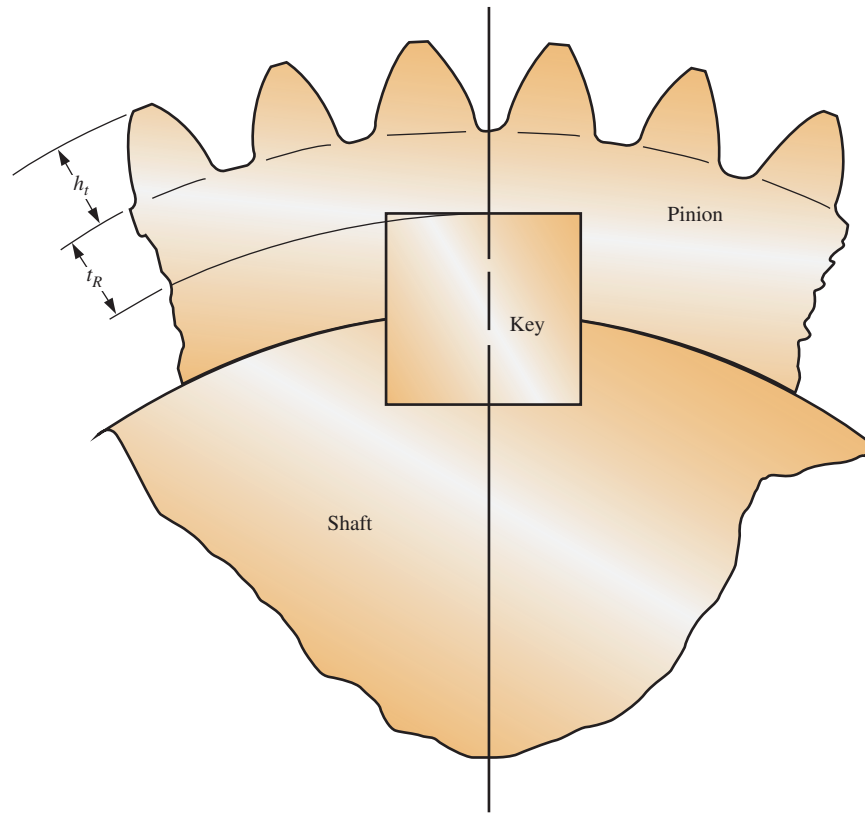


FIGURE 9-15 Pinion mounted on a shaft

If it is impractical to provide $m_B > 1.2$, it is recommended that *the pinion be machined integral with the shaft*, thus eliminating the need for a key and then using $K_B = 1.0$.

Dynamic Factor, K_v

The dynamic factor accounts for the fact that the load is assumed by a tooth with some degree of impact and that the actual load subjected to the tooth is higher than the transmitted load alone. The value of K_v depends on the accuracy of the tooth profile, the elastic properties of the tooth, and the speed with which the teeth come into contact. These factors are quantified by the use of *quality numbers* as described next.

Standards for Gear Quality

Manufacturers, designers, and users of gears must agree on standards used to determine the acceptability of gears produced. Several standard-setting organizations exist, including the AGMA (United States), ISO (International), DIN (German), JIS (Japan), and others. While the standards are similar, they are not identical. Designers and manufacturers

should become familiar with the provisions of these standards in order to communicate effectively the intent of their designs, the capability of manufacturing processes, and the acceptability of a given gear.

As indicated in Table 9-3, the current AGMA standard 2015-1-A01 (Reference 15), employs 10 accuracy grades from A2 (most precise) to A11 (least precise). An analytical measurement system [Figure 8-37] performs a total of nine elemental tangential measurements required for high-accuracy gears (A2–A5); five measurements for medium accuracy gears (A6–A9); and three for low accuracy gears (A10–A11).

The AGMA 2015-1-A01 standard replaces standard AGMA 2000-A88 that was used in the previous edition of this book and for which there are many gears in

TABLE 9-3 Accuracy Groups for AGMA 2015 Gear Quality System

Low (L)	A10–A11
Medium (M)	A6–A9
High (H)	A2–A5

TABLE 9-4 General Correlations of AGMA 2000, AGMA 2015, and ISO 1328 Gear Quality Systems

AGMA 2000	AGMA 2015	ISO 1328	AGMA 2000	AGMA 2015	ISO 1328
—Least precise—					
Q5	—	12	Q11	A6	6
Q6	A11	11	Q12	A5	5
Q7	A10	10	Q13	A4	4
Q8	A9	9	Q14	A3	3
Q9	A8	8	Q15	A2	2
Q10	A7	7	—Most precise—		

service that were made to that standard. It also replaces ISO 1328-1 that had been an interim standard modeled on the ISO methodology. For reference, Table 9-4 shows the rough relationships among the three standards although the actual values of tolerances are not identical. One notable difference is that the former AGMA 2000 standard employed classifications from Q5 (least precise) to Q15 (most precise); the order is opposite from AGMA 2015. A convenient comparison can be made by noting that the sum of the quality numbers from AGMA 2015 and AGMA 2000 is always 17. For example, Q12 and A5 are closely equivalent.

In this book, we use the AGMA 2015-1-A01 standard. Its classifications of A2 to A11 are keyed to a term called the *dynamic factor*, K_v , introduced later in this chapter. Note that the actual tables of data for tolerances are listed in terms of the metric module system with tolerance values given in micrometers (μm). Be aware that the tolerance values are quite small. For tolerance grade A5, tolerances are in the order of around $6.0\ \mu\text{m}$ (0.00024 in) for single pitch deviation for small gears (100 mm; 3.94 in) with small teeth ($m = 1\ \text{mm}$; $P_d = 25.4$) to about $22\ \mu\text{m}$ (0.00088 in) for larger gears (800 mm; 31.5 in) with large teeth ($m = 50\ \text{mm}$; $P_d = 0.51$). Gears are *very precise* mechanical components.

For some applications, often for those of the low accuracy types, the parties involved may agree to use another part of the standard, AGMA 2015-2-A06 for radial measurements based on the total radial composite deviation and the tooth-to-tooth radial composite deviation as shown in Figure 8-36. For this standard, nine classifications are called C4 (most precise) through C12

(least precise). The *double flank roll tester*, described earlier, is used to perform the testing.

Recommended Quality Numbers

All gear-type machinery should be manufactured with good levels of precision reflecting the accuracy with which the gears must operate for good performance, long life, smooth operation, and low noise. The design of the entire system, including shafts, bearings, and the housing must be consistent with the expected degree of precision. Of course, the system should not be made more precise than necessary because of cost. For this reason, manufacturers recommend quality numbers that will give satisfactory performance at a reasonable cost for a variety of applications. Table 9-5 lists several examples.

Machine tools, such as lathes, machining centers, and grinders, are included in the lower part of Table 9-5 with the recommended quality numbers keyed to the pitch line speed of the gears, as defined in Equations (9-1) and (9-2). Higher speeds require greater accuracy. *These values should be used for other kinds of precision industrial equipment not listed in the first part of the table.*

We now continue with the discussion of the dynamic factor, K_v . Figure 9-16 shows a graph of the AGMA-recommended values for K_v , where the A_v numbers are the AGMA-quality numbers described in this section. Gears in typical machine design would fall into the classes represented by curves 8 to 11 which are for gears made by hobbing or shaping with average to good tooling. If the teeth are finish-ground or shaved to improve the accuracy

TABLE 9-5 Recommended AGMA Quality Numbers

Application	Quality number	Application	Quality number
Cement mixer drum drive	A11	Small power drill	A9
Cement kiln	A11	Clothes washing machine	A8
Steel mill drives	A11	Printing press	A7
Grain harvester	A10	Computing mechanism	A6
Cranes	A10	Automotive transmission	A6
Punch press	A10	Radar antenna drive	A5
Mining conveyor	A10	Marine propulsion drive	A5
Paper-box-making machine	A9	Aircraft engine drive	A4
Gas meter mechanism	A9	Gyroscope	A2
Machine tool drives and drives for other high-quality mechanical systems			
Pitch line speed (fpm)	Quality number	Pitch line speed (m/s)	
0–800	A10	0–4	
800–2000	A8	4–11	
2000–4000	A6	11–22	
Over 4000	A4	Over 22	

of the tooth profile and spacing, curves 6 or 7 should be used. Under special conditions where teeth of high precision are used in applications where there is little chance of developing external dynamic loads, the shaded area can be used ($A_v = 2-5$). If the teeth are cut by form milling, curve 12 or higher should be used. Note that the quality 12 gears should not be used at pitch line speeds above 3000 ft/min (15 m/s). Note also that the dynamic factors are approximate. For severe applications, especially those

operating above 4000 ft/min (20 m/s), approaches taking into account the material properties, the mass and inertia of the gears, and the actual error in the tooth form should be used to predict the dynamic load. (See References 6, 19, 20, 22, and 24.)

Equations for Dynamic Factor, K_v . Reading values for K_v from either part of Figure 9-16 gives adequate accuracy because the charts are approximate and slight

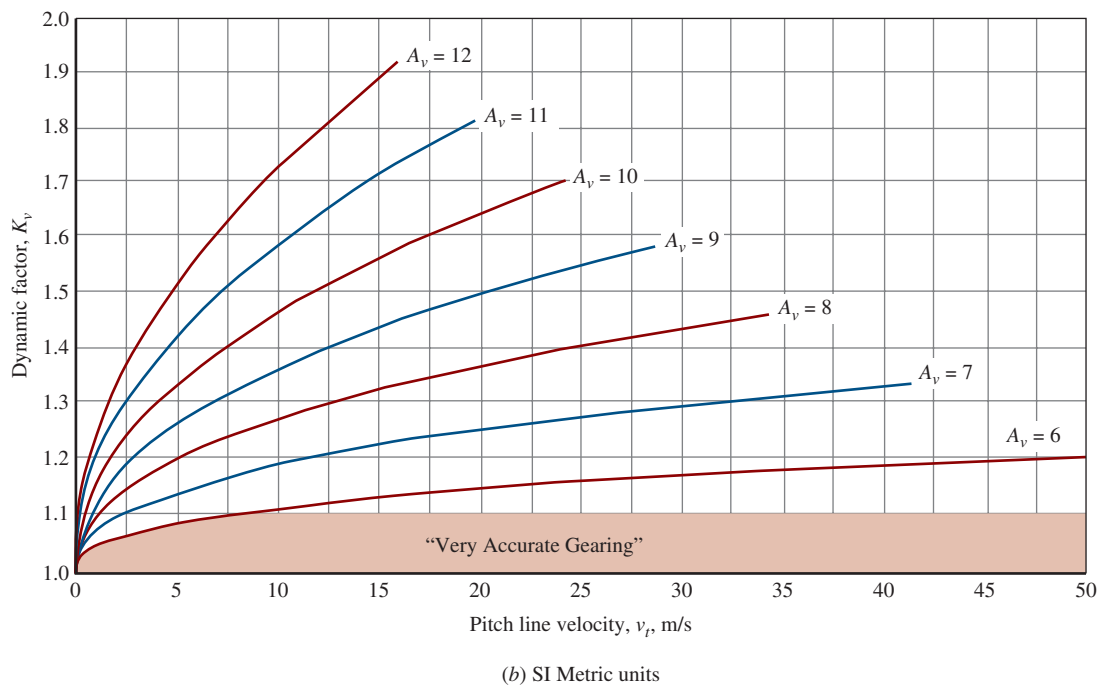
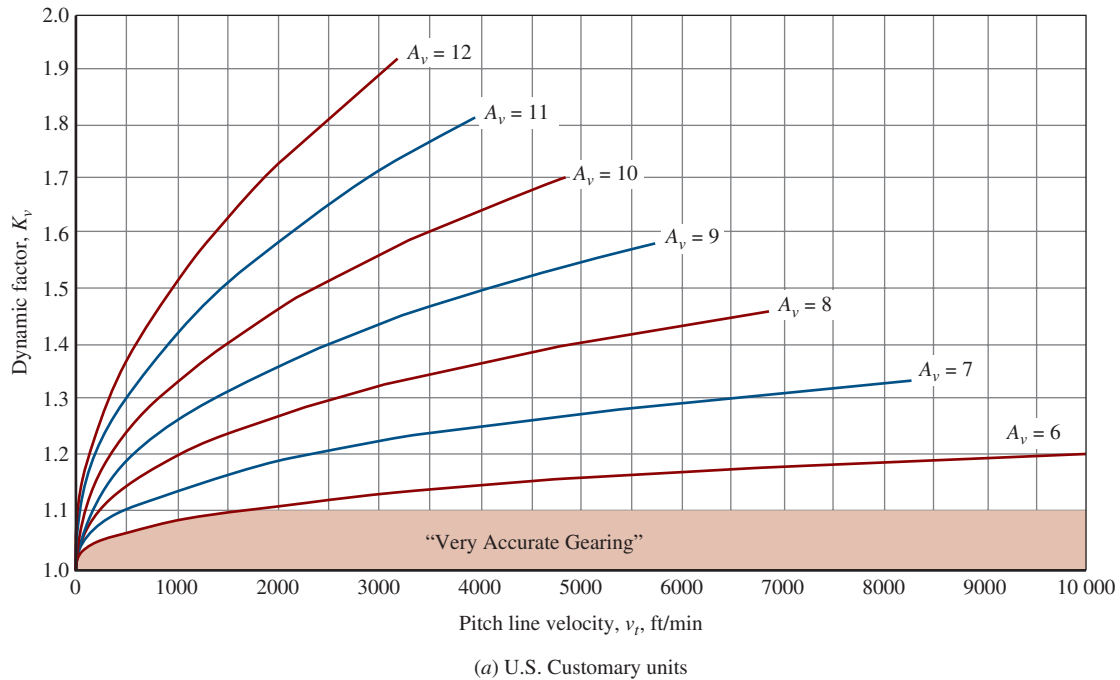


FIGURE 9-16 Dynamic factor, K_v (Adapted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

differences in reading should not cause difficulty with a gear design. However, it is recommended that an analytical method be used when performing several designs. Later in this chapter, a computer-aided approach using a spreadsheet is shown with the analytical method embedded. Table 9–6 shows equations adapted from Reference 6 and presented in both the U.S. and SI Metric unit systems. Variables involved are the dimensionless quality number (A_v) and the pitch line velocity in ft/min for the U.S. system and m/s for the SI Metric system. Both of these data are specified or computed in early stages of the gear design process. The method involves the calculation of two intermediate terms, B and C , and then computing K_v .

Completing the Calculation of Bending Stress

At this point, all of the data required for completing the calculation of the bending stress number, s_{at} , using Equation 9–16, have been defined. Example Problem 9–3 demonstrates the final calculation of that value. Note that some of the same data are used for computing the contact stress in the next section.

TABLE 9–6 Analytical Method for Computing Dynamic Factor, K_v

U.S. units		SI Metric units	
Given A_v and v_t (ft/min)		Given A_v and v_t (m/s)	
$B = 0.25(A_v - 5.0)^{0.667}$		$B = 0.25(A_v - 5.0)^{0.667}$	
$C = 50 + 56(1.0 - B)$		$C = 3.5637 + 3.9914(1.0 - B)$	
$K_v = \left[\frac{C}{C + \sqrt{v_t}} \right]^{-B}$		$K_v = \left[\frac{C}{C + \sqrt{v_t}} \right]^{-B}$	
Maximum velocity for a given A_v :		Maximum velocity for a given A_v :	
$v_{t \max} = [C + (14 - A_v)]^2$		$v_{t \max} = [C + (14 - A_v)]^2$	
A_v	$v_{t \max}$ (ft/min)	A_v	$v_{t \max}$ (m/s)
6	10 000	6	50.8
7	8239	7	41.9
8	6867	8	34.9
9	5731	9	29.1
10	4767	10	24.2
11	3937	11	20.0
12	3219	12	16.4

Example Problem 9–3

Compute the bending stress numbers for the pinion and gear of the pair of gears similar to the pair shown in Figure 9–2(a) and (b). The pinion rotates at 1750 rpm, driven directly by an electrical motor. The driven machine is an industrial saw requiring 25 hp. The gear unit is enclosed and is made to commercial standards. Gears are straddle mounted between bearings. The following gear data apply:

$$N_p = 20 \quad N_g = 70 \quad A_v = 10$$

The gear teeth are 20°, full depth, involute teeth and the gear blanks are solid. The gears will be made from steel, so Figure 9–11 can be used to select an initial value for the diametral pitch.

Solution The equation for the design power is:

$$P_{\text{design}} = K_o \cdot P$$

The required power for the industrial saw is known to be 25 hp. The overload factor is found from Table 9–1. For a smooth, uniform electric motor driving an industrial saw generating moderate shock, a reasonable value would be:

$$K_o = 1.5$$

$$P_{\text{design}} = 1.5 \cdot (25 \text{ hp}) = 37.5 \text{ hp}$$

Since the gears will be made of steel, Figure 9–11 can be used to find an initial diametral pitch based on the design power and the pinion angular velocity, $n_p = 1750$ rpm.

The diametral pitch selected is then $P_d = 6$. Recall that this value has the unit of *teeth/in* or in^{-1} . We will use this form later to ensure proper units in the calculation for stresses in the pinion and the gear.

Data that are useful to visualize the overall size of the gear pair and that serve as input to later decisions are now computed:

Pitch diameters of gear set:

$$D_p = \frac{N_p}{P_d} = \frac{20}{6} = 3.333 \text{ in}$$

$$D_g = \frac{N_g}{P_d} = \frac{70}{6} = 11.667 \text{ in}$$

Center distance of gear set:

$$C = \frac{D_P + D_G}{2} = \frac{3.333 \text{ in} + 11.667 \text{ in}}{2} = 7.500 \text{ in}$$

Velocity ratio of gear set:

$$VR = m_G = \frac{n_P}{n_G} = \frac{N_G}{N_P} = \frac{70}{20} = 3.50$$

The speed of the gear can be found from by rewriting the equation for the velocity ratio:

$$VR = \frac{n_P}{n_G}$$

$$n_G = \frac{n_P}{VR} = \frac{1750 \text{ rpm}}{3.50} = 500 \text{ rpm}$$

The pitch line speed (in ft/min or fpm) can be calculated using the pitch diameter and the angular velocity of the pinion:

$$v_t = \frac{D_P}{2} \cdot n_P = \frac{3.333 \text{ in}}{2} \cdot 1750 \text{ rev/min} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 1527 \text{ fpm}$$

We can also use the gear pitch diameter and angular velocity to calculate the pitch line speed:

$$v_t = \frac{D_G}{2} \cdot n_G = \frac{11.667 \text{ in}}{2} \cdot 500 \text{ rev/min} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 1527 \text{ fpm}$$

We can then use the principles from Section 9–3 to compute the transmitted load on the gear teeth. First find the torque on the pinion and gear:

$$T_P = \frac{\text{Power}}{n_P} = \frac{25 \text{ hp}}{1750 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 900.4 \text{ lb} \cdot \text{in}$$

$$T_G = \frac{\text{Power}}{n_G} = \frac{25 \text{ hp}}{500 \text{ rev/min}} \cdot \frac{33000 \frac{\text{lb} \cdot \text{ft}}{\text{min}}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 3151.3 \text{ lb} \cdot \text{in}$$

We can see that a smaller torque on the pinion shaft will produce a larger torque on the gear shaft.

$$\frac{T_G}{T_P} = \frac{3151.3 \text{ lb} \cdot \text{in}}{900.4 \text{ lb} \cdot \text{in}} = 3.50$$

You will notice this is the same value as the velocity ratio. The output torque will increase in proportion to the decrease in angular velocity. This means that the motor torque required is 900.4 lb-in for an output torque of 3151.3 lb-in.

We will use the Equation (9–16) to compute the bending stress number:

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v$$

Let's go through each term of this equation:

The tangential force (gear driving force) is based on the pinion torque and pinion pitch diameter:

$$W_t = \frac{T_P}{\left(\frac{D_P}{2}\right)} = \frac{900.4 \text{ lb} \cdot \text{in}}{\left(\frac{3.333 \text{ in}}{2}\right)} = 540.3 \text{ lb}$$

The tangential force can also be calculated using the gear torque and gear pitch diameter:

$$W_t = \frac{T_G}{\left(\frac{D_G}{2}\right)} = \frac{3151.3 \text{ lb} \cdot \text{in}}{\left(\frac{10.667 \text{ in}}{2}\right)} = 540.3 \text{ lb}$$

Although the radial and normal gear forces are not required for the bending stress equation, we will calculate all the forces on the spur gear teeth. The radial force (gear separating force) is:

$$W_r = W_t \cdot \tan(\phi) = 540.3 \text{ lb} \cdot \tan(20^\circ) = 196.6 \text{ lb}$$

The normal force along the line of action is:

$$W_n = \sqrt{W_t^2 + W_r^2} = \sqrt{(540.3 \text{ lb})^2 + (196.6 \text{ lb})^2} = 575 \text{ lb}$$

The nominal face width of the gears has been defined as $F = 12/P_d$ and we know $P_d = 6$.

$$F = \frac{12}{P_d} = \frac{12}{6} = 2.00 \text{ in}$$

The geometry factor, J , will be determined for both the pinion and the gear. The pressure angle is 20° so we use Figure 9–10(a).

$$\begin{aligned} J_P &= 0.335 \\ J_G &= 0.420 \end{aligned}$$

The overload factor was already determined:

$$K_o = 1.5$$

Teeth with a diametral pitch, $P_d = 6$, are relatively small so, from Table 9–2, the size factor is:

$$K_S = 1$$

The load-distribution factor, K_m , can be found from Equation (9–17) for commercial enclosed gear drives. For this design, $F = 2.00$ in and

$$\frac{F}{D_P} = \frac{2.00 \text{ in}}{3.333 \text{ in}} = 0.60$$

From Figure 9–12 we can find the approximate value for the pinion proportion factor:

$$C_{pf} \approx 0.04$$

To obtain a more precise value for the pinion proportion factor, we can use the equation that represents the curve in Figure 9–12, when $1.0 \leq F < 15$:

$$C_{pf} = \frac{F}{10D_P} - 0.0375 + 0.0125F = \frac{2.00}{10 \cdot 3.333} - 0.0375 + 0.0125 \cdot 2.00 = 0.047$$

The mesh alignment factor, C_{ma} , can be determined from Figure 9–13 using the commercial enclosed gear unit curve:

$$C_{ma} = 0.16$$

We can also use the equation that represents the commercial enclosed gear unit curve to calculate a more precise value:

$$\begin{aligned} C_{ma} &= 0.127 + 0.0158 \cdot F - 1.093 \times 10^{-4} \cdot F^2 \\ C_{ma} &= 0.127 + 0.0158 \cdot 2.00 - 1.093 \times 10^{-4} \cdot (2.00)^2 = 0.158 \end{aligned}$$

Substitute these two factors into Equation (9–17) to calculate the load distribution factor:

$$K_m = 1.0 + C_{PF} + C_{ma} = 1.0 + 0.047 + 0.158 = 1.21$$

The rim thickness factor, K_B , can be taken as 1.00 because the gears are to be made from solid blanks with no cast or machined rim.

$$K_B = 1.00$$

The dynamic factor, K_v , can be read from Figure 9–16. For a pitch line velocity, $v_t = 1527$ fpm and a gear quality number of, $A_v = 10$, the dynamic factor is:

$$K_v = 1.41$$

The bending stress can now be computed from Equation (9–16). We will compute the bending stress of the pinion first:

$$s_{tP} = \frac{540 \text{ lb} \cdot 6 \text{ in}^{-1}}{2.00 \text{ in} \cdot 0.335} \cdot 1.50 \cdot 1.0 \cdot 1.21 \cdot 1.0 \cdot 1.41 = 12\,376 \text{ psi}$$

Notice all factors in the stress equation are the same for the gear except the value of the geometry factor, J . The bending stress number for the gear is

$$s_{tG} = \frac{540 \text{ lb} \cdot 6 \text{ in}^{-1}}{2.00 \text{ in} \cdot 0.420} \cdot 1.50 \cdot 1.0 \cdot 1.21 \cdot 1.0 \cdot 1.41 = 9871 \text{ psi}$$

The stress in the pinion teeth will always be higher than the stress in the gear teeth because the value of J increases as the number of teeth increases.

9-6 CONTACT STRESS IN GEAR TEETH

In addition to being safe from bending, gear teeth must also be capable of operating for the desired life without significant pitting of the tooth form. *Pitting* is the phenomenon in which small particles are removed from the surface of the tooth faces because of the high contact stresses, causing fatigue. Refer again to Figure 9-9 showing the high, localized contact stresses. Prolonged operation after pitting begins causes the teeth to roughen, and eventually the form is deteriorated. Rapid failure follows. Note that both the driving and driven teeth are subjected to these high contact stresses equally. Reference 11 provides a comprehensive treatment of the wear and failure of gear teeth.

The action at the contact point on gear teeth is that of two externally curved surfaces. If the gear materials were infinitely rigid, the contact would be a simple line. Actually, because of the elasticity of the materials, the tooth shape deforms slightly, resulting in the transmitted force acting on a small rectangular area. The resulting stress is called a *contact stress* or *Hertz stress*. Reference 17 gives the following form of the equation for the Hertz stress,

⇒ Hertz Contact Stress on Gear Teeth

$$\sigma_c = \sqrt{\frac{W_c}{F} \frac{1}{\pi \left\{ [(1 - \nu_1^2)/E_1] + [(1 - \nu_2^2)/E_2] \right\}} \left(\frac{1}{r_1} + \frac{1}{r_2} \right)} \quad (9-18)$$

where the subscripts 1 and 2 refer to the materials of the two bodies in contact. The tensile modulus of elasticity is E and the Poisson's ratio is ν . W_c is the contact force exerted between the two bodies, and F is the length of the contacting surfaces. The radii of curvature of the two surfaces are called r_1 and r_2 .

When Equation (9-18) is applied to gears, F is the face width of the gear teeth and W_c is the normal force delivered by the driving tooth on the driven tooth, found from Equation (9-12) to be,

$$W_N = W_t / \cos \phi \quad (9-19)$$

The second term in Equation (9-18) (including the square root) can be computed if the elastic properties of the materials for the pinion and gear are known. It is given the name *elastic coefficient*, C_P . That is,

⇒ Elastic Coefficient

$$C_P = \sqrt{\frac{1}{\pi \left\{ [(1 - \nu_P^2)/E_P] + [(1 - \nu_G^2)/E_G] \right\}}} \quad (9-20)$$

Table 9-7 gives values for the most common combinations of materials for pinions and gears.

The terms r_1 and r_2 are the radii of curvature of the involute tooth forms of the two mating teeth. These radii change continuously during the meshing cycle as the contact point moves from the top of the tooth through the pitch

TABLE 9-7 Elastic Coefficient, C_P

Gear material and modulus of elasticity, E_G , lb/in ² (MPa)							
Pinion material	Modulus of elasticity, E_P , lb/in ² (MPa)	Steel 30×10^6 (2×10^5)	Malleable iron 25×10^6 (1.7×10^5)	Nodular iron 24×10^6 (1.7×10^5)	Cast iron 22×10^6 (1.5×10^5)	Aluminum bronze 17.5×10^6 (1.2×10^5)	Tin bronze 16×10^6 (1.1×10^5)
Steel	30×10^6 (2×10^5)	2300 (191)	2180 (181)	2160 (179)	2100 (174)	1950 (162)	1900 (158)
Mall. iron	25×10^6 (1.7×10^5)	2180 (181)	2090 (174)	2070 (172)	2020 (168)	1900 (158)	1850 (154)
Nod. iron	24×10^6 (1.7×10^5)	2160 (179)	2070 (172)	2050 (170)	2000 (166)	1880 (156)	1830 (152)
Cast iron	22×10^6 (1.5×10^5)	2100 (174)	2020 (168)	2000 (166)	1960 (163)	1850 (154)	1800 (149)
Al. bronze	17.5×10^6 (1.2×10^5)	1950 (162)	1900 (158)	1880 (156)	1850 (154)	1750 (145)	1700 (141)
Tin bronze	16×10^6 (1.1×10^5)	1900 (158)	1850 (154)	1830 (152)	1800 (149)	1700 (141)	1650 (137)

Source: Extracted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

Note: Poisson's ratio = 0.30; units for C_P are (lb/in²)^{0.5} or (MPa)^{0.5}.

circle, and onto the lower flank of the tooth before leaving engagement. We can write the following equations for the radius of curvature when contact is at the pitch point,

$$r_1 = (D_p/2) \sin \phi \quad \text{and} \quad r_2 = (D_G/2) \sin \phi \quad (9-21)$$

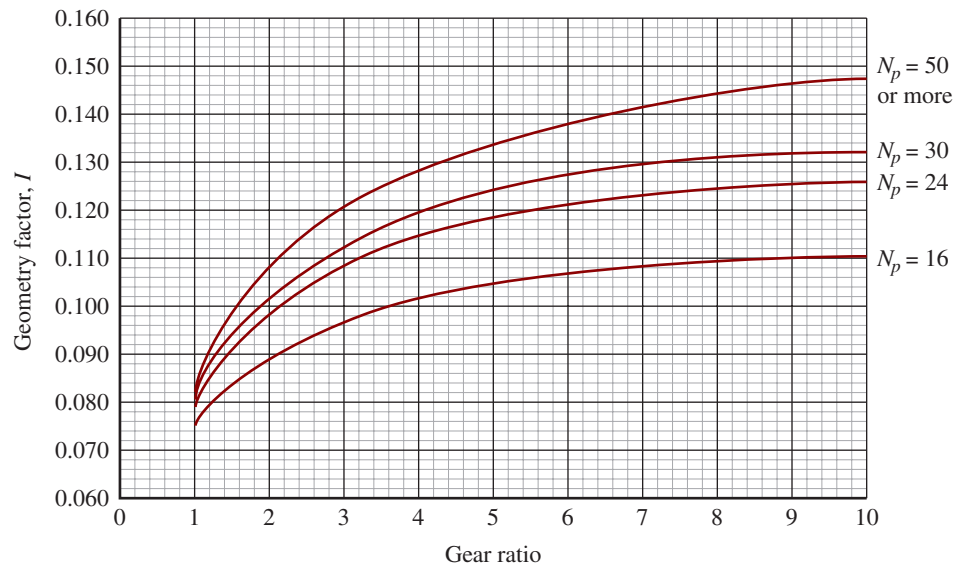
However, the AGMA calls for the computation of the contact stress to be made at the lowest point of single tooth contact (LPSTC) because above that point, the load is being shared with other teeth. Computation of the radii of curvature for the LPSTC is somewhat complex. A geometry factor for pitting, I , is defined by the AGMA to include the radii of curvature terms and the

$\cos \phi$ term in Equation (9-19) because they all involve the specific geometry of the tooth. The variables required to compute I are the pressure angle ϕ , the gear ratio $m_G = N_G/N_p$, and the number of teeth in the pinion N_p .

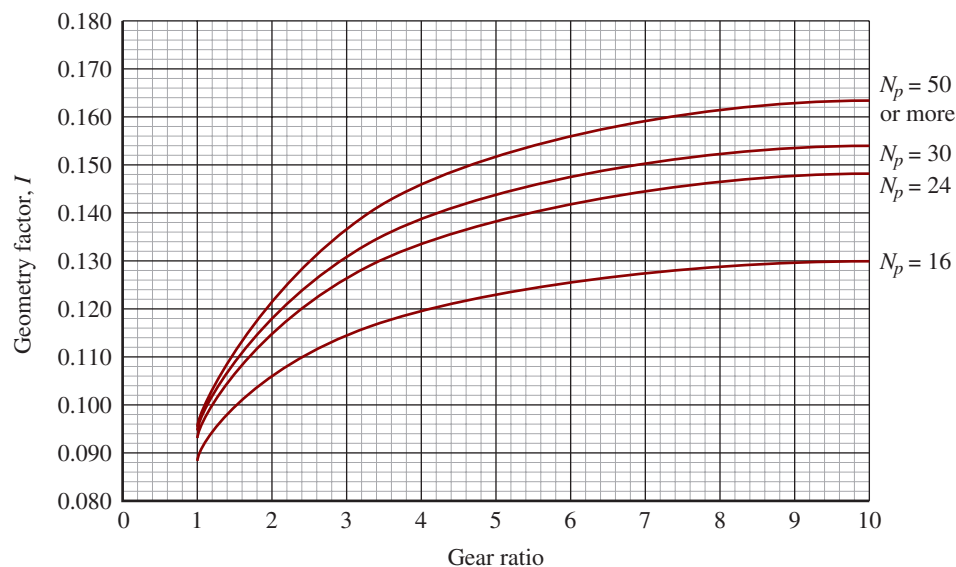
Another factor in the contact stress equation is the pinion diameter that is not included in I . The contact stress equation then becomes,

$$\sigma_c = C_p \sqrt{\frac{W_t}{FD_p I}} \quad (9-22)$$

Values for the geometry factor, I , for a few common cases are graphed in Figure 9-17 and should be used for



(a) 20° pressure angle, full-depth teeth (standard addendum = $1/P_d$)



(b) 25° pressure angle, full-depth teeth (standard addendum = $1/P_d$)

FIGURE 9-17 External spur pinion geometry factor, I , for standard center distances. All curves are for the lowest point of single-tooth contact on the pinion (Extracted from AGMA Standard 218.01, *Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

problem solving in this book. Appendix 19 provides an approach to computing the value for I for spur gears as given in Reference 3.

As with the equation for bending stress in gear teeth, several factors are added to the equation for contact stress as shown below. The resulting quantity is called the *contact stress number*, s_c :

⇒ Contact Stress Number

$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_p I}} \quad (9-23)$$

The values for the overload factor, K_o ; the size factor, K_s ; the load-distribution factor, K_m ; and the dynamic factor, K_v , can be taken to be the same as the corresponding values for the bending stress analysis in the preceding sections.

Equation 9-23 is used to compute the contact stress for both the pinion and the gear; they are equal. It is not correct to use the diameter of the gear, D_G , in this equation.

Example Problem 9-4 follows to illustrate the use of Equation 9-23. Then Section 9-7 presents more information about the types of metallic materials typically used for gears and their heat treatment. Section 9-8 summarizes the process of selecting gear materials, applying information from Sections 9-4 to 9-7.

Example Problem 9-4

Compute the contact stress number for the gear pair described in Example Problem 9-3.

Solution:

Data from Example Problem 9-3 are summarized as follows:

$$\begin{array}{lllll} N_p = 20 & N_G = 70 & F = 2.00 \text{ in} & W_t = 540 \text{ lb} & D_p = 3.333 \text{ in} \\ K_o = 1.50 & K_s = 1.00 & K_m = 1.21 & K_v = 1.41 & P_d = 6 \end{array}$$

Using the gear ratio:

$$m_G = \frac{N_G}{N_p} = \frac{70}{20} = 3.50$$

The number of teeth of the pinion and the pressure angle 20° are used to read an approximate value of pitting resistance geometry factor, I , from Figure 9-17(a):

$$I = 0.108$$

The design analysis for the bending strength indicated that two steel gears should be used. From Table 9-7, we find that:

$$C_p = 2300$$

From Equation (9-23) the contact stress number is:

$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_p I}} \\ s_c = 2300 \sqrt{\frac{540 \cdot 1.50 \cdot 1.00 \cdot 1.21 \cdot 1.41}{2.00 \cdot 3.333 \cdot 0.108}} = 100\,769 \text{ psi}$$

This value is used for both the pinion and the gear.

9-7 METALLIC GEAR MATERIALS

Gears can be made from a wide variety of materials to achieve properties appropriate to the application. From a mechanical design standpoint, strength and pitting resistance are the most important properties. But, in general, the designer should consider the producibility of the gear, taking into account all of the manufacturing processes involved, from the preparation of the gear blank, through the forming of the gear teeth, to the final assembly of the gear into a machine. Other considerations are weight, appearance, corrosion resistance, noise, and, of course, cost. This section discusses several types of metals used for gears. Plastics are covered in a later section.

Many carbon and alloy steels are used for gears and most are given heat treatments to provide controlled hardness and strength (References 8 and 26). Selection of the alloy depends, in part, on the heat-treatment process used to achieve the final properties. Table 9-8 gives some examples.

TABLE 9-8 Examples of Gear Materials

Heat treatment	Typical alloys (SAE numbers)
Through-hardened or Case-hardened by flame or induction hardening	1045, 4140, 4150, 4340, 4350
Carburizing, case-hardened	1020, 4118, 4320, 4820, 8620, 9310

It is important to recognize that alloys listed for through-hardening, flame, or induction hardening must have good hardenability to achieve the desired hardness levels described in the following section. They must also retain reasonable ductility indicated by the percent elongation in the heat-treated condition. All those listed are medium-carbon steels.

Steels that are to be carburized are typically low carbon steels because the carburization process itself infuses significant amounts of carbon into the case while the steel is at a high temperature. Then the final case hardness is achieved by quenching and tempering, leaving the core of the teeth at a lower strength but with higher ductility.

Steel Gear Materials

Through-Hardened Steels. Gears for machine tool drives and many kinds of medium- to heavy-duty speed reducers and transmissions are typically made from medium-carbon steels. AGMA Standard 2001 (Reference 6) gives data for the allowable bending stress number, s_{at} , and the allowable contact stress number, s_{ac} , for steels in the through-hardened condition. Figures 9–18 and 9–19 are graphs relating the stress numbers to the Brinell hardness number, HB, for the teeth. Notice that only knowledge of the hardness is required because of the direct relationship between hardness and the tensile strength of steels. See Appendix 17 for data

that correlate the Brinell hardness number, with the tensile strength of steel in ksi. The range of hardnesses covered by the AGMA data is from 180 to 400 HB, corresponding to a tensile strength of approximately 87 to 200 ksi. *It is not recommended to use through-hardening above 400 HB because of inconsistent performance of the gears in service.* Typically, case hardening is used when there is a desire to achieve a surface hardness above 400 HB. This is described later in this section.

The hardness measurement for the allowable bending stress number is to be taken at the root of the teeth because that is where the highest bending stress occurs. The allowable contact stress number is related to the surface hardness on the face of the gear teeth where the mating teeth experience high contact stresses.

When selecting a material for gears, the designer must specify one that can be hardened to the desired hardness. Review Chapter 2 for discussions about heat-treatment techniques. Consult Appendices 3 and 4 for representative data. For the higher hardnesses, say, above 250 HB, a medium-carbon-alloy steel with good hardenability is desirable. Examples are, listed in Table 9–8. Ductility is also rather important because of the numerous cycles of stress experienced by gear teeth and the likelihood of occasional overloads, impact, or shock loading. A percent elongation value of 12% or higher is desired.

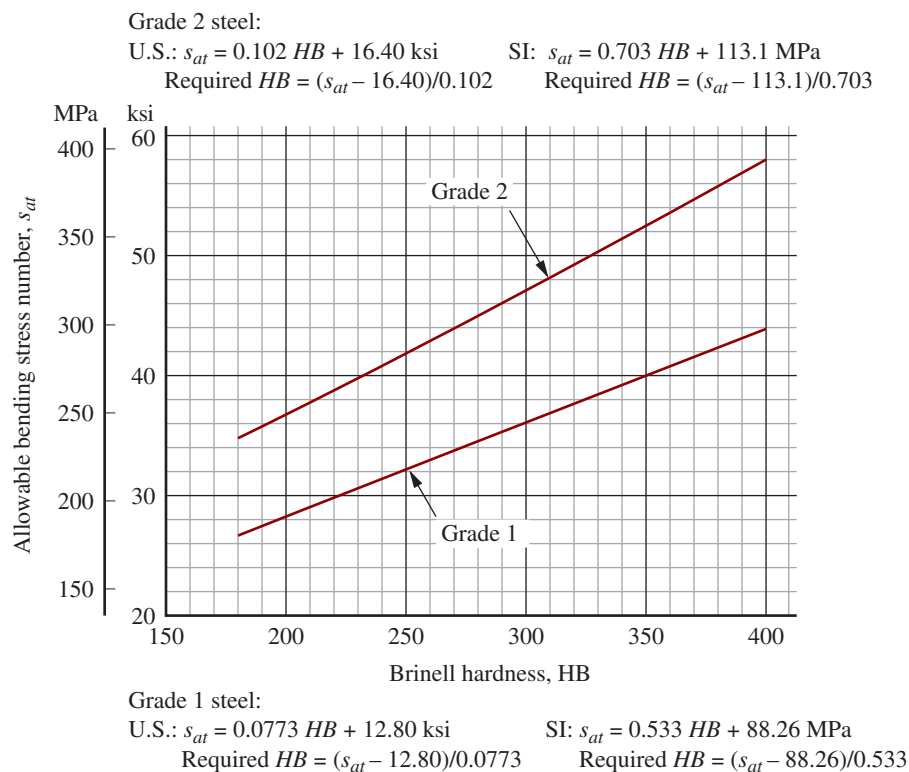


FIGURE 9–18 Allowable bending stress number for through-hardened steel gears, s_{at} (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

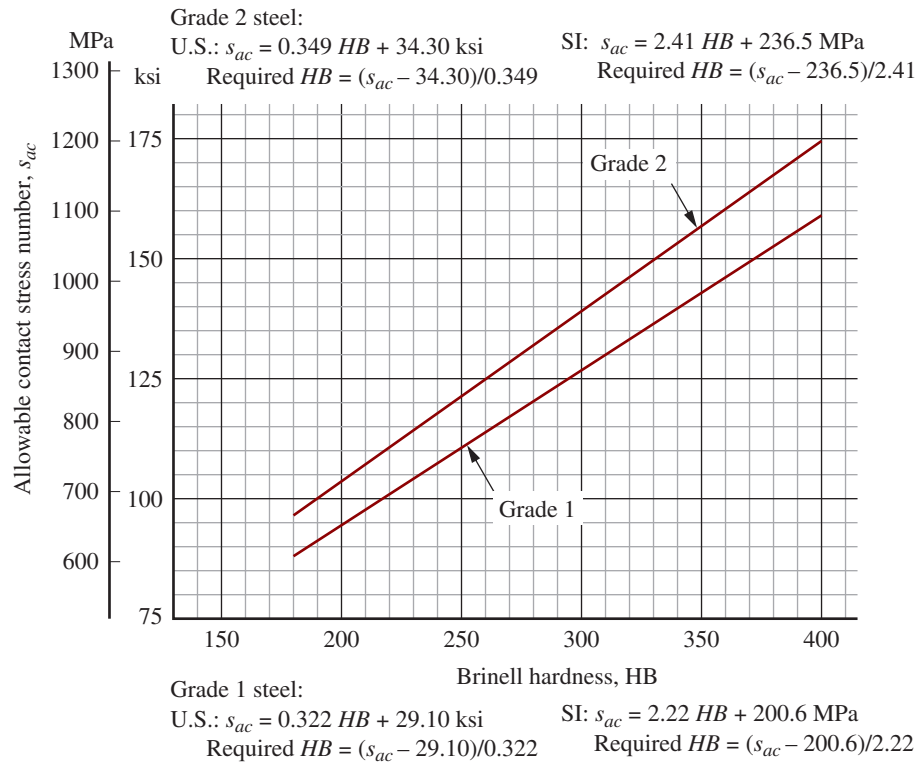


FIGURE 9-19 Allowable contact stress number for through-hardened steel gears, s_{ac} (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

The curves in Figures 9-18 and 9-19 include two grades of steel: Grade 1 and Grade 2. *Grade 1 is considered to be the basic standard and will be used for problem solutions in this book.* Grade 2 requires a higher degree of control of the microstructure, alloy composition, greater cleanliness, prior heat treatment, nondestructive testing performed, core hardness values, and other factors. See AGMA Standard 2001 (Reference 6) for details. Because of these extra requirements, cost is likely to be higher.

Case-Hardened Steels. Flame hardening, induction hardening, carburizing, and nitriding are processes used to produce a high hardness in the surface layer of gear teeth. See the related discussion in Section 2-6. These processes provide surface hardness values from 50 to 64 HRC (Rockwell C) and correspondingly high values of s_{at} and s_{ac} , as shown in Table 9-9. Special discussions are given below for each of the types of case-hardening processes.

Case-hardened steel gears can be produced to Grades 1, 2, and 3 with Grade 1 referring to typically available steels as discussed for through-hardened steels. Because of the special care required in producing Grades 2 and 3, Table 9-9 shows data for only Grade 1 steels as recommended for use in this book. Furthermore, because nitriding can be done in several ways and it is not used as frequently as carburizing, flame-, or induction

hardening, data for design values of bending or pitting resistance stresses are not listed here. Consult Reference 6 for the necessary data.

Flame- and Induction-Hardened Gear Teeth. Recall that these processes involve the local heating of the surface of the gear teeth by high-temperature gas flames or electrical induction coils. By controlling the time and energy input, the manufacturer can control the depth of heating and the depth of the resulting case. It is essential that the heating occur around the entire tooth, producing the hard case on the face of the teeth *and in the fillet and root areas*, in order to use the stress values listed in Table 9-9. This may require a special design for the flame shape or the induction heater. Refer to Reference 6.

The specifications for flame- or induction-hardened steel gear teeth call for a resulting hardness of HRC 50 to 54. Because these processes rely on the inherent hardenability of the steels, you must specify a material that can be hardened to these levels. Normally, medium-carbon-alloy steels (approximately 0.40% to 0.60% carbon) are specified. Table 9-8 and Appendices 3 and 4 list some suitable materials.

Carburizing. Carburizing produces surface hardnesses in the range of 55 to 64 HRC. It results in some of the highest strengths in common use for gears. Special

TABLE 9-9 Allowable Stress Numbers for Case-Hardened Grade 1 Steel Materials

Hardness at surface	Allowable bending stress number, s_{at}		Allowable contact stress number, s_{ac}	
	(ksi)	(Mpa)	(ksi)	(Mpa)
Flame- or induction-hardened				
50 HRC	45	310	170	1172
54 HRC	45	310	175	1207
Carburized and case-hardened				
55–64 HRC	55	379	180	1241

Source: Extracted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

carburizing steels are listed in Appendix 5. Figure 9-20 shows the AGMA recommendation for the thickness of the case for carburized gear teeth. The effective case depth is defined as the depth from the surface to the point where the hardness has reached 50 HRC.

Nitriding. Nitriding produces a very hard *but very thin* case. It is specified for applications in which loads are smooth and well known. Nitriding should be avoided when overloading or shock can be experienced, because the case is not sufficiently strong or well supported to resist such loads. Because of the thin case, the Rockwell

15N scale is used to specify hardness. See References 6 and 8 for design data for nitrided gears.

Iron and Bronze Gear Materials

Cast Irons. Two types of iron used for gears are *gray cast iron* and *ductile* (sometimes called *nodular*) iron. Table 9-10 gives the common ASTM grades used, with their corresponding allowable bending stress numbers and contact stress numbers. Remember that gray cast iron is brittle, so care should be exercised when shock loading is possible. Also, the higher-strength forms of the

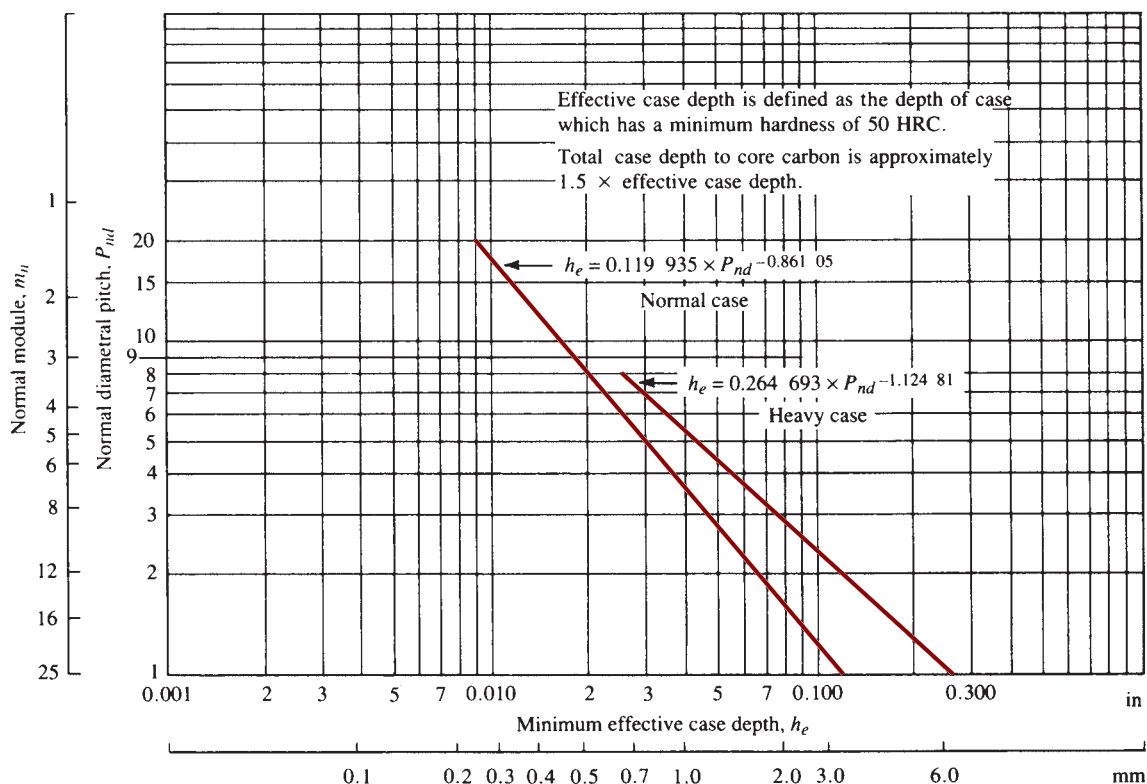


FIGURE 9-20 Effective case depth for carburized gears, h_e (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

TABLE 9-10 Allowable Stress Numbers for Iron and Bronze Gears

Material designation	Minimum hardness at surface (HB)	Allowable bending stress number, s_{at}		Allowable contact stress number, s_{ac}	
		(ksi)	(MPa)	(ksi)	(MPa)
Gray cast iron, ASTM A48, as cast					
Class 20		5	35	50	345
Class 30	174	8.5	59	65	448
Class 40	201	13	90	75	517
Ductile (nodular) iron, ASTM A536					
60-40-18 annealed	140	22	152	77	530
80-55-06 quenched and tempered	179	22	152	77	530
100-70-03 quenched and tempered	229	27	186	92	634
120-90-02 quenched and tempered	269	31	214	103	710
Bronze, sand-cast, $s_{u\min} = 40$ ksi (275 MPa)		5.7	39	30	207
Bronze, heat-treated, $s_{u\min} = 90$ ksi (620 MPa)		23.6	163	65	448

Source: Extracted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

other irons have low ductility. Austempered ductile iron (ADI) is being used in some important automotive applications. However, standardized allowable stress numbers have not yet been specified.

Bronzes. Four families of bronzes are typically used for gears: (1) phosphor or tin bronze, (2) manganese bronze, (3) aluminum bronze, and (4) silicon bronze. Yellow brass is also used. Most bronzes are cast, but some are available in wrought form. Corrosion resistance, good wear properties, and low friction coefficients are some reasons for choosing bronzes for gears. Table 9-10 shows allowable stress numbers for one bronze alloy in two common forms.

9-8 SELECTION OF GEAR MATERIALS

We now use the values of bending stress found in Section 9-5 and contact stress from Section 9-6 to specify a suitable material and its condition that will withstand those stresses without tooth breakage caused by bending stress and with good resistance to pitting on the face of the teeth caused by contact stress. We develop here an approach for specifying steel for the gears. A similar method can be used for other materials such as cast iron and bronze. Later in this chapter, we discuss the design of plastic gears. The goal of the process is to ensure that the predicted stresses are less than the allowable strength of the material. Inputs to the decisions are:

1. Estimated bending stress number, s_t , as found in Section 9-5, Equation (9-16).
2. Estimated contact stress number, s_c , as found in Section 9-6, Equation (9-23).

3. Desired service life and reliability for the gears, discussed in this section.
4. Chosen safety factor, SF , discussed in this section.
5. Allowable bending stress number, s_{at} , and allowable contact stress number for the material, s_{ac} , discussed in Section 9-7.

Note about temperature. The methods developed in this section assume that the operating temperature for the gears is less than 250°F (121°C) and above 32°F (0°C) because we rely on published data for the materials obtained within these limits. Higher temperatures can be considered if data are available on how the material strength is affected. One concern is that many steel gears are heat treated to produce desired strength and ductility and higher operating temperatures can reduce the strength due to tempering. Operating below freezing temperature can be done provided that testing is done to verify adequate impact strength using the Charpy or Izod methods discussed in Chapter 2.

General principles for specifying materials. We use the following relationships to guide the process of specifying suitable materials for gears.

$$s_t < s_{at}' \quad \text{and} \quad s_c < s_{ac}' \quad (9-24)$$

The “prime” applied to the allowable strengths indicates that adjustments are made to published data for s_{at} and s_{ac} that are presented in Section 9-7. Conditions for those data are:

- Operating temperatures are above 32°F (0°C) and below 250°F (121°C).
- The materials are exposed to 10^7 loading cycles.

- The expected reliability is 99%, less than one failure in 100.
- The safety factor, SF , is 1.00.

The adjustments to s_{at} and s_{ac} considered here take the form

$$s_{at}' = s_{at} \frac{Y_N}{(SF)(K_R)} \quad (9-25)$$

$$s_{ac}' = s_{ac} \frac{Z_N}{(SF)(K_R)} \quad (9-26)$$

where SF = safety factor

K_R = reliability factor

Y_N = bending strength stress cycle factor

Z_N = pitting resistance stress cycle factor

Safety factor: The value of SF is typically taken to be 1.0 because most of the uncertainties involved in computing the bending and contact stresses are included in the equations for s_{at} and s_{ac} by the factors K_o , K_s , K_v , and K_m . A design decision can be made for using $SF > 1.00$ for extra safety or for anticipated undesirable conditions beyond these factors. AGMA recommends

$$1.00 \leq SF \leq 1.50$$

Reliability factor: Table 9-11 shows typical values for K_R and the choice of which to apply in a given project is a design decision.

Desired life of the gear drive: Designers must evaluate each project to specify what the design life is to be. Table 9-12 shows one set of recommended design life values in hours of operation. The range from 20 000 to 30 000 hours for general industrial machines is a reasonable choice unless known conditions of the types listed exist. The same table for design life is used for rolling contact bearings in Chapter 14. Also needed is the *number of load cycles* that can be computed using this equation:

$$N_c = (60)(L)(n)(q) \quad (9-27)$$

where N_c = expected number of cycles of loading

L = design life in hours

n = rotational speed of the gear in rpm

q = number of load applications per revolution

The normal number of load applications per revolution for any given tooth is typically, of course, one. But consider the case of an idler gear that serves as both a driven and a driving gear in a gear train. It receives two cycles of load per revolution: one as it receives power from and one as it delivers power to its mating gears. Also, in certain types of gear trains, one gear may deliver power to two or more gears mating with it. Gears in a planetary gear train often have this characteristic.

As an example of the application of Equation (9-27), consider that the pinion in Example Problems 9-3 and 9-4 is designed to have a life of 20 000 h. When rotating at 1750 rpm. Then

$$\begin{aligned} N_c &= (60)(L)(n)(q) = (60)(20\,000)(1750)(1) \\ &= 2.1 \times 10^9 \text{ cycles} \end{aligned}$$

Because this is higher than 10^7 , an adjustment in the allowable bending stress number must be made.

Stress Cycle Factors: Figures 9-21 and 9-22 show recommended values of Y_N for bending stress and Z_N for contact stress for steel gears. Features of these graphs include:

- Each is plotted on log-log scales with the *number of load cycles* on the horizontal axis and the *stress cycle factor* on the vertical axis.
- The stress cycle factor is approximately 1.0 for 10^7 load cycles on both graphs.
- The stress cycle factors are less than 1.0 for load cycles higher than 10^7 and the values are independent of the type of steel or its condition. Some designs for critical service applications use values lower than those shown and the AGMA standard provides an added shaded area for use in such conditions.

TABLE 9-11 Reliability Factor, K_R

Reliability	K_R
0.90, one failure in 10	0.85
0.99, one failure in 100	1.00
0.999, one failure in 1000	1.25
0.9999, one failure in 10 000	1.50

TABLE 9-12 Recommended Design Life

Application	Design life (h)
Domestic appliances	1000–2000
Aircraft engines	1000–4000
Automotive	1500–5000
Agricultural equipment	3000–6000
Elevators, industrial fans, multipurpose gearing	8000–15 000
Electric motors, industrial blowers, general industrial machines	20 000–30 000
Pumps and compressors	40 000–60 000
Critical equipment in continuous 24-h operation	100 000–200 000

Source: Eugene A. Avallone and Theodore Baumeister III, eds. *Marks' Standard Handbook for Mechanical Engineers*. 9th ed. New York: McGraw-Hill, 1986.

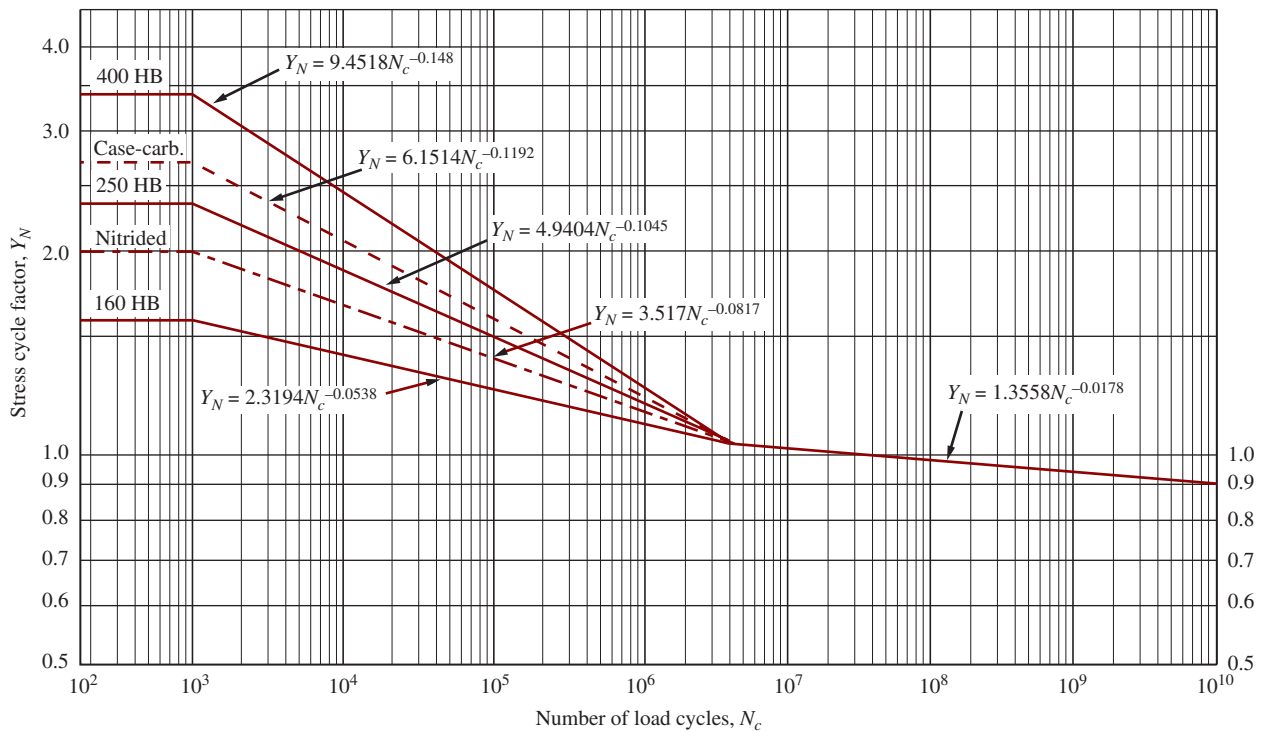


FIGURE 9-21 Bending strength stress cycle factor, Y_N (Adapted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

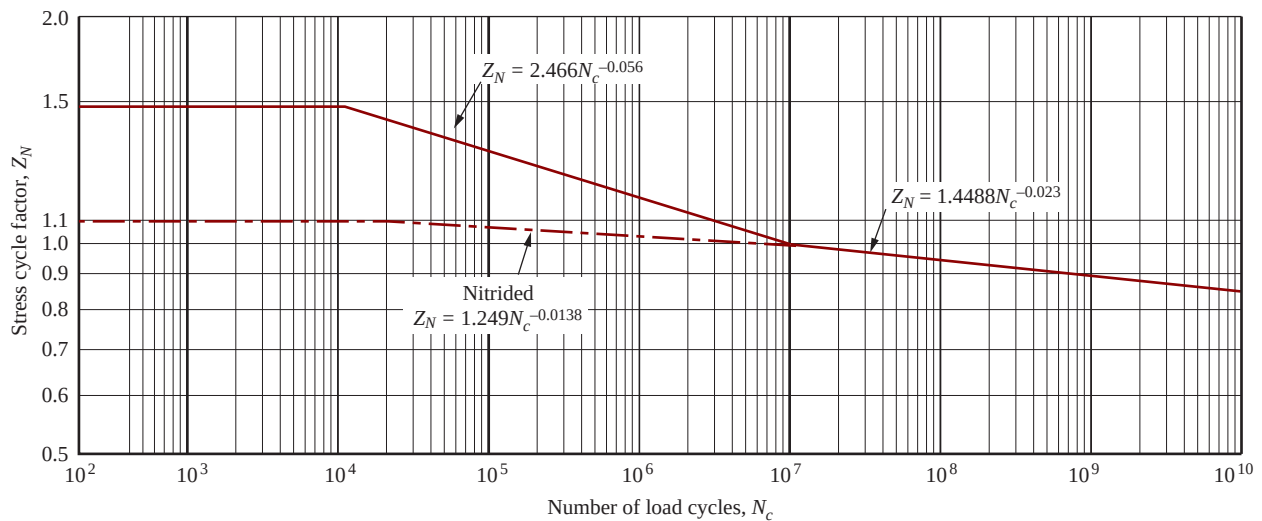


FIGURE 9-22 Pitting Resistance stress cycle factor, Z_N (Adapted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

- The stress cycle factors are greater than 1.0 for load cycles lower than 10^7 and the values are dependent on the type of steel and its condition.
- Equations of an exponential form are given in the graphs to facilitate calculation of the values of the stress cycle factors.
- These charts do not apply to cast iron or to nonferrous materials such as bronze. Values of $Y_N = 1.00$

and $Z_N = 1.00$ can be used while ensuring that the actual strengths of materials specified have a modest margin of safety.

We can now describe the general approach to specifying suitable steels or other metallic materials for gears. The goal is to access the data in Section 9-7, including Figures 9-18 and 9-19 and Tables 9-9, and 9-10, that give accepted values of s_{at} and s_{ac} for through-hardened

steels, case-hardened steels using flame or induction hardening, steels that are carburized and case hardened, cast irons, and bronzes. To facilitate this approach, we can combine Equations (9-24) to (9-26) and then solve for s_{at} and s_{ac} .

$$s_t < s_{at}' = s_{at} \frac{Y_N}{(SF)(K_R)} \quad (9-28)$$

$$s_c < s_{ac}' = s_{ac} \frac{Z_N}{(SF)(K_R)} \quad (9-29)$$

Then, solving for s_{at} and s_{ac} gives

$$\text{Required } s_{at} > s_t \frac{(SF)(K_R)}{Y_N} \quad (9-30)$$

$$\text{Required } s_{ac} > s_c \frac{(SF)(K_R)}{Z_N} \quad (9-31)$$

Both Equations (9-30) and (9-31) must be satisfied. It should be noted here that for gear drives with relatively

long design lives, the design will most often be governed by the pitting resistance and that assumption is made in the following example problems and in more general design procedures.

Both Equations (9-30) and (9-31) must be satisfied and are used when selecting a gear material. If the material is known in the problem, Equations (9-28) and (9-29) are used to solve for the safety factor. The resulting Equations (9-32) and (9-33) are listed next.

For bending stress:

$$SF = \frac{s_{at}}{s_t} \cdot \frac{Y_N}{K_R} \quad (9-32)$$

For contact stress:

$$SF = \frac{s_{ac}}{s_c} \cdot \frac{Y_N}{K_R} \quad (9-33)$$

Example Problem 9-5

Consider three different materials and calculate the safety factor for the pinion and gear using the results from Example Problem 9-3 for the computed bending stresses. Design for a reliability of 0.999, fewer than one failure in 1000. The application is a drive for an industrial saw that will be fully utilized on a normal, one-shift, five-day per-week operation. The materials proposed are:

- Ductile iron ASTM A536 type 80-55-06 quenched and tempered.
- Through hardened steel pinion and gear: SAE 4340 OQT 1000.
- SAE 8620 steel pinion and gear case hardened by carburizing to a hardness of Rockwell C 60-64 with a minimum effective case depth of 0.025 in.

Solution: The results required from Example Problem 9-3 are summarized here:

$$\begin{array}{llll} N_P = 20 & N_G = 70 & P_d = 6 & F = 2.00 \text{ in} \\ n_P = 1750 \text{ rpm} & n_G = 500 \text{ rpm} & D_P = 3.333 \text{ in} & D_G = 11.667 \text{ in} \\ s_{tP} = 12\,376 \text{ psi} & s_{tG} = 9871 \text{ psi} & & \end{array}$$

The equation relating the bending stress number with the adjusted allowable bending stress number is:

$$s_t < s_{at}' = s_{at} \frac{Y_N}{SF \cdot K_R}$$

Solving this equation for the safety factor, SF :

$$SF = \frac{s_{at}}{s_t} \cdot \frac{Y_N}{K_R}$$

Design decisions will be made on the reliability and design life. From Table 9-11, we find $K_R = 1.25$ for the desired reliability of 0.999 as specified in the problem. Because the saw will be fully utilized in an industrial environment, we choose a life of $L = 20\,000$ hours, using Table 9-12 as a guide.

Compute the number of load cycles for the pinion and the gear using Equation (9-27). Each tooth sees one load cycle per revolution, $q = 1$.

$$N_{cP} = (60) \cdot L \cdot n_P \cdot q = 60 \cdot 20\,000 \cdot 1750 \cdot 1 = 2.10 \times 10^9 \text{ cycles}$$

$$N_{cG} = (60) \cdot L \cdot n_G \cdot q = 60 \cdot 20\,000 \cdot 500 \cdot 1 = 6.00 \times 10^8 \text{ cycles}$$

From Figure 9-21, the bending strength stress cycle factor for the pinion and gear is:

$$Y_{NP} = 0.93$$

$$Y_{NG} = 0.95$$

Part (a): Let's look first at using ductile iron ASTM type A536 80-55-06. Using Table 9-10, the allowable bending stress number for this material is

$$s_{at} = 22\,000 \text{ psi}$$

Solving for the safety factor for the pinion:

$$SF = \frac{s_{at}}{s_{tP}} \cdot \frac{Y_{NP}}{K_R} = \frac{22\,000 \text{ psi}}{12\,376 \text{ psi}} \cdot \frac{0.93}{1.25} = 1.3$$

Using the same equation for the safety factor of the gear:

$$SF = \frac{s_{at}}{s_{tG}} \cdot \frac{Y_{NG}}{K_R} = \frac{22\,000 \text{ psi}}{9871 \text{ psi}} \cdot \frac{0.95}{1.25} = 1.7$$

The minimum safety factor is for the pinion and it is within the range of 1.0 to 1.5, as recommended by AGMA, so this design is considered to be satisfactory for bending stress.

Part (b) considers a through hardened steel material SAE 4340 OQT 1000 for the pinion and gear. Using Appendix 3 the properties of the material are:

$$s_u = 171\,000 \text{ psi}, s_y = 158\,000 \text{ psi}, HB = 363, 16\% \text{ elongation}$$

Using Figure 9-18, along with the material Brinell hardness of $HB = 363$, the allowable bending stress number for the through-hardened steel gear is:

$$s_{at} = 41\,000 \text{ psi}$$

Solving for the safety factor SF of the pinion and the gear:

$$SF_P = \frac{s_{at}}{s_{tP}} \cdot \frac{Y_{NP}}{K_R} = \frac{41\,000 \text{ psi}}{12\,376 \text{ psi}} \cdot \frac{0.93}{1.25} = 2.5$$

$$SF_G = \frac{s_{at}}{s_{tG}} \cdot \frac{Y_{NG}}{K_R} = \frac{41\,000 \text{ psi}}{9871 \text{ psi}} \cdot \frac{0.95}{1.25} = 3.2$$

The safety factor is well over 1.5 for both the pinion and the gear for bending stress with this through-hardened steel.

Part (c) considers a gear material, SAE 8620 case hardened by carburizing to Rockwell C 60-64 with a minimum effective case depth of 0.025 in.

Using Table 9-9, the allowable bending stress number for both the pinion and the gear is:

$$s_{at} = 54\,000 \text{ psi}$$

Solving for the safety factor of the pinion and the gear:

$$SF_P = \frac{s_{at}}{s_{tP}} \cdot \frac{Y_{NP}}{K_R} = \frac{54\,000 \text{ psi}}{12\,376 \text{ psi}} \cdot \frac{0.93}{1.25} = 3.2$$

$$SF_G = \frac{s_{at}}{s_{tG}} \cdot \frac{Y_{NG}}{K_R} = \frac{54\,000 \text{ psi}}{9871 \text{ psi}} \cdot \frac{0.95}{1.25} = 4.2$$

The safety factor is well over 1.5 for both the pinion and the gear for bending stress with a carburized, case hardened steel material.

Recall that it was stated earlier that contact stress is most often the most critical stress in the design and operation of gearing. Therefore, we need to evaluate

these proposed materials for the gears for contact stress before judging whether they are satisfactory. We do that in Example Problem 9-6.

Example Problem 9-6

Consider the same three materials used in Example Problem 9-5 and calculate the safety factor for the pinion and gear. Use the results from Example Problems 9-1, 9-2, and 9-3 for the computed contact stress and use the same reliability and number of load cycles as used in Example Problem 9-4.

- Ductile iron ASTM A536 80-55-06 quenched and tempered
- Though hardened steel pinion and gear: SAE 4340 OQT 1000
- SAE 8620 steel, case hardened by carburizing to Rc 60-64 with a minimum effective case depth of 0.025 in.

Solution: The results required from Example Problems 9–1, 9–2, 9–3 are:

$$\begin{array}{llll} N_P = 20 & N_G = 70 & P_d = 6 & F = 2.00 \text{ in} \\ n_P = 1750 \text{ rpm} & n_G = 500 \text{ rpm} & D_P = 3.333 \text{ in} & D_G = 11.667 \text{ in} \\ s_{cP} = s_{cG} = 100\,769 \text{ psi} & K_R = 1.25 & N_{cP} = 2.1 \times 10^9 \text{ cycles} & N_{cG} = 6.0 \times 10^9 \text{ cycles} \end{array}$$

The equation relating the contact stress number with the adjusted allowable contact stress number is:

$$s_c < s'_{ac} = s_{at} \frac{Z_N}{SF \cdot K_R}$$

Use this equation to solve for the safety factor, SF :

$$SF = \frac{s_{ac}}{s_c} \cdot \frac{Z_N}{K_R}$$

From Figure 9–22, we find pitting resistance stress cycle factors:

$$Z_{NP} = 0.88$$

$$Z_{NG} = 0.91$$

Part (a) Let's consider the first proposed material, ductile iron ASTM A536 type 80-55-06. Using Table 9–9, the allowable contact stress number for this material is

$$s_{ac} = 77\,000 \text{ psi}$$

Solving for the safety factor SF of the pinion and gear:

$$SF_P = \frac{s_{ac}}{s_{cP}} \cdot \frac{Z_{NP}}{K_R} = \frac{77\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.88}{1.25} = 0.54$$

$$SF_G = \frac{s_{ac}}{s_{cG}} \cdot \frac{Z_{NG}}{K_R} = \frac{77\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.91}{1.25} = 0.56$$

Both safety factors are below the minimum value of 1.00, as recommended by AGMA and are therefore considered to be unsatisfactory for this application.

Part (b) considers a through hardened steel SAE 4340 OQT 1000. Using Appendix 3, the properties of the material are:

$$s_u = 171\,000 \text{ psi}, s_y = 158\,000 \text{ psi}, \text{HB} = 363, 17\% \text{ elongation}$$

Using Figure 9–19, along with a Brinell hardness of $\text{HB} = 363$ for through-hardened steel gears material, the allowable contact stress number is:

$$s_{ac} = 146\,000 \text{ psi}$$

The safety factors, SF , of the pinion and the gear are:

$$SF_P = \frac{s_{ac}}{s_{cP}} \cdot \frac{Z_{NP}}{K_R} = \frac{146\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.88}{1.25} = 1.0$$

$$SF_G = \frac{s_{ac}}{s_{cG}} \cdot \frac{Z_{NG}}{K_R} = \frac{146\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.91}{1.25} = 1.1$$

The safety factors for the pinion and gear are both within the range of 1.0 to 1.5, as recommended by AGMA, with the pinion's factor at the lowest limit.

Part (c) considers SAE 8620 steel, case hardened by carburizing to a Rc 60-64, with a minimum effective case depth of 0.025 in. Using Table 9–9, the allowable contact stress number is:

$$s_{ac} = 180\,000 \text{ psi}$$

The safety factors SF for the pinion and the gear are:

$$SF = \frac{s_{ac}}{s_{cP}} \cdot \frac{Z_{NP}}{K_R} = \frac{180\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.88}{1.25} = 1.26$$

$$SF = \frac{s_{ac}}{s_{cG}} \cdot \frac{Z_{NG}}{K_R} = \frac{180\,000 \text{ psi}}{100\,769 \text{ psi}} \cdot \frac{0.91}{1.25} = 1.30$$

The safety factors for both the pinion and gear are well within the range of 1.0 to 1.5, as recommended by AGMA.

This completes the development of analyses for both bending and contact stresses and demonstration of how to apply them individually. Here we summarize the results and draw conclusions from them. Example Problems 9-3, 9-4, 9-5, and 9-6 were spread over many pages because the design principles and procedures were developed and demonstrated as those problems were completed. Let's now show the summary of the major design decisions and results of calculations.

Summary of the Results for Example Problems 9-3, 9-4, 9-5, and 9-6

Given data:

Application: Industrial saw operating at 500 rpm, powered by a 25 hp electric motor operating at 1750 rpm, using a gear reducer between them.
 Output speed from the gear reducer: 500 rpm; velocity ratio: 3.50
 Production operation: fully utilized; design life = 20 000 hours
 Enclosed gear reducer made to commercial standards
 High reliability for gear design: $R = 0.999$; less than one failure in 1000

Design decisions:

Diametral pitch: 6
 Pressure angle: $\phi = 20^\circ$
 Face width: $F = 2.00$ in
 Numbers of teeth: pinion = 20; gear = 70
 Quality number for gear geometry: $A_v = 10$

Calculated results:

Geometry of gear pair and forces
 Pitch diameters: $D_P = 3.333$ in; $D_G = 11.667$ in
 Center distance: $C = 7.500$ in
 Pitch line speed: $v_t = 1527$ ft/min
 Transmitted load: $W_t = 540$ lb
 Radial force: $W_r = 196.6$ lb
 Normal force: $W_n = 575$ lb

Factors in stress analyses:

Overload factor: $K_o = 1.50$
 Size factor: $K_s = 1.0$
 Load-distribution factor: $K_m = 1.21$
 Rim factor $K_B = 1.0$
 Dynamic factor: $K_v = 1.41$
 Geometry factors: $J_P = 0.335$; $J_G = 0.420$
 Geometry factor: $I = 0.108$

Stresses in gear teeth:

Bending stress: pinion $s_{tP} = 12\,376$ psi; gear $s_{tG} = 9871$ psi

Contact stress: Pinion and gear $s_c = 100\,769$ psi

Safety factor for three alternative materials for bending stress:

Ductile iron ASTM A536 80-55-06 quenched and tempered

$$SF_P = 1.30 \quad SF_G = 1.70$$

Through hardened pinion and gear: SAE 4340 OQT 1000

$$SF_P = 2.5 \quad SF_G = 3.2$$

SAE 8620 steel, case hardened by carburizing to Rc 60-64; minimum effective case depth of 0.025 in

$$SF_P = 3.20 \quad SF_G = 4.20$$

Safety factor for three alternative materials for contact stress:

Ductile iron ASTM A536 80-55-06 quenched and tempered

$$SF_P = 0.54 \quad SF_G = 0.56$$

Through hardened pinion and gear: SAE 4340 OQT 1000

$$SF_P = 1.00 \quad SF_G = 1.10$$

SAE 8620 steel, case hardened by carburizing to Rc 60-64; minimum effective case depth of 0.025 in

$$SF_P = 1.26 \quad SF_G = 1.30$$

All three materials satisfy the safety factor requirements of AGMA for bending stress.

For contact stress the ductile iron gear material had a safety factor of 0.54, which does not meet the minimum value of 1.00 recommended by AGMA. The gears made from ductile iron could be redesigned in order to lower the contact stress number. A larger number of teeth and correspondingly larger pitch diameter would lower the tangential force. The face width could also be increased. While these changes would lower the contact stress number, the gears would get larger which would increase the overall gearbox housing and related components. This gives a larger gearbox footprint and will increase the cost of all the components. This may not be a prudent design decision.

The through-hardened and case carburized gear materials have safety factors that fall within the 1.0 to 1.50 range recommended by AGMA for contact stress. The considerations then turn to material cost and availability, along with manufacturing process cost, and size requirements of the gear design. The quality number for this design, $A_v 10$, is relatively low and can be attained with a hobbing process. The gear mesh combination using the through-hardened steel pinion mating with

a through-hardened gear could be manufactured using the hobbing process and would not require a secondary operation such as grinding. This would reduce the manufacturing cost of the gear design.

The two case carburized surface hardened gears are the strongest and most durable gear meshes. The disadvantage of this hardening process is the distortion of the gear teeth after heat treating. This would require a secondary grinding process to remove distortions and finish the teeth within specifications. The secondary process would add cost to the gear manufacturing process. Case carburized gears are typically designed to be smaller in size compared to a through-hardened gear set. This decrease in gear size will reduce the overall footprint size of the gearbox which may offset the extra cost for heat treatment and finish grinding.

Final decisions on which material to use and other details of the gear designs are part of the responsibility of the team creating the complete industrial saw.

9-9 DESIGN OF SPUR GEARS TO SPECIFY SUITABLE MATERIALS FOR THE GEARS

In previous sections, the main objective was the *analysis* of the performance of a gear pair with a specific geometry using three proposed materials and heat treatments. This section approaches the gear design process differently. First the geometric characteristics are specified, and then the required material properties, particularly the hardness of the gear teeth, are computed. Finally, a suitable material is specified that meets those requirements.

In designs involving gear drives, normally the required speeds of rotation of the pinion and the gear and the amount of power that the drive must transmit are known. These factors are determined from the application. Also, the environment and operating conditions to which the drive will be subjected must be understood. It is particularly important to know the type of driving device and the driven machine, in order to judge the proper value for the overload factor.

The designer must decide the type of gears to use; the arrangement of the gears on their shafts; the materials of the gears, including their heat treatment; and the geometry of the gears: numbers of teeth, diametral pitch, pitch diameters, tooth form, face width, and quality numbers.

This section presents a design procedure that accounts for the bending fatigue strength of the gear teeth and the pitting resistance, called *surface durability*. This procedure makes extensive use of the design equations presented in the preceding sections of the chapter and of the tables of material properties in Appendices 3 through 5, 8, and 12.

You should understand that there is no one best solution to a gear design problem; several good designs

are possible. Your judgment and creativity and the specific requirements of the application will greatly affect the final design selected. The purpose here is to provide a means of approaching the problem to create a reasonable design.

Design Objectives

Some overall objectives of a design are listed below. The resulting drive should

- Be compact and small
- Operate smoothly and quietly
- Have long life
- Be low in cost
- Be easy to manufacture
- Be compatible with the other elements in the machine, such as bearings, shafts, the housing, the driver, and the driven machine.

The major objective of the design procedure is to define a safe, long-lasting gear drive. General steps and guidelines are outlined here to produce a reasonable initial design. However, because of the numerous variables involved, several iterations are typically made to work toward an optimum design. Details of the procedure are presented in Example Problem 9-7.

Table 9-13 is a combination of a glossary of terms and a reference for finding equations, tables, or figures required to complete a gear design. The procedure given below uses U.S. Customary units and the diametral pitch, P_d , system. Section 9-10 adjusts this procedure using SI metric units and the metric module, m , system. You should refer back to earlier sections of this chapter and to Chapter 8 for details. Several of the unit-specific formulas developed in Section 9-3 are applied in this section to minimize unit manipulations. *You should understand fully the terms used and the how units were handled when these formulas were developed.*

PROCEDURE FOR DESIGNING A SAFE AND LONG-LASTING GEAR DRIVE ▼

1. From the design requirements, identify the input speed of the pinion, n_P , the desired output speed of the gear, n_G , and the power to be transmitted, P .
2. Choose the type of material for the gears, such as steel, cast iron, or bronze.
3. Considering the type of driver and the driven machine, specify the overload factor, K_o , using Table 9-1. The primary concern is the expected level of shock or impact loading.
4. Specify a trial value for the diametral pitch. When steel gears are used, Figure 9-11 provides initial guidance.

TABLE 9-13 Summary and Glossary of Terms Used in Gear Design

Term by cluster	Symbol	Description of use	Reference location
Recommended quality numbers	A_v	Specify value of A_v	Table 9-5
Allowable bending stress number—steel	s_{at}	Specify material	Figure 9-18
Allowable contact stress number—steel	s_{ac}	Specify material	Figure 9-19
Allowable stresses—case-hardened steel	s_{at} and s_{ac}	Specify material	Table 9-9
Allowable stresses—nonferrous	s_{at} and s_{ac}	Specify material	Table 9-10
Geometry factors for bending stress	J_P and J_G	Bending stress	Figure 9-10
Overload factor	K_o	Bending and contact stress	Table 9-1
Size factor	K_s	Bending and contact stress	Table 9-2
Alignment factor: $K_m = 1.0 + C_{pf} + C_{ma}$	K_m	Bending and contact stress	Equation 9-17
Proportion factor:	C_{pf}	Bending and contact stress	Figure 9-12
Mesh alignment factor:	C_{ma}	Bending and contact stress	Figure 9-13
Rim thickness factor	K_B	Bending and contact stress	Figure 9-14
Dynamic factor	K_v	Bending and contact stress	Figure 9-16
Equations for K_v	K_v	Bending and contact stress	Table 9-6
Elastic coefficient	C_P	Contact stress	Table 9-7
Geometry factor for contact stress	I	Contact stress	Figure 9-17
Reliability factor	K_R	Bending and contact stress	Table 9-11
Recommended life in hours	L	Bending and contact stress	Table 9-12
Bending stress cycle factor	Y_N	Bending stress	Figure 9-21
Contact stress cycle factor	Z_N	Contact stress	Figure 9-22
Design pitch selection aid		Specify trial P_d or m	Figure 9-11
Primary equations used in gear design:			
Bending stress number	S_t	$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v$	Equation 9-16
Required allowable bending stress number	S_{at}	Required $s_{at} > s_t \frac{(SF)(K_R)}{Y_N}$	Equation 9-28
Contact stress number	S_c	$s_c = C_P \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_P I}}$	Equation 9-23
Required allowable contact stress number	S_{ac}	Required $s_{ac} > s_c \frac{(SF)(K_R)}{Z_N}$	Equation 9-31

The graph of design power transmitted versus the pinion rotational speed was derived for selected pitches and pinion diameters. Design power, $P_{des} = K_o P$. Steel that is through hardened to HB 300 is used. Because of the numerous variables involved, the value of P_d read from the figure is only an initial target value. Subsequent iterations may require considering a different value, either higher or lower.

- Specify the face width within the following recommended range for general machine drive gears, described earlier:

➤ Nominal Face Width

$$8/P_d < F < 16/P_d$$

$$\text{Nominal value of } F = 12/P_d$$

- Specify or compute the following values:

- Number of teeth in the pinion (N_P) and the gear (N_G) to achieve the desired output speed of the gear.
- Compute the actual output speed of the gear and ensure that it is satisfactory.
- Compute key geometric features: pitch diameters D_P , D_G , and center distance, C . Judge that they are acceptable.
- Compute the pitch line speed, v_t , and the transmitted load, W_t .
- Determine the geometry factors for bending stress, J_P and J_G , and the geometry factor for contact stress, I .

- Specify the quality number for the teeth of the gears using Table 9–5 as a guide.
 - Determine values for all of the factors in Equation (9–16) for bending stress and Equation (9–23) for contact stress.
7. Compute the bending stress and the contact stress on the pinion and gear teeth. Judge whether the stresses are reasonable (neither too low nor too high) in terms of being able to specify a suitable material. If not, select a new pitch or revise the number of teeth, pitch diameter, or face width. Typically the contact stress on the pinion is the limiting value for gears designed for a long life.
 8. Iterate the design process to seek more optimum designs. It is not unusual to make several trials before settling on a particular design. Using computer aids, such as the spreadsheets described in Sections 9–11 and 9–12, can make successive trials quickly.

acceptable by consulting Figure 9–18; they will almost always be acceptable.

9. Using the required hardness HB for the pinion computed in Step 7, specify a suitable material and its heat treatment using Appendices 3 and 4. The charts in Appendix 4 are recommended because they give the widest range of tempering temperatures and corresponding hardness values from which to choose. Examine the lower line of data for HB to find a value just larger than the minimum found from Step 7 and specify the tempering temperature that will produce that hardness level. Table 9–8 recommends several steel alloys that are typically used for gears that are through-hardened; SAE 1045 (similar to 1040), 4140, and 4340 are included and charts for these three alloys are shown in Appendix 4.
10. If Step 8 is not successful, either redesign the gears or consider using case-hardened steel with the values for s_{ac} taken from Table 9–9.
11. If the design is using cast iron or bronze, consult Table 9–10 for values for s_{ac} .

Guidelines for Specifying Metallic Materials for Gears ▼

1. Given the bending stress, s_b , the contact stress, s_c , and the rotational speed of both the pinion, n_P , and the gear, n_G from analyses similar to those in Example Problems 9–3 and 9–4. Note that this also required that the material of choice is to be steel, cast iron, or bronze.
2. Decide on the safety factor; typically $SF = 1.00$ unless unusual conditions exist. AGMA recommends values for SF from 1.0 to 1.5.
3. Decide on the desired reliability and use Table 9–11 to determine K_R ; typically use $K_R = 1.00$ for a reliability of 0.99 (one failure in 100).
4. Specify the desired life in hours for the gears using Table 9–12 as a guide,
5. Compute the expected number of load cycles for both the pinion and the gear using Equation (9–27).
6. Use Figure 9–21 to determine Y_N for the pinion and Figure 9–22 to determine Z_N for the gear. The values may be read from the graphs or computed from the equations given in the figures.
7. Use Equation (9–31) to evaluate the required s_{acP} for the pinion because it typically has the most critical value for s_{ac} . Then consult Figure 9–19 to determine if through-hardened Grade 1 steel can be used. Enter the chart from the value of s_{ac} on the vertical axis; project horizontally to the Grade 1 line; then project vertically down to the Brinell hardness axis to read the minimum acceptable HB hardness. Note that the value should be between HB 180 and HB 400. The most desirable portion of this range is about HB 250 to HB 400. If the value is lower, then a more compact gear size can usually be designed. No value greater than HB 400 should be considered for through-hardened steels. Only Grade 1 steels are recommended.
8. If Step 7 produces a reasonable result, proceed to compute s_{acG} for the gear using Equation (9–31) and compute s_{at} for both the pinion and the gear using Equation (9–30). Check that the bending stress numbers are

Guidelines for Adjustments in Successive Iterations ▼

The following relationships should help you determine what changes in your design assumptions you should make after the first set of calculations to achieve a more optimum design:

- Decreasing the numerical value of the diametral pitch results in larger teeth and generally lower stresses. Also, the lower value of the pitch usually means a larger face width, which decreases stress and increases surface durability.
- Increasing the diameter of the pinion decreases the transmitted load, generally lowers the stresses, and improves the surface durability.
- Increasing the face width lowers the stress and improves the surface durability, but to a generally lesser extent than either the pitch or the pitch diameter changes discussed previously.
- Gears with more and smaller teeth tend to run more smoothly and quietly than gears with fewer and larger teeth.
- Standard values of diametral pitch should be used for ease of manufacture and lower cost (see Table 8-3).
- Using high-alloy steels with high surface hardness results in the most compact system, but the cost is higher.
- Using very accurate gears (with ground or shaved teeth) results in a higher quality number, lower dynamic loads, and consequently lower stresses and improved surface durability, but the cost is higher.
- The number of teeth in the pinion should generally be as small as possible to make the system compact. But the possibility of interference is greater with fewer teeth. Check Table 8–7 to ensure that no interference will occur. (See Reference 23.)

**Example Problem
9-7**

Design a pair of spur gears to be used as a part of the drive for a chipper to prepare pulpwood for use in a paper mill. Intermittent use is expected. An electric motor transmits 3.0 horsepower to the pinion at 1750 rpm and the gear must rotate between 460 and 465 rpm. A compact design is desired.

**Solution and General
Design Procedure**

Step 1. Considering the transmitted power, P , the pinion speed, n_P , and the application, refer to Figure 9-11 to determine a trial value for the diametral pitch, P_d . The overload factor, K_o , can be determined from Table 9-1, considering both the power source and the driven machine.

For this problem, $P = 3.0$ hp and $n_P = 1750$ rpm, $K_o = 1.75$ (uniform driver, heavy shock driven machine). Then $P_{des} = (1.75)(3.0 \text{ hp}) = 5.25$ hp. Try $P_d = 12$ for the initial design.

Step 2. Specify the number of teeth in the pinion. For small size, use 17 to 20 teeth as a start.

For this problem, let's specify $N_P = 18$.

Step 3. Compute the nominal velocity ratio from $VR = n_P/n_G$.

For this problem, use $n_G = 462.5$ rpm at the middle of the acceptable range.

$$VR = n_P/n_G = 1750/462.5 = 3.78$$

Step 4. Compute the approximate number of teeth in the gear from $N_G = N_P(VR)$.

For this problem, $N_G = N_P(VR) = 18(3.78) = 68.04$. Specify $N_G = 68$.

Step 5. Compute the actual velocity ratio from $VR = N_G/N_P$.

For this problem, $VR = N_G/N_P = 68/18 = 3.778$.

Step 6. Compute the actual output speed from $n_G = n_P(N_P/N_G)$.

For this problem, $n_G = n_P(N_P/N_G) = (1750 \text{ rpm})(18/68) = 463.2$ rpm. OK.

Step 7. Compute the pitch diameters, center distance, pitch line speed, and transmitted load and judge the general acceptability of the results.

For this problem, the pitch diameters are:

$$D_P = N_P/P_d = 18/12 = 1.500 \text{ in}$$

$$D_G = N_G/P_d = 68/12 = 5.667 \text{ in}$$

Center distance:

$$C = (N_P + N_G)/(2P_d) = (18 + 68)/(24) = 3.583 \text{ in}$$

$$\text{Pitch line speed: } v_t = \pi D_P n_P / 12 = [\pi(1.500)(1750)] / 12 = 687 \text{ ft/min}$$

$$\text{Transmitted load: } W_t = 33\,000(P)/v_t = 33\,000(3.0)/687 = 144 \text{ lb}$$

These values seem to be acceptable.

Step 8. Specify the face width of the pinion and the gear.

For this problem: Lower limit $= 8/P_d = 8/12 = 0.667$ in.

Upper limit $= 16/P_d = 16/12 = 1.333$ in

Nominal value $= 12/P_d = 12/12 = 1.00$ in. Use this value.

Step 9. Specify the type of material for the gears and determine C_p from Table 9-7.

For this problem, specify two steel gears. $C_p = 2300$.

Step 10. Specify the quality number, A_v , using Table 9-5 as a guide. Determine the dynamic factor from Figure 9-16.

For this problem, specify $A_v = 11$ for a wood chipper. $K_v = 1.35$.

Step 11. Specify the tooth form, the bending geometry factors for the pinion and the gear from Figure 9-10 and the pitting geometry factor from Figure 9-17.

For this problem, specify 20° full depth teeth. $J_P = 0.325$, $J_G = 0.410$, $I = 0.104$.

Step 12. Determine the load-distribution factor, K_m , from Equation (9-17) and Figures 9-12 and 9-13. The precision class of the gear system design must be specified. Values may be computed from equations in the figures or read from the graphs.